Gewu Insight

User Guide

Version 1.0.1 (Build 1)

Your personal research workspace, keeping papers, evidence, and files connected over time.

Designed and developed by Lingyuan Kong and Zhengran Wu,
with coding assistance from OpenAI Codex.

Copyright © 2026 Lingyuan Kong and Zhengran Wu. All Rights Reserved.

About this manual

This PDF is the fixed-layout edition of the in-app Gewu Insight User Guide. It contains the same product guidance as the searchable Help window, arranged for comfortable reading, printing, and archival use. Interface names remain in English so that the manual matches the application.

Three complementary reading paths

Operate the workspace
Parts I and III explain the interface and everyday workflows: navigation, papers and files, importing, monitors, settings, recovery, and transfer.

Understand the product
Parts II and IV explain the local-first philosophy, the visible-workspace model, SQLite's role as the source of truth, and the product's visual identity.

Inspect the technical contract
Part V documents storage, database relationships, network behavior, migration, troubleshooting, and other details needed for maintenance or development.

These paths intentionally share one manual. The user-facing behavior, the product model, and the implementation constraints can therefore be read independently without drifting into separate and contradictory documents.

Contents

About this manual 2

I. Orientation 7

1-1. Welcome 8
1-1.1. Why this workspace exists . . . 8
1-1.2. Why the name 格物洞见 . . . 8
1-1.3. Five design commitments . . . 9
1-1.4. Cross-platform core, platform-specific integration . . . 9
1-1.5. What this guide covers . . . 10
1-1.6. What Gewu Insight is—and is not . . . 10
1-1.7. Six main tabs . . . 10
1-1.8. One continuous research loop . . . 10
1-1.9. Choose a path through this guide . . . 11
1-1.10. Open this guide again . . . 11
1-1.11. Contact Us . . . 11

1-2. Guided Tour 12
1-2.1. Read the interface as a sentence . . . 12
1-2.2. The Library window . . . 12
1-2.3. Follow selection from broad to specific . . . 12
1-2.4. Tabs across the top . . . 13
1-2.5. Watch one paper move through the workspace . . . 13
1-2.6. Resize a pane . . . 13
1-2.7. Recognize stable and temporary state . . . 14

1-3. Your First Session 15
1-3.1. Open your database first . . . 15
1-3.2. Choose your starting route . . . 15
1-3.3. Add and inspect a paper . . . 15
1-3.4. Understand what may now exist . . . 16
1-3.5. Verify the identity before enriching it . . . 16
1-3.6. Complete the first Paper Files bundle . . . 16
1-3.7. Optional: verify shared identity . . . 17
1-3.8. Next useful actions . . . 17
1-3.9. Use the collection while writing . . . 17
1-3.10. Finish with a safe baseline . . . 17

II. Product Concepts and Architecture 18

2-1. System Architecture at a Glance 19
2-1.1. How the parts cooperate . . . 19
2-1.2. What each persistent layer answers . . . 19
2-1.3. Why this separation matters . . . 20

2-2. The Persistent Paper Object 21
2-2.1. One identity can accumulate evidence . . . 21
2-2.2. Do not collapse four different things . . . 21
2-2.3. What happens when the same paper is found again . . . 21
2-2.4. Identity corrections should preserve continuity . . . 22
2-2.5. Publication updates stay with the original work . . . 22

2-3. Why SQLite Is the Core 23
2-3.1. Four families of structured state . . . 23
2-3.2. Why a folder scan is not an equivalent database . . . 23
2-3.3. Transactions, relationships, and WAL . . . 23

2-3.4. Durable preferences and shipped defaults . . . 23
2-3.5. Exports and provenance . . . 24
2-3.6. Paper-level Monitor provenance . . . 24
2-3.7. What the app does during an upgrade . . . 24

2-4. Visible Research Files 25
2-4.1. One paper, one file bundle . . . 25
2-4.2. Managed and linked resources . . . 25
2-4.3. How the default file is resolved . . . 25
2-4.4. Fingerprints and annotations . . . 26
2-4.5. Why Finder and the UI do not look identical . . . 26

2-5. The Personal Research Workspace 27
2-5.1. Views are relational projections . . . 27
2-5.2. Library placements and shortcuts . . . 27
2-5.3. Discovery does not require immediate file management . . . 27
2-5.4. Why the same action can have different scope . . . 27

2-6. From Discovery to Durable Memory 28
2-6.1. A paper can change without losing its history . . . 28
2-6.2. Operational state is part of scientific memory . . . 28
2-6.3. Background work should not seize the foreground . . . 29

2-7. State, Scope, and Safe Change 30
2-7.1. Eight invariants behind normal operations . . . 30
2-7.2. Read an action by its scope . . . 30
2-7.3. Recovery is part of the design . . . 31
2-7.4. Back up one coherent research asset . . . 31

III. User Manual 32

3-1. Workspace Navigation 33
3-1.1. Library Layout . . . 33
3-1.2. Navigate with Mouse and Keyboard . . . 34
3-1.3. Folders, Shortcuts, and Spaces . . . 35
3-1.4. Search, List View, and Table View . . . 38

3-2. Papers and Files 41
3-2.1. Add Papers and Files . . . 41
3-2.2. Use Paper Details . . . 41
3-2.3. Edit DOI and arXiv Identifiers . . . 43
3-2.4. Refresh Info: One Paper, a Folder, or the Library . . . 45
3-2.5. Understand Paper Files . . . 48
3-2.6. Preview Files with Quick Look . . . 50
3-2.7. Choose File Types and the Default File . . . 53
3-2.8. Add and Classify Attachments . . . 54
3-2.9. Download and Sync Versions . . . 55
3-2.10. Move a Standalone File into Paper Files . . . 57
3-2.11. Delete and Recover Items . . . 58

3-3. Import and Transfer 60
3-3.1. Use Folder Loader . . . 60
3-3.2. Import a Zotero Library . . . 62
3-3.3. Make a Portable Desktop Copy . . . 65
3-3.4. Back Up or Move to Another Mac . . . 66

3-4. My Publications 69
3-4.1. My Publications . . . 69

3-5. Journal Monitor **72**
3-5.1. Journal Monitor . . . 72

3-6. Subject Monitor **76**
3-6.1. Subject Monitor . . . 76

3-7. Author Monitor **79**
3-7.1. Author Monitor . . . 79

3-8. Browser **83**
3-8.1. Browse, Bookmark, and Revisit Websites . . . 83

3-9. Settings and Safety **93**
3-9.1. Literature Data Source . . . 93
3-9.2. arXiv Scope Settings . . . 94
3-9.3. Field Settings . . . 96
3-9.4. Quick Look Settings . . . 97
3-9.5. Default Add to Library Folder . . . 99
3-9.6. Website Links . . . 100
3-9.7. AI Access . . . 101
3-9.8. Naming and PDF Rename Rules . . . 113
3-9.9. How Gewu Insight Stores Files . . . 114
3-9.10. Local Data and Privacy . . . 115

3-10. Zotero Integration **117**
3-10.1. One Workspace, Two Interfaces . . . 117
3-10.2. Install, Update, and Remove the Plugin . . . 119
3-10.3. Save Papers and Preview PDFs . . . 120
3-10.4. Connection Settings and Troubleshooting . . . 122

IV. Product Identity 124

4-1. Visual Identity and Cultural Elements **125**

4-2. The calligraphic name **126**
4-2.1. Historical sources of the four characters . . . 126

4-3. The app icon and the seal-script watermark **127**
4-3.1. The app icon . . . 127
4-3.2. The seal-script watermark . . . 127

4-4. Why the three marks are used differently **127**

V. Technical and Operational Reference 128

5-1. Keyboard Shortcuts **129**
5-1.1. Monitor keyboard navigation . . . 129
5-1.2. Browser keyboard context . . . 129
5-1.3. Selection and unread state in Discovery . . . 130

5-2. Publisher and Network Access **131**
5-2.1. If a publisher page is blocked . . . 131
5-2.2. Why figures may arrive later . . . 131
5-2.3. Browser and your system network . . . 131
5-2.4. Check access to the actual main PDF . . . 131
5-2.5. Secure-connection failures and VPN interruptions . . . 131
5-2.6. Service rate limits . . . 132

5-3. Startup, Monitor Cache, and Text Size **133**
5-3.1. Before the first launch . . . 133
5-3.2. Find the database you are using . . . 133
5-3.3. Keep the current App in the Dock . . . 135
5-3.4. Normal startup . . . 135
5-3.5. Startup is not a Library cleanup . . . 135
5-3.6. Automatic local-server recovery . . . 135
5-3.7. When automatic updates run . . . 136
5-3.8. If the app cannot start . . . 136
5-3.9. Clear a stale interface . . . 136
5-3.10. Choose a comfortable text size . . . 136

5-4. Troubleshooting **137**
5-4.1. An upgrade appears to open a different or empty Library . . . 137
5-4.2. The database cannot be opened . . . 137
5-4.3. A PDF cannot be found after a move . . . 137
5-4.4. Paper Files shows Missing File . . . 137
5-4.5. Quick Look is slow on the first open . . . 137
5-4.6. Markdown stays on Loading Markdown . . . 137
5-4.7. Markdown begins with unexpected metadata . . . 138
5-4.8. PDF switching flashes in built-in Quick Look . . . 138
5-4.9. The wrong file is the default . . . 138
5-4.10. A duplicate appears after download . . . 138
5-4.11. A corrected DOI still shows old metadata . . . 138
5-4.12. Refresh complete reports partial results . . . 138
5-4.13. A normal Library item points into Recycle . . . 138
5-4.14. Search misses the phrase you expected . . . 138
5-4.15. A monitor shows no papers . . . 139
5-4.16. Author Monitor shows a red Scholar pause . . . 139
5-4.17. A journal count and the visible cards disagree . . . 139
5-4.18. A journal disappeared from the configured list . . . 139
5-4.19. LaTeX remains visible as source text . . . 139
5-4.20. Folder Loader looks incomplete . . . 139
5-4.21. A default file cannot be deleted . . . 139
5-4.22. Change where website links open . . . 139
5-4.23. Home works offline but a website will not load . . . 140
5-4.24. A publisher stays on security verification . . . 140
5-4.25. AutoFill does not appear or does not fill . . . 140
5-4.26. A WebCrypto keychain prompt keeps appearing . . . 140
5-4.27. Science displays a PDF but Save shows Article only . . . 140
5-4.28. The in-app guide and Print Guide look like different editions . . . 140
5-4.29. A long task keeps returning as Continue . . . 141
5-4.30. Two Paper Files cards seem to be the same file . . . 141
5-4.31. Contact support . . . 141
5-4.32. Download reports supplements but no main PDF . . . 141
5-4.33. A journal's unread dot returns after changing fields . . . 141
5-4.34. A correction or retraction is missing . . . 141
5-4.35. A mathematical note shows formula source . . . 141

5-5. Storage Reference **142**
5-5.1. App and machine locations . . . 142
5-5.2. Associated Files layout . . . 142
5-5.3. Database and exports . . . 143
Copyright and product notice . . . 144

Part I.

Orientation

1-1 Welcome

Gewu Insight is a local workspace for collecting papers, preserving their files and versions, and following journals, subjects, authors, and your own publications.

Version 1.0.1 · Build 1 · October 27, 2026 · Zotero import edition

Source-available research software. The Gewu Insight Academic and Non-Commercial Research License permits downloading, installing, building, running and modifying the software for academic/non-commercial research. Original and modified versions, derivative works and patches may be shared free of charge for those purposes under the same research-use terms, with retained notices and clearly identified modifications. Public research forks and pull requests within those terms need no separate permission. Commercial use, commercial redistribution, sublicensing and integration into commercial products or services require separate prior written authorization. Copyright © 2026 Lingyuan Kong and Zhengran Wu. All rights reserved except for those express permissions. Read the complete LICENSE, also included with the local distribution and inside the App resources, before use. This is not an open-source license.

1-1.1 Why this workspace exists

Research rarely arrives as one clean collection. A paper may first appear in a feed, later acquire an arXiv version and a DOI, be downloaded under several filenames, reappear through an author's profile, and eventually become part of a project or your own publication record. Publisher pages, alerts, PDFs, folders, notes, and citation services each preserve only part of that history.

Gewu Insight is designed to keep those fragments connected. It treats discovery, identity, files, versions, organization, reading state, and provenance as parts of one continuing research process. The aim is not to accumulate the largest possible list. The aim is to preserve enough context that a useful paper remains understandable and usable months or years after it was first found.

1-1.2 Why the name 格物洞见

The name draws on the Chinese idea of 格物致知 and expresses a way of doing research shaped by physics: begin with real-world phenomena and experimental observations, and seek to understand what lies behind them.

Element	Role
格物 · Gewu	Investigate things as they occur in the real world. Let concrete phenomena, experimental observations, and questions from nature give research its starting point.
洞见 · Insight	Seek the patterns and mechanisms behind phenomena through physical pictures, intuition, imagination, and insightful guesses. Mathematics is a tool for developing that understanding.

The name expresses the research outlook that this workspace supports.

Gewu Insight keeps papers, versions, supporting files, and research notes connected so you can return to experimental results, compare physical pictures, and pursue emerging ideas. Read the full explanation in Visual Identity and Cultural Elements.

> Investigate nature through experimental phenomena; seek insight through physical pictures and intuition.（从实验现象中格物，以物理图像和直觉求洞见。）

1-1.3 Five design commitments

Principle	What it means
Local ownership	Your live database and managed research files remain in your selected local Gewu Database folder, separate from the App. Online services contribute evidence; they do not own the workspace.
Visible evidence	PDFs and attachments remain real files that Finder, Preview, and other applications can inspect.
Continuity over time	A preprint, later DOI, publication version, annotations, and personal contexts can accumulate around one persistent paper object.
Explicit state and provenance	Completed, empty, partial, failed, downloaded, imported, and retried are different states. Sources and operational outcomes should remain visible.
Researcher control	Automation assists discovery and maintenance, but classification, interpretation, organization, and scientific judgment remain yours.

Local-first does not mean offline-only
The durable workspace is local, while supported discovery, metadata, profile, and download features may require network access and remain subject to source, publisher, and institutional availability. Verify uncertain matches and versions; the app does not bypass access controls.

1-1.4 Cross-platform core, platform-specific integration

Gewu Insight separates its portable Python, SQLite, and web-interface core from the native desktop layer. Paper identity, database migrations, monitor state, resource relationships, and most interface logic are designed without making macOS the definition of the product.

Platform layer	Current status
Portable core	Python, SQLite, the local application service, and the web-based workspace form the shared architectural foundation.
macOS native release	This is the current primary release and the platform on which the complete desktop workflow is developed and tested, including Finder, Preview, Quick Look, Dock, Trash, and native drag and drop.
Windows development branch	A separate experimental Windows branch is under development. Its Explorer, PDF viewer, Recycle Bin, packaging and native integration have their own implementation and validation. This manual documents the macOS baseline.

Current distribution status
This guide does not claim that a supported Windows package is available today. Instructions that name Finder, Preview, Quick Look, the Dock, or macOS Trash describe the currently tested macOS integration. Consult the Windows branch for its own build instructions and current limitations; macOS screenshots do not establish Windows feature parity.

This edition adds Zotero library import and manual DOI conflict review to the companion guide. The App version and Build identify the host application; the companion plugin has its own version. The Zotero chapters describe the local 0.4.0 preview, tested with Zotero 10.0.6 on macOS. It does not imply that the preview is included in older published installers. Check **Plugin > Zotero Integration** for the package included with your App. Updating the guide or adding the plugin does not require a fresh database or another Library import.

Use the same Gewu workspace in Zotero
The standalone App and the Zotero companion are two interfaces to one Gewu Insight installation and the same selected Gewu Database. Start with One Workspace, Two Interfaces, then follow Install, Update, and Remove the Plugin. To copy Zotero's own collection into Gewu, use the separate Import a Zotero Library workflow.

1-1.5 What this guide covers

This guide follows the work you do in the app: put papers into the Library, organize them, manage publication versions and attachments, review new literature, and keep the database and files safe. It describes the current interface rather than the history of the project.

1-1.6 What Gewu Insight is—and is not

It is	It is not only
A local research workspace that connects discovery, identity, files, versions, organization, and durable state.	An RSS reader that forgets a result after it leaves the feed.
A visible file and resource system built around persistent paper identities.	A PDF folder whose filenames must carry every piece of meaning.
A personal research memory shared by Library, Spaces, publications, and monitors.	Five unrelated tools that happen to occupy the same window.
A system that records evidence and assists repeatable work.	A black-box ranking or summary engine that replaces scientific judgment.

1-1.7 Six main tabs

- **Library** — your managed papers, folders, spaces, versions, and related files.
- **My Publications** — your publication list, citation metrics, author roles, and export tools.
- **Journal Monitor** — papers published on a selected date, organized by journal and field.
- **Subject Monitor** — topic-based discovery from supported online sources and feeds.
- **Author Monitor** — monitored researchers, profile-linked discovery, and newly found work.
- **Browser** — Google search, websites, online PDFs, hierarchical bookmarks, and local browsing history.

1-1.8 One continuous research loop

Element	Role
Acquire **Find**	Journal, Subject, and Author Monitor introduce relevant candidates.
Identify **Recognize**	DOI, arXiv, title, authors, and source evidence are connected to a shared record.
Resolve **Preserve**	Metadata, files, versions, roles, paths, and provenance are made explicit.
Remember **Understand**	Library, Spaces, My Publications, reading state, and personal organization retain meaning.

What you save, read, classify, and follow becomes context for the next cycle. The five research views share this loop; Browser supplies a separate place to explore websites.

> **Important**
> SQLite is the source of truth for Gewu Insight. Do not replace, rebuild, or edit the database by hand. Use the app to move, classify, or delete managed items.

1-1.9 Choose a path through this guide

This is one comprehensive manual with three complementary reading paths. The **operational path** explains how to use the app; the **concept and architecture path** explains why the workspace is organized this way; and the **technical reference path** documents storage, database, network, migration, and troubleshooting details. They remain together so visible behavior, product principles, and implementation constraints do not drift apart.

If you want to	Continue with
Start using the app now	Read **Part I · Orientation**, then use **Part III · User Manual** as the operational guide.
Understand the product idea and architecture	Read **Part II · Product Concepts and Architecture**, followed by **Part IV · Product Identity**.
Maintain, migrate, or inspect the system	Use **Part V · Technical and Operational Reference**, with Part II for the underlying model.
Look up one operation	Use the topic list or press `⌘F` to search this guide.
Prepare an upgrade or move to another Mac	Read Back Up or Move to Another Mac before changing the installation.

1-1.10 Open this guide again

Choose **Help > User Guide**. The guide opens in its own window and works without an internet connection. Search filters the topic list; choose a result to read it, or use Previous and Next to follow the manual. Select **Print Guide** to open the complete, fixed-layout PDF in macOS Preview, where you can print or save a copy. In a separately hosted web UI, that button opens the PDF in a browser tab. It opens the whole guide, not just the topic currently on screen.

1-1.11 Contact Us

Choose **Help > Contact Us** to open a draft in your Mac's default email application, addressed to **info@gewu-insight.com** with the subject **Gewu Insight — Feedback and Support**. This uses Mail only if Mail is your default; another configured email application is supported. Review and send the draft yourself. Include the version/build and the steps that led to the problem; omit passwords and connection tokens.

Official website: gewu-insight.com.

See also: Guided Tour · Your First Session · System Architecture at a Glance · How Gewu Insight Stores Files

1-2 Guided Tour

The top chooses a work mode; the panes move from context on the left to the active object and its evidence; the bottom preserves navigation and progress without reshaping the workspace.

1-2.1 Read the interface as a sentence

Gewu Insight uses a consistent left-to-right grammar. The exact number of panes changes with the task, but their meaning remains stable: choose a research context, narrow it to an object, inspect its evidence, then act. This lets a dense workspace remain predictable without hiding the underlying files and relationships.

Element	Role
Top **Choose a mode**	Library, My Publications, one of the three discovery monitors, or Browser.
Left **Set the context**	A root, folder, date, journal, subject, author, or filter defines the working scope.
Center **Select the object**	Browse folders, paper lists, search results, or the resources belonging to one paper.
Right **Inspect and act**	Read metadata and evidence; open, download, classify, refresh, export, or reorganize.

The bottom edge is reserved for progress, breadcrumbs, and view controls so background work does not repeatedly resize the main content.

1-2.2 The Library window

Area	Purpose
1. Source tabs	Library, shortcuts, and independent Spaces.
2. Folder columns	Move from broad folders to the selected item.
3. Item column	Papers, files, and child folders in the current location.
4. Paper Files	One flat list of resources with independent type, annotation, source, and Default status.
5. Details	Metadata, identifiers, actions, citations, and folder membership.

1-2.3 Follow selection from broad to specific

A selected item does not always mean “open this file.” In a folder column, selection establishes the next level of context. In the item column, it identifies the active paper or standalone file. In Paper Files, it identifies one resource inside the active paper bundle. Paper Details then describes the shared work rather than merely repeating the selected filename.

Selection	What becomes active	What remains shared
A Library folder	Its child folders and items.	The papers keep their identities, resources, and other placements.
A paper in the item column	Its Paper Files bundle and Paper Details.	The same work can still belong to other folders, monitors, or My Publications.
A resource in Paper Files	That version or attachment for preview, classification, or file action.	The parent paper remains the active scholarly object.
A Monitor result	Its discovery evidence, report, and available actions.	Adding it to Library reuses the cataloged paper record when identity has been resolved.

1-2.4 Tabs across the top

Click a top tab to change workspaces and focus that tab. Left/Right then switch tabs; Down enters the page. Monitors show calendars, filters and papers. Browser has a Bookmarks/History sidebar beside a native webpage. Progress messages stay in reserved status areas.

The compact, flat search control at the top right expands on hover or focus. Each research workspace retains its own query; clear it with ×. On macOS, double-click only blank title-bar space to zoom or restore the window. Tabs, search and other controls do not resize it.

Drag the same blank top area to move the window using macOS window movement. With a neighboring desktop available, hold the dragged window at the left or right screen edge to let macOS move it between desktops. Desktop switching and edge tiling follow your macOS configuration. Start from blank space, not a tab, search field or button.

Area	Primary question	Typical next action
Library	What have I chosen to keep, and how is it organized?	Read, classify files, move, copy, attach, or recover.
My Publications	Which works are mine, and what authorship or citation context do they carry?	Verify roles and metrics, refresh, order, or export.
Journal Monitor	What appeared in selected journals on a given date?	Filter, inspect, download, or add to Library.
Subject Monitor	What matches a defined scientific topic across supported sources?	Review source evidence and preserve useful results.
Author Monitor	What is new in the work of a researcher I follow?	Inspect the author relationship, open the work, or add a Library placement.
Browser	What can I find on the web without leaving this workspace?	Search Google, read a website or online PDF, save a bookmark, or revisit history.

1-2.5 Watch one paper move through the workspace

Element	Role
Discovered	A Monitor records where and when the work was observed.
Cataloged	Identity evidence connects the observation to a persistent paper record.
Organized	A Library folder, Space, or My Publications adds personal context.
File-backed	Downloaded or attached resources become available in Paper Files and Finder.

These states can overlap. A cataloged Monitor result may have no local file, and a downloaded file does not have to belong to a Library folder immediately.

1-2.6 Resize a pane

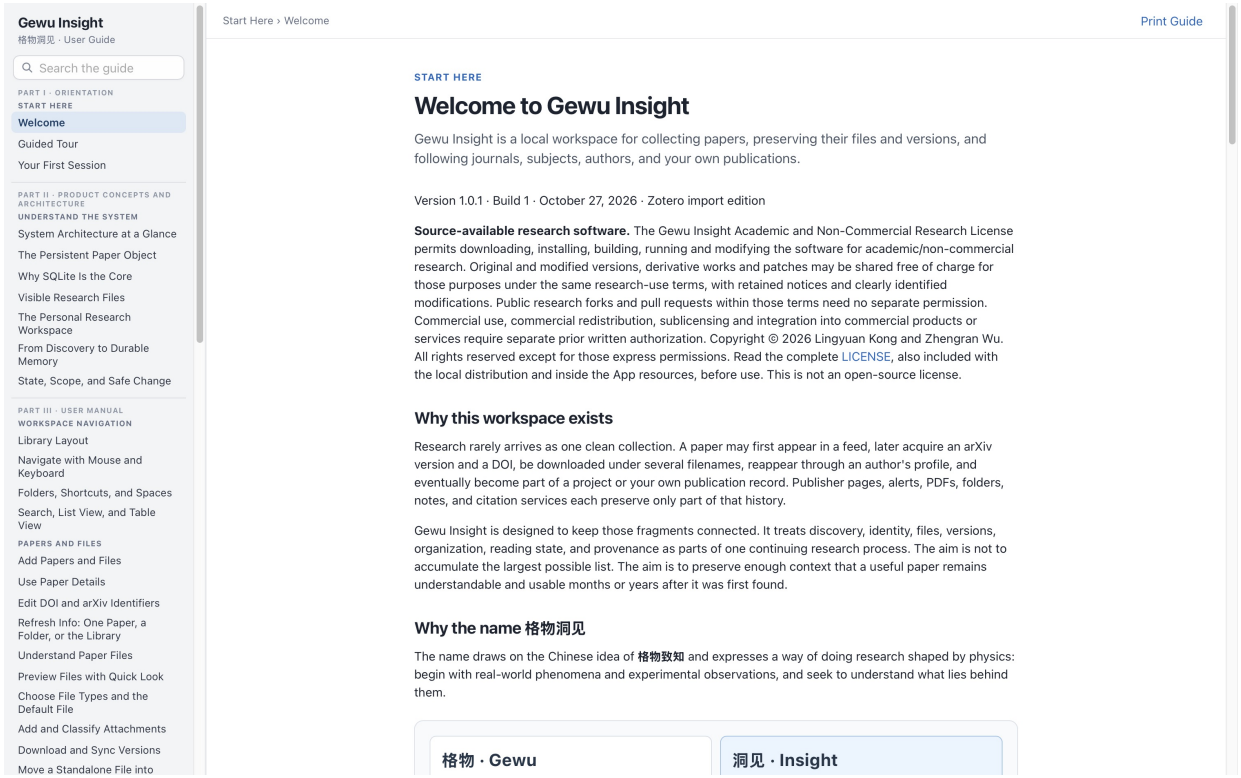
1. **Point to a divider.** Move the pointer to the vertical line between two panes.
2. **Drag the divider.** Move it left or right until the useful content fits.
3. **Release.** The app uses the new working width without changing the underlying files.

> **Note**
> The right-side Paper Details area scrolls independently. A long author list or citation section does not move the Library columns.

1-2.7 Recognize stable and temporary state

Selection, keyboard focus, menus, and scroll position are temporary interface state. Paper identity, placements, resource properties, read state, provenance, and completed operations persist when saved. A breathing progress message means work is active, not that a paper or folder was created.

> **Keep the active context visible**
> Before moving, deleting, or attaching, check the focused pane and selected Paper Details. Similar names can identify different targets: a paper placement, one Paper Files resource, or a standalone file.



The User Guide window. Help opens a separate searchable manual. The sidebar selects topics; Print Guide opens the complete fixed-layout PDF.

See also: Library Layout · Navigate with Mouse and Keyboard · Keyboard Shortcuts

1-3 Your First Session

Start with one paper, confirm its identity and context, then inspect or add a local file when the work is file-backed.

1-3.1 Open your database first

1. **Open Gewu Insight from Applications.** A configured installation remembers its database and opens it automatically. Updating the App does not require another Library import.
2. **If Choose Your Gewu Database appears, select the appropriate action.** Use **Open Existing Database** for an existing collection and select its enclosing `Gewu Database` folder, which contains data/paper_monitor.sqlite. Select the whole folder, not the SQLite file or the inner `data` folder.
3. **Use Create New Database only to start a new collection.** Choose a new folder name at a local location you control, such as `~/Desktop/Gewu Database`. The chooser does not replace an existing folder. This is separate from creating a Space inside an existing Library.
4. **Check Gewu Insight > Database Location….** It displays the active research folder; **Show in Finder** reveals it. Keep that folder together when making a backup.

If the remembered database is unavailable, **Locate Your Gewu Database** offers to open the existing folder or quit. Reconnect its disk or locate the intended folder; the App does not silently create an empty replacement. For an older installation whose data is still inside the source project, follow the one-time migration first. Moving a populated database to a different path needs its file references migrated too.

1-3.2 Choose your starting route

You do not need to prepare a perfect record before beginning. Start from the strongest evidence you already have, then let the paper object become richer over time.

You already have	Start here	What Gewu Insight does first
A local PDF or another research file	Drag it into the target Library folder, or use the folder's add command.	Preserves a visible file, creates a catalog record, and attempts to extract or resolve identity evidence.
A DOI, arXiv identifier, supported URL, or title	Choose **Add Paper** in the target folder and enter the evidence.	Creates or reuses the metadata record. Download or attach a PDF separately when needed.
A useful result in Journal, Subject, or Author Monitor	Select the result and choose **Add to Library**.	Adds a Library placement for the cataloged paper while retaining its discovery provenance.

> **External originals stay separate**
> Folder Loader and ordinary file import are designed to preserve the external source rather than silently reorganize it. The managed copy and the source path can both remain part of the provenance.

1-3.3 Add and inspect a paper

1. **Open Library.** Select an existing folder, or create one from the context menu.
2. **Add a paper.** Drag a PDF into the folder, or open the target folder, Control-click blank space in its contents, choose **Add Paper**, and enter a DOI, arXiv identifier, supported URL, or title.
3. **Select the new item.** Paper Details appears at the right. Gewu Insight attempts to identify the work. A local file is recorded only when you imported one or a supported download succeeded.
4. **Check DOI and arXiv.** Edit either identifier and press `Return`. Wait for the green **Identifier updated** or **Identifier saved** message, then use **Refresh Info** to retrieve metadata.
5. **Inspect the available evidence.** If a local PDF exists, choose **Open in Preview**, or enter Paper Files and press `Space` for Quick Look. If no file exists yet, use **Open Website** or download a supported version when one is available.

Result. The paper now has a cataloged paper record, a stable Gewu Key, and a Library placement. A cataloged paper may have no local file; that state is valid. A file resource exists only after a file is imported, downloaded, or attached. An independent non-PDF file instead uses an internal storage key without a public Gewu Key in Paper Details.

1-3.4 Understand what may now exist

Element	Role
Always **Catalog record**	Title, authors, identifiers, source evidence, and a stable internal paper identity.
Always **Library placement**	A relationship that makes the paper visible in the folder you selected.
When file-backed **File record**	Type, annotation state, source, stored path, fingerprints, and Default status when known.
When file-backed **Visible local file**	The actual PDF or attachment that Finder and Preview can open.

A cataloged-only paper legitimately has the first two objects and no local file. A file-backed paper also has resource metadata and real bytes. These objects cooperate, but they are not interchangeable.

1-3.5 Verify the identity before enriching it

1. **Compare the title and authors with the PDF.** A similar title is useful evidence but not sufficient proof.
2. **Check the DOI and arXiv fields.** A work can have both. Correct an inaccurate identifier before refreshing more metadata or downloading another version.
3. **Read the source and journal information.** Confirm that a publication record belongs to this work rather than a cited paper or a similarly named article.
4. **Keep uncertainty visible.** It is safer to leave a field empty than to confirm a plausible but wrong identifier.

1-3.6 Complete the first Paper Files bundle

A cataloged-only paper may have no local file, and nothing is missing merely because Paper Files is empty. Continue with these steps after you import, download, or attach a file and the paper becomes file-backed.

1. **Select the paper to show Paper Files.** Cards initially show File type and size. Click a card's size footer to show or hide Annotation and Source.
2. **Classify the available file.** Use its small type, annotation, and source controls. Do not infer these facts from the filename alone.
3. **Add related material if you have it.** Add a supplement, annotated copy, note, presentation, data file, code file, or other attachment. Import first, then classify the resource.
4. **Check the default file. Open in Preview** and Library preview use the resolved default when applicable. Selecting **Set as Default** preserves the resource identity and content. The managed Default becomes the main file, including for non-PDF formats; the other files remain in Associated Files. An automatic fallback is available even before you choose a Default manually.
5. **Reveal one file in Finder.** Choose **Show in Finder** and confirm the real managed file exists where the database expects it.

Principle	What it means
Cataloged and organized	The shared paper record and Library placement are durable even when no local file has been added.
File-backed	After a file is added, its independent file properties and physical bytes can be inspected separately from paper identity and folder context.

1-3.7 Optional: verify shared identity

Copy the paper to a second Library folder, then select it there. The title, Gewu Key, Paper Files, and metadata should remain the same because the second folder contains another placement for the shared paper—not an unrelated duplicate. Delete only the extra placement when you are finished with the check.

1-3.8 Next useful actions

- Use **Download** when a better publication version is available.
- Use **Paper Files** to classify imported PDFs or add notes, data, slides, and supplements.
- Add the paper to another folder without creating a second independent work record.
- Back up the project before a large import or migration.

1-3.9 Use the collection while writing

1. **Collect a working set.** Create a project folder or Space and copy relevant papers into it. Copies share the same Gewu Key and Paper Files.
2. **Check the version you cite.** Correct DOI/arXiv if needed, use Refresh Info, and inspect the actual Default PDF. Metadata refresh does not download a publication PDF.
3. **Keep reading notes with the paper.** Add your Markdown notes through **Add File**, or explicitly ask an authorized AI connection to append an analysis. Review the note against the source.
4. **Reuse references.** Use **Copy Code** for BibTeX or **Copy Reference** where shown. Check names, publication year and version in the resulting manuscript. **Export List** in My Publications exports your publication list; it is not a general manuscript editor.
5. **Preserve your baseline.** Make a closed, complete backup before large maintenance operations. Normal reading and writing do not require repeatedly refreshing the whole Library.

1-3.10 Finish with a safe baseline

Your first session is complete when you can answer two questions for every paper: What is this work, and why is it in this folder? If the paper is file-backed, answer two more: Which file is the default, and where are the real bytes? Paper Details, Paper Files, and Finder should answer the applicable questions without contradicting one another.

Principle	What it means
Identity checked	The title, authors, DOI, and arXiv evidence do not point to different works.
Placement understood	The Library folder expresses personal organization, not a second scholarly identity.
Resource classified when present	If the paper is file-backed, its files can have independent type, annotation, and source descriptions.
File visible when present	If the paper is file-backed, the expected managed file opens from the app and can be revealed in Finder.

> **Create the backup habit early**
> Before a major import, upgrade, or migration, quit Gewu Insight and back up both the SQLite database and the managed file roots. A database-only copy lacks some bytes; a files-only copy lacks the relationships and state that make the workspace intelligible.

See also: Add Papers and Files · Use Paper Details · Understand Paper Files · Back Up or Move to Another Mac

Part II.

Product Concepts and Architecture

2-1 System Architecture at a Glance

Gewu Insight is a local research system that coordinates structured state, visible files, and a personal workspace. It is not simply a web page placed in front of a folder.

2-1.1 How the parts cooperate

Element	Role
Native desktop shell	Starts the app, owns Help and Browser, and connects Preview, Finder, and native file actions to the portable core.
Personal Research Workspace	Library, Spaces, My Publications, and the three Monitor views you see and operate.
External research sources	Publisher pages, feeds, metadata services, scholarly profiles, and preprint services.
Local application service	Interprets your actions, resolves identities, schedules network work, and coordinates database and file changes.
SQLite state core	Owns identity, relationships, resource descriptions, preferences, provenance, and durable workflow state.
Visible Research Files	PDFs, versions, supplements, notes, data, and other files that remain inspectable in the native file manager and document applications.

The interface sends an intention to the local service. The service validates it and coordinates SQLite with the relevant files; the UI does not directly treat a Finder folder as the database.

The desktop window presents a local interface served from the project on the current computer. In today's primary release, a macOS shell supplies Finder, Preview, Quick Look, Dock, and native drag-and-drop integration. The Python, SQLite, and web-interface core is designed to support other desktop shells without changing the meaning of the research data. Network sources contribute evidence and downloadable material, but the durable workspace stays local.

2-1.2 What each persistent layer answers

Browser is a separate web surface. Its native WebKit views load websites directly, not through the Python service. The local service stores bookmarks and history in SQLite; WebKit manages a separate persistent website profile. External pages do not receive the Library's local API bridge. Visiting a page, saving a bookmark, or downloading through Browser does not create a paper or assign a Gewu Key; import the paper explicitly.

Layer	Questions it answers	What it does not decide alone
SQLite	What is this paper? Where is it referenced? Which resources belong to it? What happened in the last operation?	It does not contain the bytes of your PDFs and attachments.
Filesystem	Which real files can the native file manager, PDF reader, or another application open?	A filename or folder position alone does not define paper identity, version priority, or monitor membership.
Personal Research Workspace	How should the shared state be presented for discovery, reading, organization, and action?	It is a projection of saved state, not a separate copy of the collection.

System relationship
Persistent research workspace = structured SQLite state + visible research files, coordinated by the local application.

2-1.3 Why this separation matters

A paper can appear in a Monitor, the Library, a Space, and My Publications without becoming four unrelated papers. It can also acquire a DOI after first arriving as an arXiv preprint, or acquire a publication PDF while retaining earlier versions. The database preserves continuity across those changes; visible files preserve user ownership and inspectability.

> **Terminology**
> This guide uses **Personal Research Workspace** for the on-screen projection and **Visible Research Files** for assets visible to the native filesystem. These are explanatory terms, not menu names. Together they form the visible workspace, but they serve different roles.

See also: The Persistent Paper Object · Why SQLite Is the Core · Visible Research Files

2-2 The Persistent Paper Object

Gewu Insight treats a scholarly work as an object that persists through discovery, publication, file changes, and personal organization—not as a temporary search result or a single PDF.

2-2.1 One identity can accumulate evidence

Element	Role
Identity evidence	DOI, arXiv identifier, title, authors, journal, dates, and source identifiers.
Resources	Publication PDFs, preprints, accepted manuscripts, supplements, annotations, and attachments.
Contexts	Library folders, Spaces, My Publications, and Monitor relationships.
Durable state	Read status, metadata quality, provenance, download state, retries, and timestamps.
One shared paper identity · one stable Gewu Key	The internal identity stays stable when an external identifier, filename, folder location, or preferred PDF changes.

Identifiers help resolve and enrich a paper, while the Gewu Key provides continuity inside Gewu Insight.

The central record uses an internal paper identity. A DOI or arXiv identifier is important evidence attached to that identity, but neither is a safe universal primary key: an early preprint may have no DOI, an imported PDF may initially have no reliable identifier, and an incorrect identifier may later be corrected.

2-2.2 Do not collapse four different things

Object	Meaning	Example
Paper record	The shared scholarly identity and normalized metadata.	The work that later connects arXiv and a journal DOI.
Placement or relation	Why and where the paper appears in your workspace.	A Library folder entry, a shortcut, a My Publications role, or an Author Monitor association.
Resource record	A described file or retrievable version belonging to the paper.	Preprint v2, published version, supplementary material, peer review file, or data file.
Physical file	The bytes stored at a real path on disk.	A PDF under data/Library or an associated attachment.

> System relationship
> Paper record ≠ workspace placement ≠ resource record ≠ physical file.

2-2.3 What happens when the same paper is found again

Journal, Subject, and Author Monitor can encounter the same work from different directions. Identity resolution is designed to reuse one shared paper record and add source or context relationships. Copying a paper to another Library folder likewise creates another placement rather than a second independent identity. Its metadata, Gewu Key, versions, and attachments remain shared.

When the available evidence is insufficient or contradictory, separate candidates can remain until their identifiers or metadata are corrected. Automatic metadata enrichment retains conflicting evidence instead of overwriting a saved identity or forcing a merge. File import follows a distinct preservation rule: a PDF matched to an existing paper is kept in its Paper Files and shown in the chosen folder, even if its detected metadata conflicts. Review that assignment and use the file reassignment commands when necessary. The shared-object model is the resolution target; it does not justify merging uncertain works merely because their titles look similar.

A network observation can be transient while acquisition is running. Once a result is cataloged as a shared paper record, it has a Gewu Key even if a particular Monitor view does not display that key. A Monitor result may still have no local Paper Files because no file has been downloaded or attached; adding it to Library creates a visible placement rather than inventing a new scholarly identity.

Gewu Insight also manages independent non-PDF files, such as Markdown overviews, spreadsheets and images. These ordinary files have an internal storage key so that their location and attachments remain trackable, but the App does not display a scholarly Gewu Key for them. A PDF paper can have a public Gewu Key before its DOI or arXiv identity is known; a standalone Markdown note does not become a paper merely by being imported.

2-2.4 Identity corrections should preserve continuity

Editing a DOI or arXiv identifier changes evidence on the existing paper. It should not silently make an unrelated second paper, rename every file, or discard your folder placements and read history. When evidence conflicts, inspect the title, authors, venue, and files before confirming the correction.

> **Important**
> Similar titles are not proof of identity, and matching filenames are not proof that two resources are the same. Use identifiers, metadata, file fingerprints, and provenance together.

2-2.5 Publication updates stay with the original work

A correction, erratum, retraction, expression of concern, withdrawal or addendum is a notice about an article, not a second research-paper identity. Its own DOI, title and date are retained as linked notice metadata under the original Gewu Key. A notice PDF is an associated file and does not become the automatic Default. A corrected full article is a different file type and may become Default through the normal version rules. See Check an article for publication updates.

See also: Edit DOI and arXiv Identifiers · Understand Paper Files · The Personal Research Workspace

2-3 Why SQLite Is the Core

SQLite is the source of truth for Gewu Insight because it preserves the identity, relationships, resource registry, and operational memory that neither a filename nor an exported report can express.

2-3.1 Four families of structured state

Element	Role
SQLite state authority	A local transactional database stored with the project.
Identity and metadata	Shared papers, external identifiers, normalized title and authors, dates, fields, and Gewu Key.
Relationships and context	Library placements, folder tree, Spaces, My Publications, and Monitor associations.
Resource registry	File types, paths, source evidence, hashes, annotation state, and default preferences.
Operational memory	Fetch runs, import batches, source tasks, reads, settings, failures, and retries; also Browser bookmarks and history.

Table names and schema versions can evolve; these four responsibilities are the stable user-level model.

The database does not store the PDF bytes themselves. It stores enough structured information to understand the files: which paper they belong to, what file type they represent, where they are, where they came from, whether they are annotated, and which one should be used by default.

2-3.2 Why a folder scan is not an equivalent database

Finder can show that `paper.pdf` exists. It cannot reliably say whether it is a published version or a preprint, whether it is annotated, where it came from, whether the same work is also in My Publications, or which file is the manual default. Those facts live in SQLite.

2-3.3 Transactions, relationships, and WAL

Database mechanism	User-facing purpose
Transactions	Keep a multi-step state change together so a partly applied edit is less likely to become the visible truth.
Foreign-key relationships	Connect a resource or placement to the correct shared paper instead of relying on a coincidentally similar filename.
WAL (write-ahead logging)	Lets foreground reading and background updates coexist more safely than repeated full-file rewrites.
Schema migration	Adds new structures and backfills required state during an upgrade rather than asking the user to rebuild the database.

WAL improves normal concurrency; it is not a substitute for a coherent backup. Close Gewu Insight before copying its live database, and back up the managed files with it.

2-3.4 Durable preferences and shipped defaults

Journal subscriptions are durable SQLite state as well. The database preserves which journals are enabled, their identifiers and source settings, and their display order. The journal catalog shipped with a newer app version is a set of defaults, not an instruction to overwrite your personal list.

On the first upgrade that introduces durable subscriptions, Gewu Insight seeds the database from the existing journal configuration and history. After that one-time seed, adding, disabling, restoring, or reordering a journal changes the database record. Replacing source code or rebuilding the app does not silently re-enable a journal you removed or discard journals you added.

> **Cross-layer changes**
> SQLite transactions protect database state. A file move and its database update cannot be one native SQLite transaction, so the local service coordinates the steps and uses validation or compensating recovery when a filesystem operation fails.

2-3.5 Exports and provenance

JSON, HTML, Markdown, Word, LaTeX, and PDF outputs are derived artifacts. They can be regenerated, shared, or inspected, but editing them does not alter the source records. Raw source evidence may also be retained inside SQLite for provenance; this does not turn an external JSON export into a second source of truth.

2-3.6 Paper-level Monitor provenance

Normalized paper-source observations are the authority for how a canonical paper was encountered. Each observation identifies a source kind, provider name, and stable source key, and keeps first-seen and last-seen times. The same paper and effective source update one observation instead of increasing the provenance count indefinitely.

Journal Monitor also connects each observed paper to the relevant fetch run. The run supplies the journal, requested date window, status, and timing; the paper-level observation records that the reporting source saw this particular work. Subject and Author Monitor relationships can point to the same canonical paper, so three discovery routes add context without manufacturing three scholarly identities.

A bibliographic metadata refresh can add a newly proven source, but it cannot erase an earlier Journal, Subject, or Author observation. Display counts and the older raw-source list are compatibility projections of the normalized observations rather than an independent truth. When two paper identities are safely merged, their source observations move to the surviving canonical paper and duplicate observations converge.

> **Discovery is not storage**
> A provenance relationship does not by itself create a Library placement or a local PDF. It records how the work entered view. Organization and file acquisition remain separate, explicit actions.

2-3.7 What the app does during an upgrade

On startup, the storage layer reads the database schema version, checks for required tables and fields, applies additive migrations in order, ensures required roots exist, and backfills stable keys or resource records where older data needs them. A normal migration preserves paper identities, folder placements, monitored-journal choices, fetch provenance, and file paths. This is why a project upgrade should preserve the existing database and files rather than replace them with an empty database.

> **Do not edit the live database manually**
> A seemingly small manual change can violate a relationship that is not visible in one table. Use the app for normal work and make a closed, complete backup before any diagnostic intervention.

See also: Back Up or Move to Another Mac · Local Data and Privacy · Storage Reference

2-4 Visible Research Files

Your PDFs and attachments remain real local files. Gewu Insight adds a resource model around them so several versions and related materials can belong to one paper without hiding them inside an opaque cloud library.

2-4.1 One paper, one file bundle

Element	Role
Persistent Paper Object	Owns the resource relationships and decides which usable file is the default.
Default file	The automatically ranked or explicitly preferred valid file used by Open in Preview and Library preview.
Associated Files	Earlier versions, supplements, annotated copies, presentations, notes, data, code, images, and other attachments.
Paper Files	Flat peer records with independent type, annotation state, and source.
Supplements	Publisher or author-provided supporting files.
Files and figures	User material such as notes, slides, data, code, and extracted or attached images.

The Paper Files pane presents this as a compact bundle rather than another Finder hierarchy.

2-4.2 Managed and linked resources

Storage mode	Meaning	Safe expectation
Managed	The active stored path is inside the Gewu Insight project data area.	Move, classify, replace, or delete it through the app so paths and relationships change together.
Linked	The active stored path remains outside the managed research workspace.	Availability depends on that external path, and the app should not silently reorganize it.

A resource can be managed while retaining a separate external source path as provenance. The existence of that source path does not by itself make the managed copy a linked resource, and the external original remains protected.

Folder Loader is a migration-oriented workflow. It imports a visible collection as a collection instead of asking you to classify every incoming file as an attachment before identity resolution. The original source path and the managed result remain conceptually distinct.

2-4.3 How the default file is resolved

An explicit manual default takes precedence while it remains available and valid. Otherwise, usable files are ranked as Corrected published version, Published version, Accepted manuscript, then Preprint. If none is available, Unknown or Other files provide a stable fallback: prefer the existing main file, otherwise the earliest PDF, otherwise the earliest remaining eligible file. No publication type is inferred from this fallback.

Correction notices, supplements, peer review files, Perspective, News and Views, Summary, Analysis, Source Data and Source Code do not become the automatic default. You may choose an available valid resource manually. Missing files, Invalid file and unresolved identity-conflict files are excluded from automatic selection; some bundles can therefore have no eligible automatic Default.

> **Published files**
> The user-facing type is **Published version**. A later formal correction can be classified as **Corrected published version**; the correction notice itself remains a separate peer.

2-4.4 Fingerprints and annotations

In the normal App PDF import, attachment and download workflows, exact SHA-256, content-oriented fingerprint, annotation count, and annotated state are computed as the resource is registered. They do not depend on a later refresh. Exact SHA-256 answers "are these bytes identical?" Adding a highlight or note normally changes those bytes. The content fingerprint is designed to recognize the same underlying document while discounting annotation-only changes, allowing clean and annotated copies to be related without treating them as blindly interchangeable.

Gewu Insight counts user-created comments and markup—such as highlights, notes, drawings, stamps, and form edits—as annotations. Structural PDF elements such as publisher hyperlinks, internal navigation links, and ordinary page actions do not make a freshly downloaded publisher or arXiv PDF **Annotated**.

An exact duplicate can be reused instead of stored twice. A clean PDF and an annotated copy may have different exact hashes but the same content fingerprint; both can be retained when the annotation state is meaningful.

2-4.5 Why Finder and the UI do not look identical

The filesystem must expose practical paths and folders; the UI must expose the meaning of one paper bundle. A Library paper can therefore retain a compatibility primary file in the visible topic folder while related files live under a mirrored Associated Files path. Paper Files deliberately presents all resource records as flat peers instead of mirroring those physical directories.

> **Important**
> Do not manually move or rename managed resources in Finder. The bytes may still exist, but the saved path and live workspace can disagree. Use an app-supported move, repair, or restore; do not assume a restart will rediscover an arbitrary manual move.

The separate MCP import workflow requires metadata from the AI and performs file integrity checks without the full PDF identification or image-fingerprint workflow. It never starts Resolve as part of import.

See also: Understand Paper Files · Choose File Types and the Default File · How Gewu Insight Stores Files

2-5 The Personal Research Workspace

The five research views—Library, My Publications, Journal Monitor, Subject Monitor, and Author Monitor—share research objects. Browser is the sixth main tab, with its own website session. Moving between research views does not reset a paper's identity.

Element	Role
Journal Monitor	Discovers work by publication venue and date.
Subject Monitor	Discovers work through topic definitions and supported feeds or searches.
Author Monitor	Discovers work through a researcher relationship.
One shared paper identity	Metadata, identifiers, resources, provenance, and state accumulate here.
Library and shortcuts	A durable managed collection with logical placements and fast entry points.
Spaces	Independent organizational roots for projects, courses, or collaborations.
My Publications	Your authorship, role, metrics, ordering, and export context on the same paper.

The same paper can participate in several contexts. Each context adds meaning; it does not require an unrelated duplicate record.

2-5.1 Views are relational projections

A Library folder answers "where do I organize this?" My Publications answers "how does this relate to my authorship?" Author Monitor answers "how does this relate to this researcher?" These are different relationships to the same possible work. The interface queries those relationships and projects the relevant controls, status, and files for the current task.

2-5.2 Library placements and shortcuts

A paper can have more than one Library placement. Copying it to another folder reuses the shared identity and resources. A shortcut tab is another entry point to a Library location, not another paper collection. A Space is an independent organizational root, but papers placed there can still reuse the same identity model.

2-5.3 Discovery does not require immediate file management

A Monitor result may remain a discovery relationship with metadata and status but no local Paper Files. A cataloged work already has a Gewu Key, which Author Monitor also shows. Downloading or attaching a file creates a resource relationship; adding the paper to Library creates a managed placement. Neither action should manufacture a second work merely because the user entered from another tab.

2-5.4 Why the same action can have different scope

Removing an author from Author Monitor removes that monitored-author relationship. Removing a shortcut removes an entry point. Moving a Library placement changes organization. Deleting a managed resource changes the paper's file bundle. The selected view supplies scope, while SQLite preserves what remains shared.

> **A useful mental model**
> The UI shows a personal map of a shared object graph. It is "visible" because you can navigate and act on it, but SQLite—not pixels on the screen—preserves the graph.

See also: Folders, Shortcuts, and Spaces · My Publications · State, Scope, and Safe Change

2-6 From Discovery to Durable Memory

Gewu Insight connects monitoring, identity resolution, version handling, and personal organization into one repeatable research loop.

Element	Role
1. Acquire	Journal, Subject, Author, Folder Loader, or a direct identifier introduces a candidate.
2. Identify	DOI, arXiv, title, authors, source evidence, and existing records are reconciled.
3. Resolve	Metadata, file types, annotation states, sources, paths, hashes, and a default preference are established.
4. Remember	Placements, authorship, reading state, provenance, completion, and failures persist.

The loop is continuous because a work can acquire better evidence, a new publication version, or a new personal context later.

2-6.1 A paper can change without losing its history

1. **Acquire an early work.** Subject Monitor finds an arXiv preprint. Its source and discovery context are stored.
2. **Identify the work.** The title, authors, and arXiv identifier connect it to one shared paper identity.
3. **Resolve its first resource.** A preprint PDF becomes the automatic default because no higher-priority file is available.
4. **Remember your context.** You add it to a Library folder, read it, and attach notes or data.
5. **Acquire new evidence later.** Journal Monitor or a metadata refresh finds the published DOI. Use Download, or open the publisher PDF in Browser and choose Save to Gewu, to acquire the file.
6. **Resolve without erasing.** The published file becomes the automatic default unless a manual choice is sticky; the preprint and its provenance remain peer records.

2-6.2 Operational state is part of scientific memory

A result count alone is ambiguous. The app distinguishes work that was never attempted from work that completed with no results, partially completed, failed, or entered a cooldown before retry. It also remembers imports, downloads, metadata refreshes, source tasks, and read state so the next action can be informed rather than blindly repeated.

State	Meaning	Reason to preserve it
Empty and complete	The requested source or date was checked and returned no matching records.	Prevents “zero” from being mistaken for “not run.”
Partial	Some sources completed while another did not.	Allows a targeted retry instead of discarding useful completed work.
Failed	An operation produced a specific error.	Makes the failure visible and avoids pretending the result is authoritative.
Cooldown or retry scheduled	A remote service asked the app to wait or a transient error needs backoff.	Reduces repeated traffic and protects normal foreground use.

2-6.3 Background work should not seize the foreground

Fetches, metadata refreshes, and imports report progress in reserved status areas and save their state. You should still be able to browse already stored material. Where callers share an online service, service-specific pacing, locks, cooldowns, and retry policies coordinate their requests. These controls reduce unnecessary traffic; they cannot guarantee that a publisher will never rate-limit or block a network address.

> **The loop is user-directed**
> Gewu Insight preserves the consequences of what you monitor, save, read, ignore, classify, and publish. It does not silently replace your research judgment with an invisible global keyword filter.

See also: Journal Monitor · Subject Monitor · Author Monitor

2-7 State, Scope, and Safe Change

A safe local research workspace changes only the object and relationship you intended while preserving shared identity, resources, provenance, and recoverable state.

2-7.1 Eight invariants behind normal operations

Principle	What it means
One shared identity	The same scholarly work should not fragment merely because several sources or folders found it.
Many contexts, not blind copies	Folders, Spaces, monitors, and My Publications add relationships around the shared paper.
Deletion has scope	Removing one relationship should not destroy unrelated relationships or shared resources.
State survives restart	Saved reading and workflow state, retries, failures, versions, and provenance are durable.
Repeated operations converge	Refreshing, importing, or downloading the same evidence should not create endless duplicates.
Good metadata is protected	A weak or empty response should not silently erase stronger saved evidence.
Database and files agree	Saved resource paths should point to the intended real files after an app-managed move.
Failure remains inspectable	Deletion, import, move, and network failure should preserve enough state to recover or retry.

2-7.2 Read an action by its scope

Action	Intended scope	What should remain
Remove an Author Monitor profile	The monitored-author relation and its profile context.	Shared papers, Library placements, My Publications, and managed files.
Remove a shortcut	That fast entry point.	The underlying Library folder and its papers.
Copy a paper to another folder	Add another placement.	One shared paper identity and resource bundle.
Move a paper in the app	Change its managed placement and corresponding stored paths.	Gewu Key, metadata, versions, and associations.
Make another file default	Change the reading preference; the selected managed file becomes the main file.	Paper and resource identity, file contents and all other resources. The previous main file is retained in Associated Files, including when the new Default is not a PDF.
Delete an associated file	That resource relationship and its managed file, subject to confirmation.	The paper identity and unrelated resources.

Recycle follows active ownership. As long as any normal Library, Space or My Publications placement uses a paper, its original must remain outside the local Recycle directory. Removing a folder applies this rule to its whole subtree. Re-adding a recycled paper to a normal location brings its original back out, with SQLite paths and shortcuts updated together. See Delete and Recover Items for the separate rules for placements, individual attachments and permanent deletion.

2-7.3 Recovery is part of the design

Recycle keeps ordinary removals recoverable. Import batches and operation status provide evidence about what ran. Resource paths, hashes, and provenance help diagnose missing or duplicate files. The app can migrate older storage structure forward, but migration is not a reason to discard the previous database or original files.

2-7.4 Back up one coherent research asset

A reliable backup contains the SQLite database and the managed file roots from the same closed application state. Copying only the database preserves descriptions without every artifact; copying only folders preserves bytes without all identities, relationships, classifications, and operation history. A portable desktop copy is useful for sharing visible folder material, but it is not automatically a complete live-workspace backup.

> **Before a major update or migration**
> Quit Gewu Insight, preserve the complete selected Gewu Database and any external linked originals, verify the backup can be read, then update the software while keeping that database. Do not create a fresh empty database and attempt to reconstruct the workspace from export files.

See also: Delete and Recover Items · Back Up or Move to Another Mac · Troubleshooting

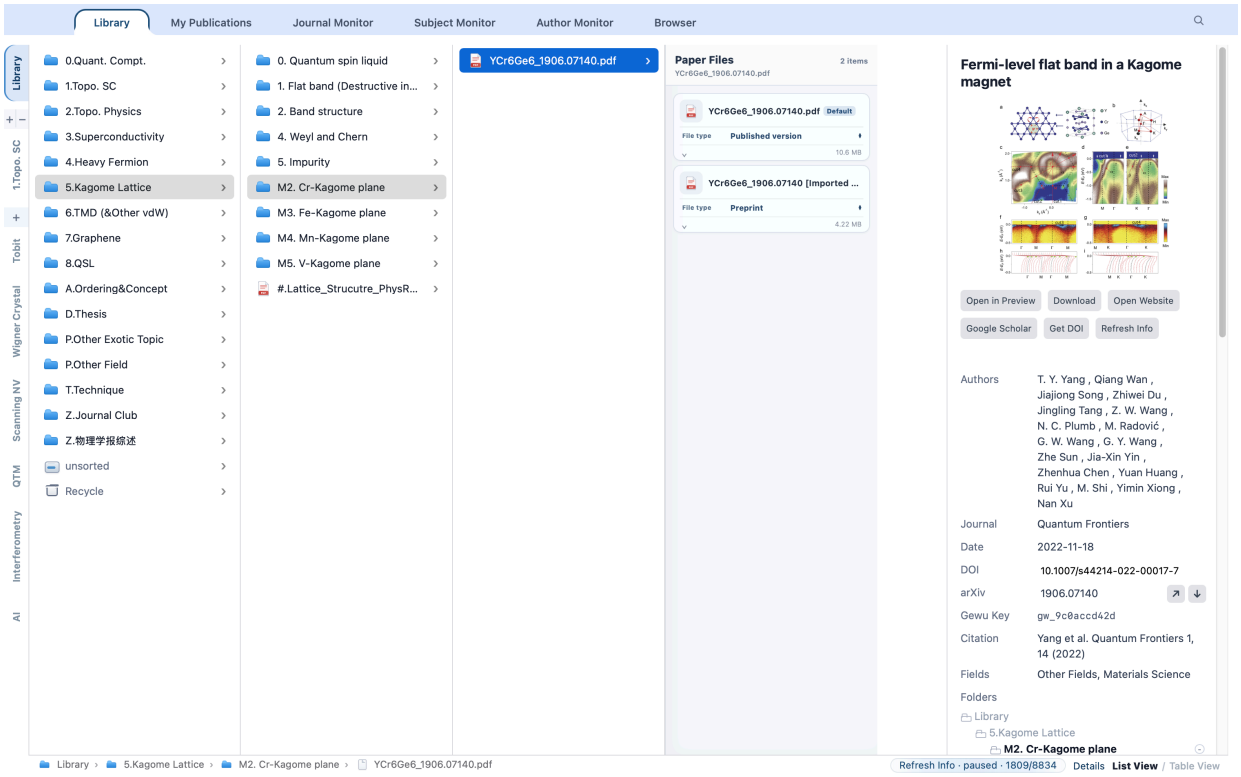
Part III.

User Manual

3-1 Workspace Navigation

3-1.1 Library Layout

The Library uses a Finder-style column view, but the database—not the visible folder alone—defines the paper and its relationships.



Library and Paper Files. This real paper has two already-classified files: Preprint and Published version. The published file is marked Default. Its folder placement, related files, and Paper Details remain connected.

Read the columns from left to right

1. **Source:** the main Library, a shortcut, or a Space.
2. **Folders:** each selection opens its children in the next column.
3. **Items:** the papers and other files in the selected folder.
4. **Paper Files:** shown only when a paper or managed file can expose related files.
5. **Paper Details:** metadata and actions for the selected work.

Use the breadcrumb

The breadcrumb at the bottom displays the full managed path. A shortcut shows the real ancestry from Library or its independent Space root, including folders above the shortcut target. Its alias does not replace that path. Hover over the path to read it in full when space is limited. The segments are informational; use the columns or Left Arrow to navigate to a parent. The **List View / Table View** control at the lower right changes the representation of items without moving them.

Follow background work in the bottom bar

Folder Loader, batch Refresh Info, and individual downloads or refreshes have separate activity indicators. A long batch stays visible alongside a smaller operation instead of combining their messages. Indicators share the available footer width, shorten long text with an ellipsis, and keep navigation controls available. If many tasks no longer fit, scroll the indicator area horizontally. Hover over an indicator to read its full message and

available task details. The activity dot marks work in progress; a paused task remains visible without that running animation.

The footer's **Details** opens the current Loader operation, or an active/paused Refresh Info task when no Loader notice is present. For a particular saved refresh, use **Loader > Refresh All Metadata** and select it under **Refresh tasks**. These task details are separate from the right-side **Paper Details**. See the complete close, return and stop workflow.

Folder Details

When you select a folder, the right side shows its name, type, and the number of child folders and items. This is informational; it is not another folder level.

See also: Folders, Shortcuts, and Spaces · Search, List View, and Table View · Understand Paper Files

3-1.2 Navigate with Mouse and Keyboard

Library and Space columns form one keyboard path with Paper Files. Focus moves into Paper Files with the Right Arrow and returns to the selected paper with the Left Arrow.

Move through folders and papers

- Press ↑ or ↓ to change the selection in the focused column.
- Press → on a folder to enter it.
- Press ← to return to its parent.
- Double-click a Library paper row to open its Default file: PDFs open in macOS Preview; other formats use their macOS application. An unclassified legacy import falls back to its available current file.

Move between the source rail and Library columns

1. **Return to the leftmost Library column.** Press ← once more to move focus to the vertical source rail.
2. **Press ↑ or ↓ on the rail.** Move through Library, shortcuts, and Spaces. The highlighted source opens immediately; you do not need to press Return.
3. **Press →.** Focus enters the first visible Library column for that source. Its active row changes from the quieter background selection to the stronger blue content selection.

While the source rail owns focus, the current row in the adjacent column remains visible in a quieter blue so you do not lose your place. The rail keeps focus while you move between sources and releases it only after you press →. Once you are navigating Library content, pressing ↑ at the first row stays in the Library instead of unexpectedly moving to the app tabs across the top.

Enter and leave Paper Files

1. **Select a paper in the main item column.** Paper Files appears to its right. No Paper Files card is highlighted yet.
2. **Press →.** Focus enters Paper Files and the active card gains a light-blue selection. The main paper remains visible with a quieter selection color.
3. **Press ↑ or ↓.** Move through the flat Paper Files list.
4. **Press ←.** Focus returns to the main paper. Paper Files no longer shows an active card.

The same focus rules apply in the main Library, a shortcut, and every Space. Paper Files is a database-backed view of the selected work rather than a physical child folder, but the keyboard transition is intentionally continuous: → enters the bundle, ← returns to the same paper, and the next ↑ or ↓ moves in the main item column.

Preview while navigating

In Paper Files, press `Space` to open Quick Look. While Quick Look is open, use ↑ and ↓ to follow adjacent files. Press `Space` again or close Quick Look to dismiss it.

Use the Quick Look window

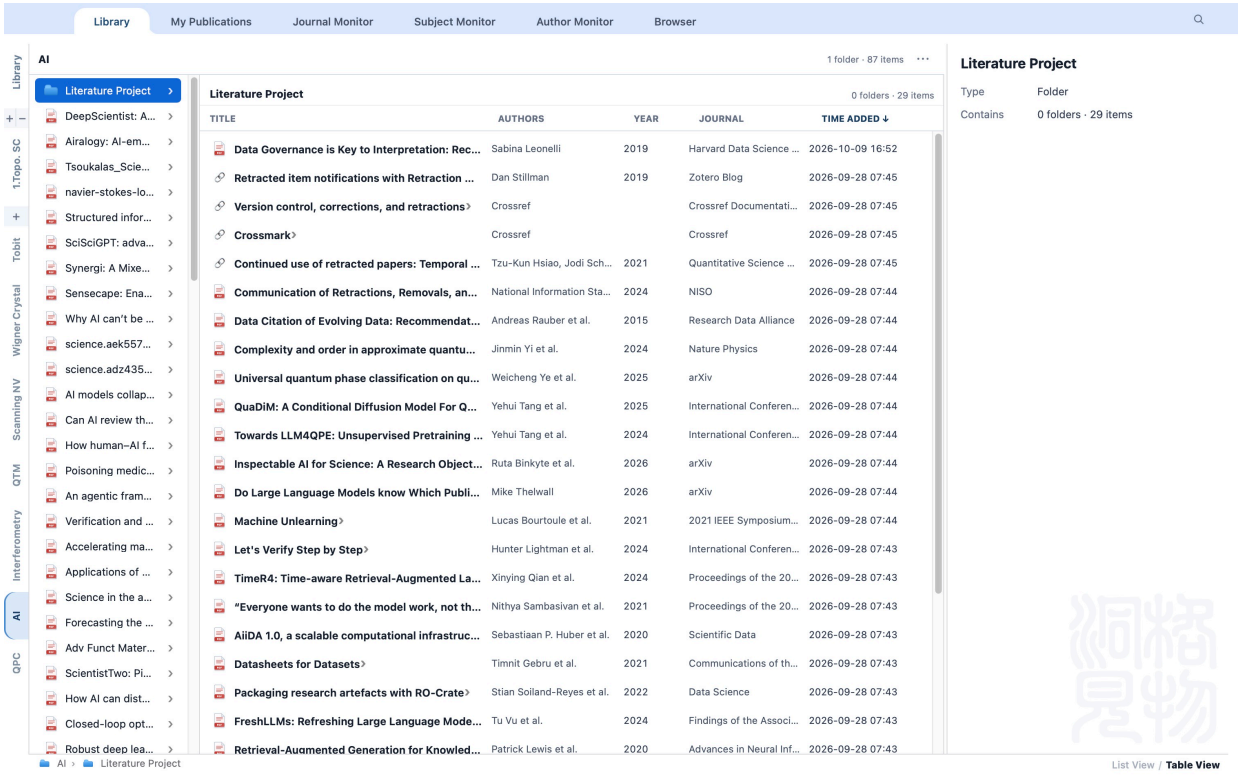
- For a PDF, use the thumbnail strip to jump to a page and scroll the page area independently.
- Drag the title bar to reposition the preview. Drag its resize corner when you need more reading space.
- Choose **Open in Preview** when you want the full macOS Preview application.
- When Quick Look is already open, changing the Paper Files selection updates the preview without requiring another open action.

> **Note**
> Single-click selects a Paper Files item. Double-click opens it. This prevents an accidental external app launch while you are classifying files.

See also: Understand Paper Files · Preview PDFs with Quick Look · Keyboard Shortcuts

3-1.3 Folders, Shortcuts, and Spaces

Folders organize a root. Shortcuts provide another entrance to the Library. Spaces keep an independent root for a project or research context.



A Space with an open folder. AI remains the heading across its folder and content columns. Literature Project has its own heading beneath it. The Space heading stops at the Details divider; the right-hand Details pane keeps its full height.

Renaming a managed folder or Space also renames its local storage. Use the App's Rename command so its main directory, mirrored Associated Files paths and shortcuts stay consistent. Existing paper identities, file contents, classifications and Default choices are retained. A busy-file message means the operation could not finish yet; allow the active file operation to finish and retry. Renaming a Shortcut changes its label rather than its target folder.

Choose the right structure

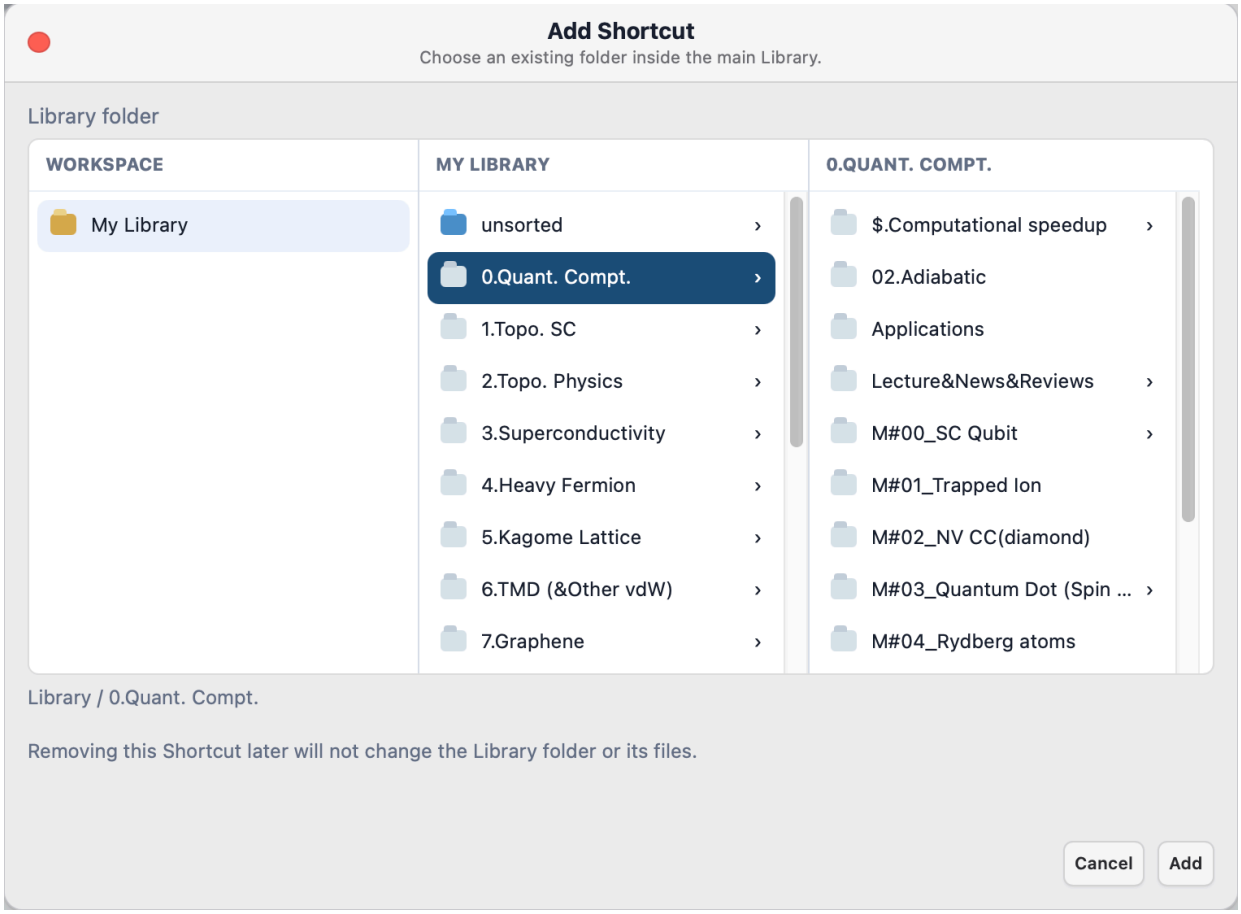
Structure	Use it for	What it changes
Library folder	A durable topic hierarchy inside the main collection.	Creates or moves a managed path.

Structure	Use it for	What it changes
Shortcut tab	Fast access to a Library location you use often.	Adds another entry point; it does not duplicate the paper record.
Space	An independent project, course, collaboration, or temporary collection.	Uses its own managed root under data / Spaces.

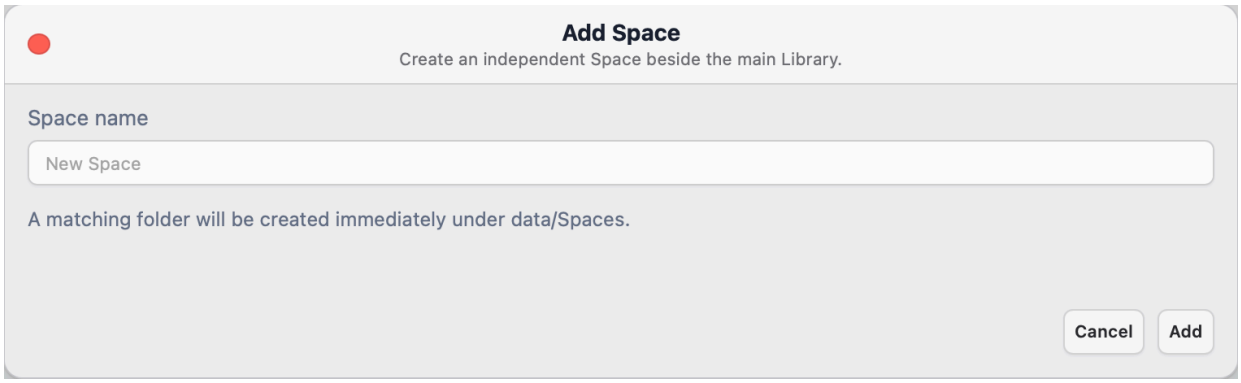
Create, rename, or move

1. **Control-click the relevant folder area.** Choose **New Folder**, **Rename**, **Copy**, or **Paste**.
2. **For a move, drag the item to its destination.** Wait for the destination to indicate it can receive the item, then release.
3. **Check the breadcrumb and Paper Details.** After the operation completes, the visible path and folder membership reflect the new location.

Shortcut and Space tabs can be dragged to change their order. Their inactive colors distinguish the tab groups; the active tab uses the common selected style. With the keyboard, press ← from the leftmost content column to focus this rail, use ↑ and ↓ to open another source, then press → to enter its content.



Add Shortcut. Choose an existing Library folder through the column picker. Add creates a navigation shortcut to that folder.



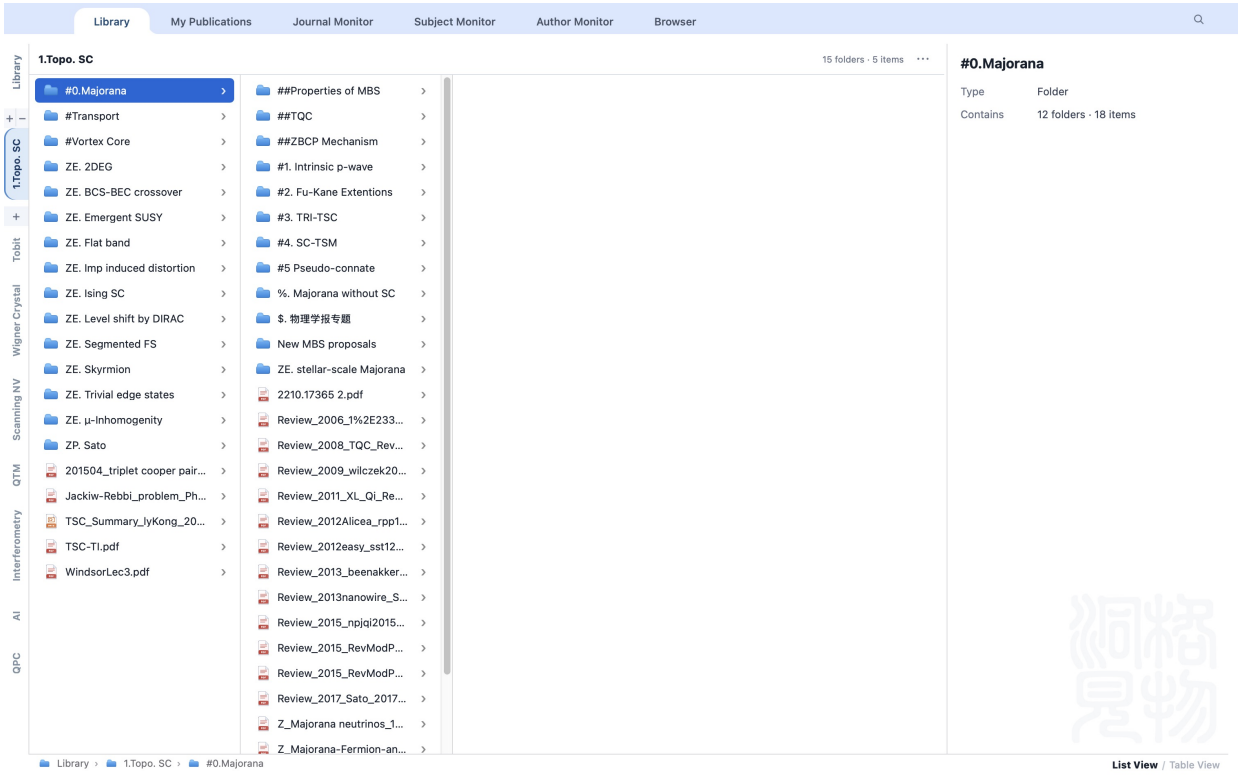
Add Space. Give the independent workspace a name, then choose Add. Cancel closes this form without creating a Space.

Use the Space or Shortcut heading

The persistent heading identifies the source in both List View and Table View, even after you open a nested folder. It spans that source's folder and content area, including Paper Files when visible, and stops before Paper Details. The nested folder keeps its own heading below it.

Open the **…** menu at the right of the heading, or Control-click the source tab, for **Rename** and source actions. Renaming a shortcut changes its navigation label, not the underlying folder. Spaces also offer **Move Space to Library** here. These commands replace the former toolbar of large buttons.

To add a paper inside the current folder, Control-click blank space in that folder and choose **Add Paper**. Library, shortcuts and Spaces use the same inline input. **Add Paper** from the source heading menu targets that source's root; check the destination before submitting.



A shortcut retains its full path. The footer starts at Library and includes the shortcut target and selected folder. The heading identifies the shortcut while Paper Details remains independent.

Color folders and remove navigation entries

Control-click an ordinary folder and choose **Color** to change its folder color. Protected roots do not offer this command. On the source rail, **Remove Shortcut** removes only the shortcut, leaving its folder and papers intact. **Remove Empty Space** works only after both its managed contents and remaining files are empty; move the Space into Library if you want to keep its hierarchy.

Work with several entries

Command-click to add or remove individual entries from a selection. Shift-click, or hold Shift while using the arrow keys, to extend a range in the current folder. Copy the selected entries with `⌘C`, paste into the active destination with `⌘V`, or press `Delete` to apply the normal reference-aware deletion rules.

To include an existing Library paper in your own publication list, select one or more paper rows, Control-click, and choose **Add to My Publications**. This adds the publication relationship to the same paper and Gewu Key; it does not remove the Library placement. Review authorship roles in My Publications afterward.

To update the information for one project or topic, Control-click its active folder and choose **Refresh Info**. The task includes every subfolder once per Gewu Key and follows the same cooldowns as the all-Library task. See Refresh Info: One Paper, a Folder, or the Library.

Copy a paper to another folder

A Library copy creates another reference to the same paper record rather than a second unrelated identity. On macOS, the secondary main-file location uses a native Finder alias; its secondary Associated Files location points to the canonical bundle through a directory alias. Paper versions and attachments remain associated with the work. See managed file locations.

Move an entire Space into Library

Use this when a temporary or project Space should become part of the permanent Library hierarchy.

1. **Select the Space on the left rail.** Open the Space that you want to keep as a normal Library folder.
2. **Open the heading's … menu and choose Move Space to Library.** You can also Control-click the Space tab. The move sheet explains that the Space will become a folder with the same name.
3. **Choose Destination in My Library.** Select the Library folder that should contain the moved Space.
4. **Confirm the move.** The complete Space hierarchy, including its files and subfolders, moves together.

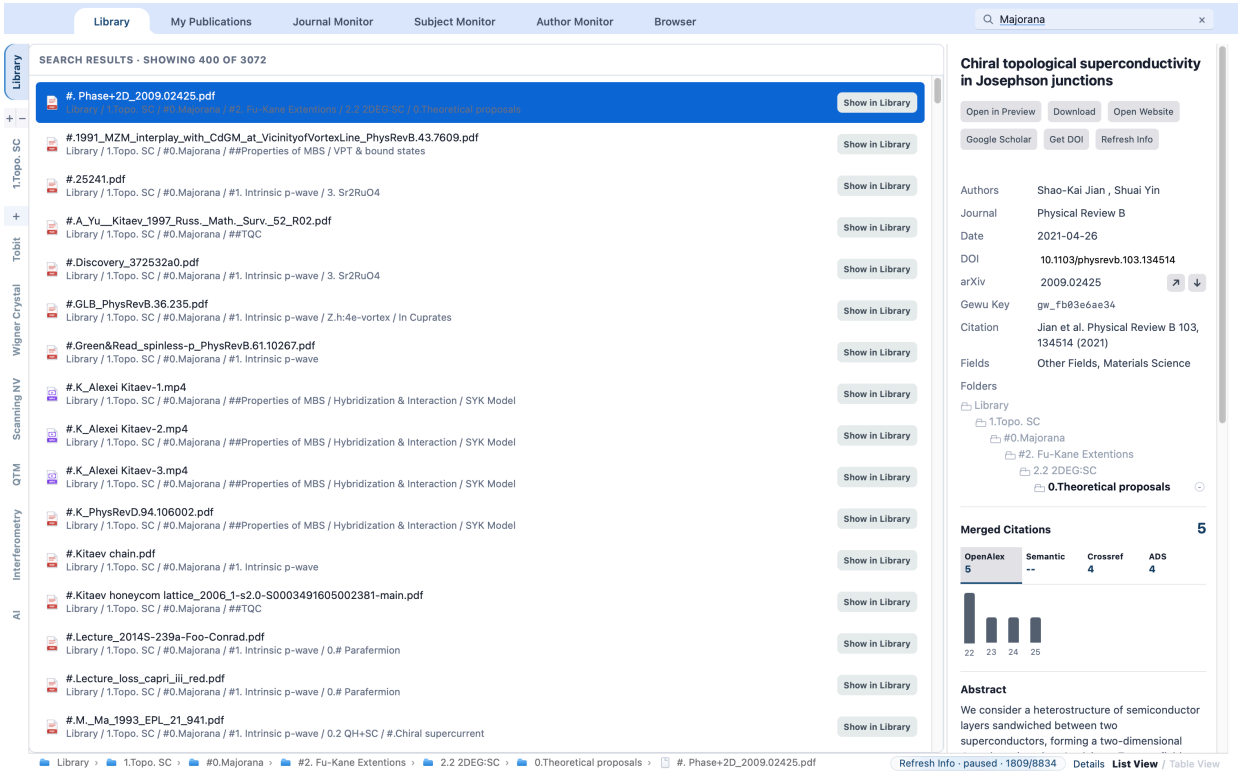
When the move finishes, the Space tab is removed and Library opens the moved folder. Canonical paper identities, folder placements, and Paper Files resource associations remain connected; the operation does not create duplicate paper records.

> **Important**
> Move managed papers in Gewu Insight, not directly in Finder. The app updates the visible file, associated bundle, and live database paths together.

See also: How Gewu Insight Stores Files · Delete and Recover Items

3-1.4 Search, List View, and Table View

Search locates items across the managed collection. List View emphasizes browsing; Table View emphasizes metadata comparison.



Search Library. Library search finds saved items across Library and Spaces. Select a result or use Show in Library to reveal its folder location.

Search the collection

1. **Select the search control in the top-right corner.**
2. **Enter a title, author, journal, identifier, or filename.** Results may come from the Library and Spaces.
3. **Select a result.** Its Paper Details and managed path become available.
4. **Clear the search.** Return to the previous Library location.

Edit the query and use fuzzy matching

Double-click a word, drag to select characters, or use ⌘A to select the focused query. Type to replace it or press Delete. ⌘F focuses workspace search; each research tab retains its own query.

Library matching tolerates small spelling errors, partial words and word endings, with exact matches ranked first. Server and visible-result matching share the same rules. It searches saved titles, authors, journals, identifiers and filenames, not every PDF's full text. Try a surname or title fragment if a long query finds nothing.

This is fuzzy text matching; it does not generate Chinese/English translations, acronym expansions or concept-level synonyms. Subject Monitor discovery uses separate, literal keywords. Library searches do not change those keywords or request online discovery.

Switch views

Choose **View > List View** or **View > Table View**. You can also use ⇧⌘V. Changing the view does not rename, move, or rewrite files. Active Library search uses its result list and disables Table View. Clear the search to use Table View for the current folder again.

In either view, Control-click the folder contents and choose **Sort by Name** or **Sort by Modified Time** for folder ordering. These commands clear any active table-column sort.

Sort by a table column

1. **Click Title, Authors, Year, Journal, or Time added.** The first click sorts ascending: displayed text alphabetically, Year numerically, or Time added from oldest to newest.
2. **Click again to reverse.** The header arrow shows the direction.
3. **Continue navigating.** Library, shortcuts and Spaces retain the chosen column and direction. Keyboard movement and range selection follow the visible order.

Folders stay above papers, and missing metadata stays below known values in either direction. Sorting preserves the selected paper, its identity and its files.

Time added shows when that folder or paper placement was added, in local date and time. It is separate from the publication date and the file's modification time. A Shortcut shows the original placement's time; renaming, moving or refreshing metadata does not reset it. A new copied placement has its own addition time. Click again for newest first. In a narrow window, scroll the table horizontally to reach all five headers; the headers and rows move together.

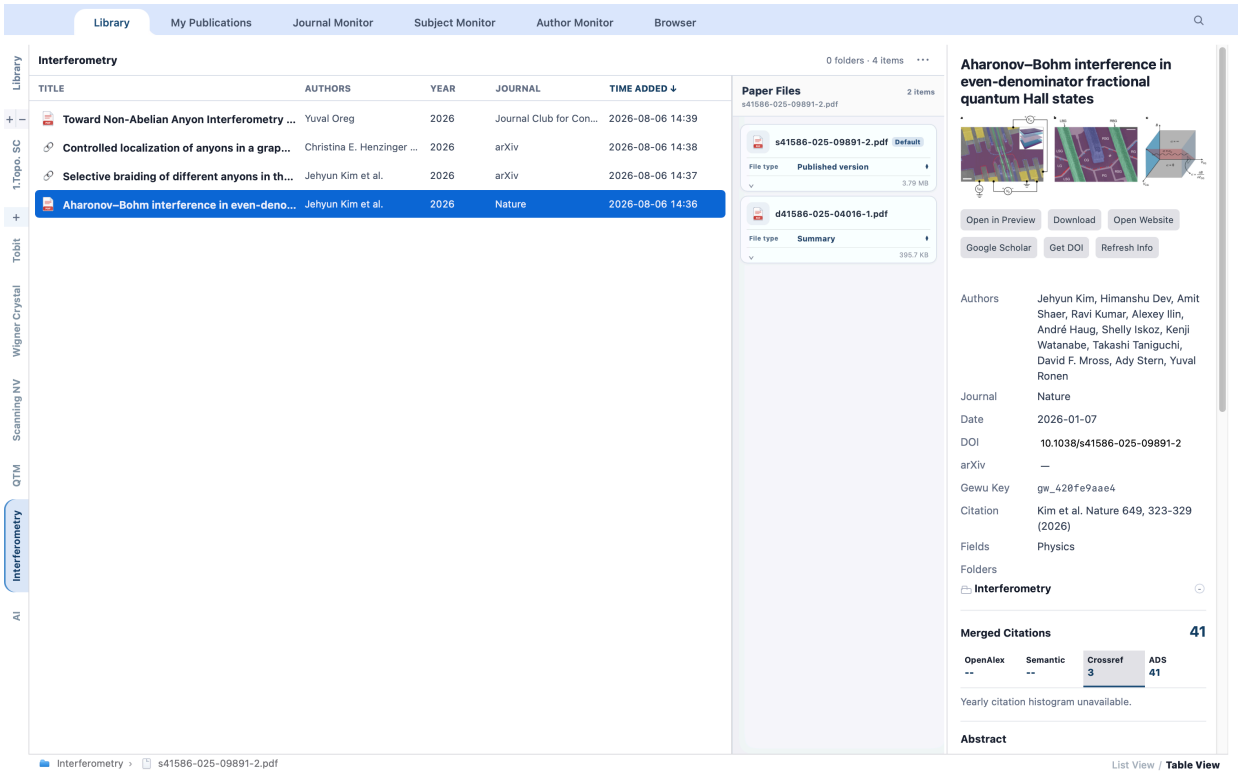


Table View sorting. The Time added arrow shows newest first. All five sortable headers are shared by Library, Spaces and shortcuts. This existing paper's Published version remains Default, with a separate Summary in Paper Files.

See also: Library Layout · Keyboard Shortcuts

3-2 Papers and Files

3-2.1 Add Papers and Files

A normal Library import creates a managed local copy and a database record. The original source file remains where it was.

Drag a paper into a folder

1. **Open the destination folder in Library.**
2. **Drag a PDF from Finder into the item area.** Drop it on the folder or in its contents.
3. **Wait for the item to appear.** The app copies the file into managed storage and begins local identification.
4. **Review Paper Details.** Confirm the title, DOI, arXiv identifier, and file classification.

Add a paper by DOI, arXiv ID, URL, or title

1. **Open the destination folder.** The resolved paper will be placed in that Library, Shortcut or Space folder.
2. **Choose Add Paper.** Control-click blank space in the folder contents and choose the command to open the inline lookup field.
3. **Enter a DOI, arXiv identifier, supported article URL, or title.** A DOI or arXiv identifier is the strongest identity clue.
4. **Press Return or click Add Paper once.** The app resolves the query and adds the paper to the displayed destination. A failed lookup leaves an error in the inline form.
5. **Review the added paper in Details.** Check its title, authors, identifiers, journal and year, especially after using a title query.

If the identity already exists, Gewu Insight reuses the canonical paper and adds the selected folder placement rather than creating an unrelated duplicate.

> **Note**
> A paper record does not require a local PDF. **Add Paper** creates or reuses the metadata record; it does not automatically download a PDF. Use **Download** or **Add File** separately when you want a managed local file.

> **Note**
> A file added to an existing paper's **Paper Files** bundle is an attachment or version of that paper. A normal folder drop creates or reuses a paper placement in that folder. When identity matching finds an existing paper, the incoming file joins its Paper Files; different PDFs are retained even when they share a DOI.

Folder drops and Folder Loader use the same import rules. An identity mismatch keeps the file with the matched paper, preserves its existing metadata and Default, and records the filename and identifiers in **Details**. It does not pause for a review decision. Import status stays in the bottom bar until dismissed; failed or stopped imports also refresh Library so completed items remain visible. A temporary progress connection failure triggers reconnection rather than reporting that the import stopped.

See also: Add and Classify Attachments · Use Folder Loader · Move a Standalone File into Paper Files

3-2.2 Use Paper Details

Paper Details combines actions, identifiers, metadata, and citation counts. Library, My Publications, and Author Monitor start with the paper title, without a separate heading bar. The available actions occupy two rows: Preview, Download and Website first; Google Scholar, Get DOI and Refresh Info second. Each button label stays on one line.

Top actions

Action	What it does
Open in Preview	Opens the preferred local PDF in macOS Preview.
Download	Retrieves the best available paper file without overwriting a different preserved version.
Open Website	Opens the publisher, DOI, arXiv, or best known work page using Website Links: Gewu Insight Browser by default, or the system default browser.
Google Scholar	Searches for the work in Google Scholar.
Get DOI	Examines local paper files for DOI or arXiv evidence.
Refresh Info	Refreshes paper information using its saved DOI or arXiv ID. Hover over the button to see the latest result below it.

Check an article for publication updates

1. Check that the article's saved DOI is correct, then choose **Refresh Info**.
2. The refresh checks Crossref for explicit publication-update relations to that DOI, including notices published after you saved the article. Review the refresh result for source failures or conflicts.
3. Inspect the notice summaries in **Paper Files**. Expand one to read its kind, date, title and DOI, then follow the DOI to the publisher's notice.

Refresh saves the notice relationship; it does not automatically download a notice PDF, replace the original article, change its Gewu Key or change your Default. A notice can therefore appear without a local file. When you obtain its PDF, retain it under the original article and choose the appropriate notice file type.

> **Coverage and timing**
> Saved Library, Space and My Publications papers are not periodically checked merely because they are saved. Request Refresh Info for a paper or a folder when you need a fresh check. Detection depends on the publisher supplying an explicit relation and Crossref making it available. A successful check with no notice is not proof that the article has never been corrected or retracted; consult the publisher when that status matters. Missing or ambiguous original-article evidence requires review.

File management

Library uses the adjacent **Paper Files** column; Paper Details does not repeat a Files box. My Publications keeps its collapsible **Files · n** management panel. Author Monitor shows a compact local filename with the full path on hover; **Get DOI** appears when a local PDF is available.

Metadata rows

The title appears once at the top. The rows below include authors, journal, date, DOI, arXiv, Gewu Key, citation, fields, and folder membership when available. **Gewu Key** is the stable, human-visible storage handle for a cataloged work record. It is not a DOI, arXiv identifier, or the database's relational primary key.

Compare citation sources

Merged Citations brings together the counts returned by OpenAlex, Semantic Scholar, Crossref, NASA ADS, and saved Google Scholar data when those services have a match. Select a source label to show its saved count in the headline; this does not start a refresh. The merged estimate compares published, preprint, and already-merged records rather than simply adding every provider's total. Separate published/preprint estimates can be combined, so this is not a verified count of unique citing papers. The yearly histogram uses OpenAlex data and does not switch provider when you select another count. PubMed Details supplies identifiers and indexing information, not another citation-count source.

Inspect influential citations

When Semantic Scholar reports an **Influential** count, select it to inspect citing papers that Semantic Scholar classifies as especially meaningful to the cited work. This is Semantic Scholar's classification, not a score calculated by Gewu Insight. The list can include citation context, intent, citing authors, venue, year, and citation counts when the service supplies them.

Influential Citations is fetched on demand and cached locally for several hours. A saved API key gives Gewu Insight an individual Semantic Scholar allowance, but it is still limited to about one request per second and can be temporarily throttled. If the service returns a rate-limit message, the app pauses repeat requests for a few minutes; the paper record, existing citation totals, and any previously displayed result remain unaffected.

Read and reuse the record

- A representative first figure, when available, provides a visual checkpoint above the actions.
- **Abstract** keeps the paper summary in the independently scrolling details pane.
- **BibTeX** and **Copy Code** provide a citation record for reference software or LaTeX.
- **Nature Reference** and **Copy Reference** provide a formatted reference string.
- **Folders** lists every Library location for the shared work. Remove a location there only when you intend to remove that reference.

> **Note**
> Gewu Key is assigned when a work is first cataloged in SQLite, before Add to Library. Its row remains visible in Author Monitor, including while details load; a missing value shows `(undefined)`. Adding to Library reuses the existing key and updates the view without creating a second identity.

See also: Edit DOI and arXiv Identifiers · Understand Paper Files · How Gewu Insight Stores Files

3-2.3 Edit DOI and arXiv Identifiers

DOI and arXiv can identify the same work across publication stages. Correct either value in Paper Details before downloading or synchronizing versions.

Save or correct either identifier

1. **Select the DOI or arXiv value.** Enter the identifier, including in an empty row. DOI URLs are accepted; arXiv accepts forms such as `arXiv:2106.14849` and valid legacy IDs.
2. **Press `Return`.** Wait for the server to save it, or review the two records if the DOI is already assigned elsewhere. After a normal save, the field leaves editing and the green message above the metadata reads **Identifier updated**, or **Identifier saved** if unchanged. Invalid input stays editable with an error.
3. **Choose Refresh Info.** Saving alone does not request remote metadata. Refresh uses the saved DOI and/or arXiv ID; a DOI is not required for an arXiv-only paper. A changed DOI still in the editor is saved before refresh.
4. **Inspect the result.** Hover over **Refresh Info** for its information box below the button: changed fields, last check, source outcomes and conflicts. Keyboard focus also reveals it; pointer departure or Escape dismisses it.

To clear either identifier, save an empty value with Return. Both rows remain available for later edits. The arXiv row also has an arrow to open its website and a download control for the latest available PDF.

Replace an incorrectly identified work

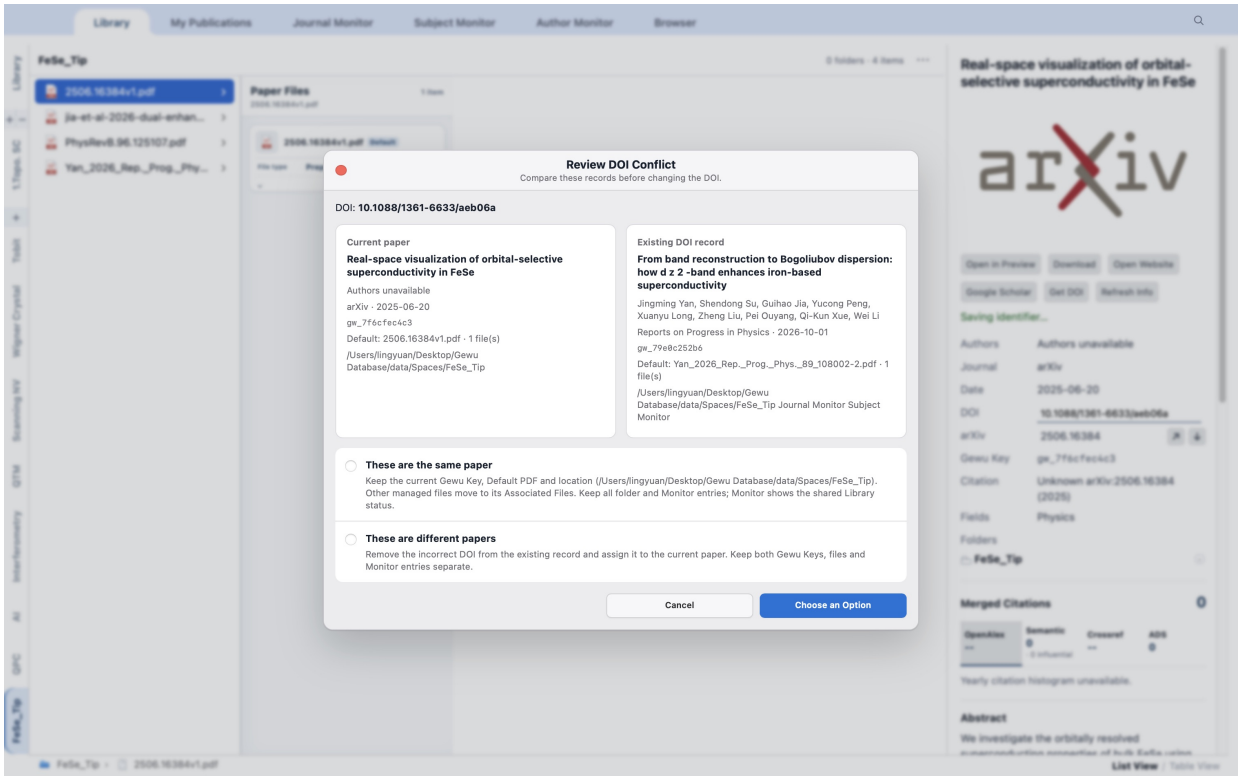
If an automatically recognized DOI belongs to the wrong article or book, replace it and press Return first. Clearing and then entering the correct value also records a correction. On the next Refresh Info, a primary source must return that exact corrected identifier before the old automatic bibliography can be replaced. The old information is retained in correction history; missing or mismatched responses leave it available while you retry. This is why an ordinary automatic conflict guard does not prevent a verified manual correction.

An ordinary correction keeps the same Gewu Key, paper locations, files, classifications, Default and user notes. A manual DOI edit that conflicts with another record opens the explicit review below before changing either record. After a correction is applied, later refreshes use the normal conflict checks again.

Review a DOI already assigned to another record

This review applies only when you manually enter or correct a DOI in Paper Details. It does not enable automatic merging during discovery, import, Refresh Info or arXiv editing. A matching DOI is a reason to compare the records, not proof that they represent the same paper.

1. Open the paper from the Library or Space location you want to retain as its main location. Check its Default PDF before editing the DOI.
2. Enter the verified DOI and press Return. **Review DOI Conflict** compares **Current paper** with **Existing DOI record**, including titles, authors, Gewu Keys, Default files, file counts, locations and Monitor sources.
3. Choose the appropriate option only after checking the actual articles, then read and confirm the action. **Cancel** leaves both records and their files unchanged.



Compare before choosing. This capture shows a deliberately unconfirmed entry between two different articles. Here the entered DOI belongs to the existing record, so Cancel is appropriate. Neither merge nor reassignment was applied; the picture is not a recommendation to move this DOI.

What you have verified	Choice and result
Both records represent the same paper	Select **These are the same paper**, then **Merge Records**. Keep the current Gewu Key, selected main location and current Default PDF. Other managed files, including the other record's former main PDF, join its Associated Files bundle. Folder memberships and Monitor sources remain attached to the shared paper.
They are different papers, and the existing record wrongly owns the DOI that belongs to the current paper	Select **These are different papers**, then **Correct DOI Assignment**. Clear the incorrect DOI on the existing record and assign it to the current paper. Both records keep their separate Gewu Keys, files and Monitor entries.
The new entry is wrong, or the evidence is uncertain	Cancel. Do not use reassignment merely because the titles differ; it is only for a DOI verified to belong to the current paper.

A confirmed merge updates managed disk paths as well as the visible Library. The main PDF stays at the chosen current location, and the other managed files follow the corresponding Associated Files path. Other folder references are retained using the existing storage rules; externally linked originals are not moved or deleted.

If the existing record came from a Monitor, it remains visible there after merging. Its shared status shows that the paper is already in Library; the Monitor entry does not disappear. After either operation, inspect Paper Files and Paper Details. Metadata refresh remains a separate action.

Follow a preprint into publication

Explicit links from arXiv, Crossref, OpenAlex, Semantic Scholar or configured ADS can connect a preprint and its published DOI. A newly linked DOI loads detailed publication metadata while retaining the arXiv ID. Title similarity alone cannot bind identifiers. Manual corrections and cleared values are preserved; a DOI owned by another Gewu Key is reported without merging records.

Saved Library, Space and My Publications papers are refreshed when you request it, including through a batch Refresh Info task. Saving a paper does not schedule periodic metadata refreshes. Author Monitor separately completes missing identifiers and abstracts for monitored publications, with bounded attempts and a daily allowance. This completion work does not download PDFs, change Paper Files or Default, create placements, or move your selection and focus.

The bottom bar shows a running refresh, then green **Refresh complete**. Partial results preserve successful updates; the hover report explains unavailable sources and retry times. For batch scope and shared cooldowns, see Refresh Info: One Paper, a Folder, or the Library.

> **Important**
> Confirm an identifier before running a version sync. A wrong identifier can attach files from a different work.

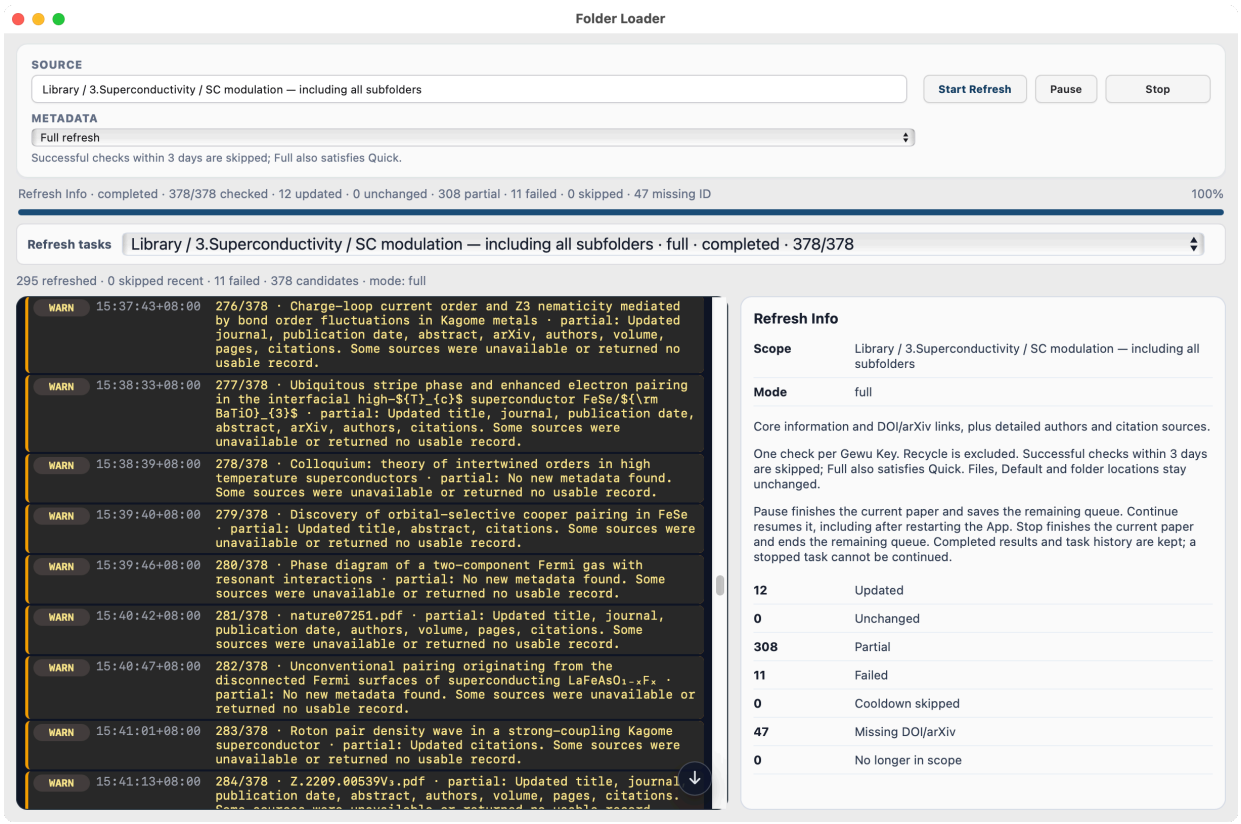
See also: Download and Sync Versions · Literature Data Source

3-2.4 Refresh Info: One Paper, a Folder, or the Library

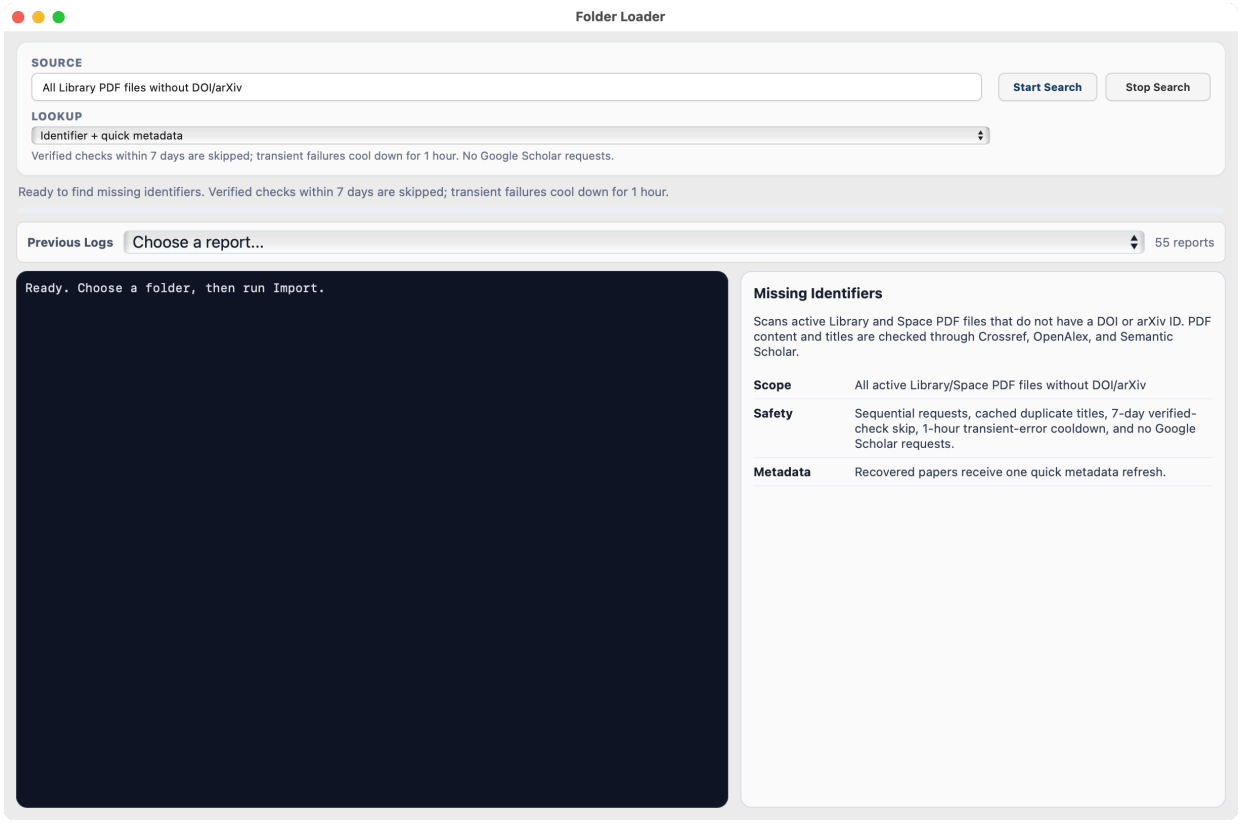
Use one information-refresh service for a single paper, a selected folder, the active Library collection, or the metadata stage of an import. Choose the scope before starting.

Choose the operation you need

Entry point	Scope and behavior
Paper Details > Refresh Info	One paper, using Full. Either saved DOI or arXiv is sufficient; see identifier correction.
Folder context menu > Refresh Info	The selected active Library/Space folder and all descendants, once per Gewu Key. Choose Quick or Full; default Full.
Loader > Refresh All Metadata	Active Library and Space papers, excluding Recycle. It is not a refresh of every Monitor-only database record. Choose Quick or Full; default Full.
Folder Loader metadata mode	Imported papers use the chosen Skip, Quick or Full setting after file import and identifier recovery. Finder folder drops retain their separate import behavior.
Automatic background checks	Bounded completion of missing identifiers and abstracts in Author Monitor. Saved papers do not enter a periodic refresh queue.



Refresh All Metadata. This completed folder task shows its scope, Quick/Full mode, task history and per-paper log. Pause and Stop are separate controls. Partial results mean some sources did not supply usable information; completed means the task has ended.



Find All Missing Identifiers. Start Search checks eligible Library PDFs for missing identifiers. The right pane explains the source policy and retry cooldowns.

Refresh known works

1. **Choose the scope.** Use **Loader > Refresh All Metadata**, or Control-click a Library/Space folder and choose **Refresh Info**. A folder task includes all its subfolders. Recycle is excluded.
2. **Choose Quick or Full.** Folder and all-Library tasks default to Full. Quick updates core information and DOI/arXiv links; Full adds detailed authors and citation sources.
3. **Choose Start Refresh.** Each Gewu Key is checked once, even when it has several files or locations. The bottom status bar shows the task while you continue browsing.
4. **Review the log.** Counts distinguish updated, unchanged, partial, failed, cooldown skipped and missing DOI/arXiv records. Missing identifiers remain visible; use the separate identifier search when needed.
5. **Pause, continue, or stop. Pause** finishes the current paper and saves the remaining queue. Select the task under **Refresh tasks** and choose **Continue**, including after reopening or restarting. **Stop** ends the remaining queue permanently for that task while keeping completed results and history. Interrupted running refresh tasks reopen paused; stopped tasks cannot be continued.

Batch and import work skips compatible successful checks from any entry point within three days. A changed identifier does not inherit the old identifier's cooldown. Full satisfies a later Quick task; Quick does not block Full. Partial retries reuse recent successful source responses and retry failed sources. A direct Paper Details refresh can check again immediately, while shared source rate limits still apply. Author Monitor completion retains its separate retry limits and daily allowance; Google Scholar protections are unchanged.

Refreshing a shared Gewu Key updates its information in every location. Folder locations, PDF files, Paper Files classifications and Default selections stay unchanged. Multiple tasks wait in a queue; a direct single-paper refresh has priority between papers. The task log shows the most recent 200 entries; totals and the remaining queue are retained.

Pausing means the current paper is still finishing; it changes to **Paused** when the saved queue can resume later. Keep browsing while it settles. A completed refresh uses green **Refresh complete** in the bottom bar, including partial results. This reports that the operation ended, not that every source answered. Detailed outcomes remain in the task window and, for a single paper, below **Refresh Info** on hover. A transport failure or rejected operation still reports its error.

Jump from a log entry to Library, then return

1. **Click a paper entry in the scrolling log.** Entries linked to a paper offer **Show in Library** on hover. You can also focus an entry and press `Return` or `Space`. General process messages and older entries without a saved paper link are plain text.
2. **Inspect the selected paper.** The task window closes after the paper is revealed in Library. Its folder and paper selection are restored for this destination. Closing the window does not pause or stop the background job.
3. **Return through the bottom bar's Details.** While the Loader or refresh task remains active or paused, reopen its progress window and continue reading the log. For completed refreshes, reopen **Loader > Refresh All Metadata** and select the saved task in **Refresh tasks**.

The same log navigation is available in Folder Loader after an entry has been associated with a saved paper. Selecting text in the log lets you copy it without opening the paper. If the referenced item is no longer available, the window shows an error instead of silently navigating to another work.

Close, pause, and stop are different

Action	What happens
Close the task window	Hide the window and keep working. The task continues while the app is running; Details returns to it.
Pause / Continue	Refresh Info finishes the current paper, saves the remaining queue and can resume it later, including after an app restart.
Stop	End a running or paused Refresh Info task. Stopping lets the current paper finish safely. Completed changes and history stay, but the remaining queue will not resume. Start Refresh creates another task if needed.

Action	What happens
Stop Import / Stop Search	Folder Loader or identifier search stops at its next safe checkpoint. Completed imports and identifier changes stay. Stop Import also ends that import's linked metadata queue; it is not Delete Import.

A stopped task no longer leaves a resumable progress indicator in the footer. Folder import and identifier search use their own Stop controls; they do not offer the batch refresh's Pause/Continue pair.

Find missing identifiers

1. **Choose Loader > Find All Missing Identifiers.**
2. **Choose Start Search.** The app checks active Library and Space PDFs using Crossref, OpenAlex, and Semantic Scholar where available.
3. **Review uncertain matches.** Refresh or recover DOI evidence as needed, and edit an incorrect arXiv identifier inline before downloading versions.

Verified “not found” checks and transient failures use cooldowns so a repeated run does not immediately send the same requests. Bulk missing-identifier work does not use Google Scholar.

Include later corrections and retractions

Quick and Full refresh both check explicit Crossref publication updates when the paper has a DOI. A folder or all-Library refresh can therefore discover a later notice for an older saved article. Existing refresh cooldowns, scope and partial-result reporting still apply. Linked notices appear in Paper Files; refreshing does not fetch their PDFs or replace existing files. See the single-paper workflow and its coverage limits.

See also: Literature Data Source · Publisher and Network Access · Edit DOI and arXiv Identifiers

3-2.5 Understand Paper Files

A paper is one work record that can own several PDFs and related files. Paper Files shows that bundle without pretending it is another Library folder level.

Compact cards for one flat bundle

The Paper Files column is a flat peer list, not a hierarchy of version or attachment folders. Each card shows its filename, **File type**, and size by default. Click the size footer or its chevron to expand **Annotation** and **Source** into separate rows; click again to collapse them. The chevron points down when collapsed and up when expanded. Each card expands independently, and the choice lasts for the current session. File type remains visible because it identifies a preprint, accepted manuscript, published version, supplement, or other resource. The **Default** marker identifies the preferred file.

Select, open, or act on a file

- Single-click a card to select it.
- Drag a card's file icon or filename to Finder or the Desktop to copy it, or onto a compatible application icon in the Dock or on the Desktop to open the original file with that application. Library item file icons support the same native drag. The target application decides which file types it accepts.
- Double-click a card to open it.
- Double-click the paper row to the left to open the paper's Default file, independently of the highlighted resource card. PDFs open in macOS Preview. For unclassified files, automatic selection uses a stable Unknown/Other fallback when available; you do not have to choose a Default manually first.
- Press `Space` to use Quick Look.
- Control-click a card for **Open**, **Quick Look**, **Show in Finder**, default-file controls, **Change File Type**, or **Delete from Paper Files** where allowed. Control-click blank space in Paper Files for **Add File** and, when the paper has an arXiv identifier, **Sync Versions**. Those bundle-wide commands do not appear on an individual file card.
- Press `Delete` or `Backspace` while Paper Files has keyboard focus to remove only the selected removable resource after confirmation. This does not delete the parent paper or its other files.

The current main file remains protected from direct deletion. Choosing **Set as Default** makes that resource the preferred file for opening and preview. For a privately managed paper with an active folder placement, it becomes the main file and the other resources belong in Associated Files. This applies to PDFs and non-PDFs alike. File IDs, classification and bytes are preserved; unrelated same-name files are not overwritten. **Use Automatic Selection** releases a manual choice and returns to the priority and fallback rules in Default selection.

Library Paper Details no longer repeats this bundle. In My Publications, which has no adjacent Paper Files column, the collapsible **Files** panel keeps its file-management controls. **Show in Finder** reveals a physical file.

Deleting a private managed attachment moves its bytes to macOS Trash before removing its resource record. Shared files and external originals are preserved. A failed database deletion restores the moved file when macOS returns its Trash path. The interface shows progress and updates only after success; path indexes keep the shared-file safety check fast even in large libraries.

Attachment deletion is shared. Every placement of the same Gewu Key uses this Paper Files bundle. Deleting one attachment here removes it from that paper everywhere; it does not merely hide it from the current folder. The current Default and protected primary file must be changed before they can be removed. This differs from deleting a Library placement, which follows the reference-aware Recycle rules.

Correct a file's paper assignment

If an imported PDF was matched to the wrong paper, Control-click its card and choose **Move to Another Paper**. Search the destination in Library by title, DOI, or Gewu Key, select it, and choose **Move File**. To make it independent, choose **Create Paper from File**, enter a title, and choose **Create Paper**. The new paper appears in the current Library folder with a new Gewu Key and no guessed DOI.

Only the selected file changes ownership when using these card commands. Its bytes, file ID, classification and source information remain intact; a name collision may require a unique filename. A source with remaining files keeps its identity and selects another eligible Default. An emptied, unshared source is removed as an independent Library item rather than left as a fileless link. Protected shared relationships can prevent that removal. For a PDF, the new independent paper receives a new Gewu Key and uses the file as its Default. A detached non-PDF remains an ordinary standalone file with an internal storage key and no public Gewu Key in the App. Managed paths follow the change: an attachment goes into the destination bundle in Associated Files; a newly independent file goes into its main folder. Explicit reassignment is respected on repeat imports. Historical analysis text is not rewritten.

AI Access can perform file ownership moves and rename files only with the separate move or rename permissions described in AI Access. Reading and adding analyses do not grant these permissions. Existing file contents cannot be edited or overwritten through AI Access.

Understand a Missing File row

Missing File means that SQLite still has a resource record, but the file is not present at the recorded local path. Gewu Insight keeps genuine missing resources visible so you can identify and repair them; it hides only an older untyped migration alias when it can prove that the alias has already been replaced by the same paper's live canonical file.

A missing attachment is not another copy of the default PDF merely because both rows have a similar name. Check **Show in Finder**, the paper's Associated Files bundle, and Recycle before deleting or importing anything. The default resource and explicitly typed attachments remain visible even when their paths need repair.

Read linked publication notices

Expandable notice summaries appear alongside the file bundle when a correction, retraction or another supported update has been linked to the article. Expand a summary to read the notice title, date and DOI. This is notice metadata, so a summary can exist even when no notice PDF has been saved. A saved notice PDF remains an ordinary associated file with its own file type.

The original article, its bibliography, files and Gewu Key remain. Notices never become the automatic Default; a corrected full article follows its separate version rule. The same summaries are available in My

Publications' expanded Files panel. Use Refresh Info to check for later notices, or add a discovered notice from Journal Monitor.

See also: Preview Files with Quick Look · Choose File Types and the Default File · Add and Classify Attachments · Navigate with Mouse and Keyboard

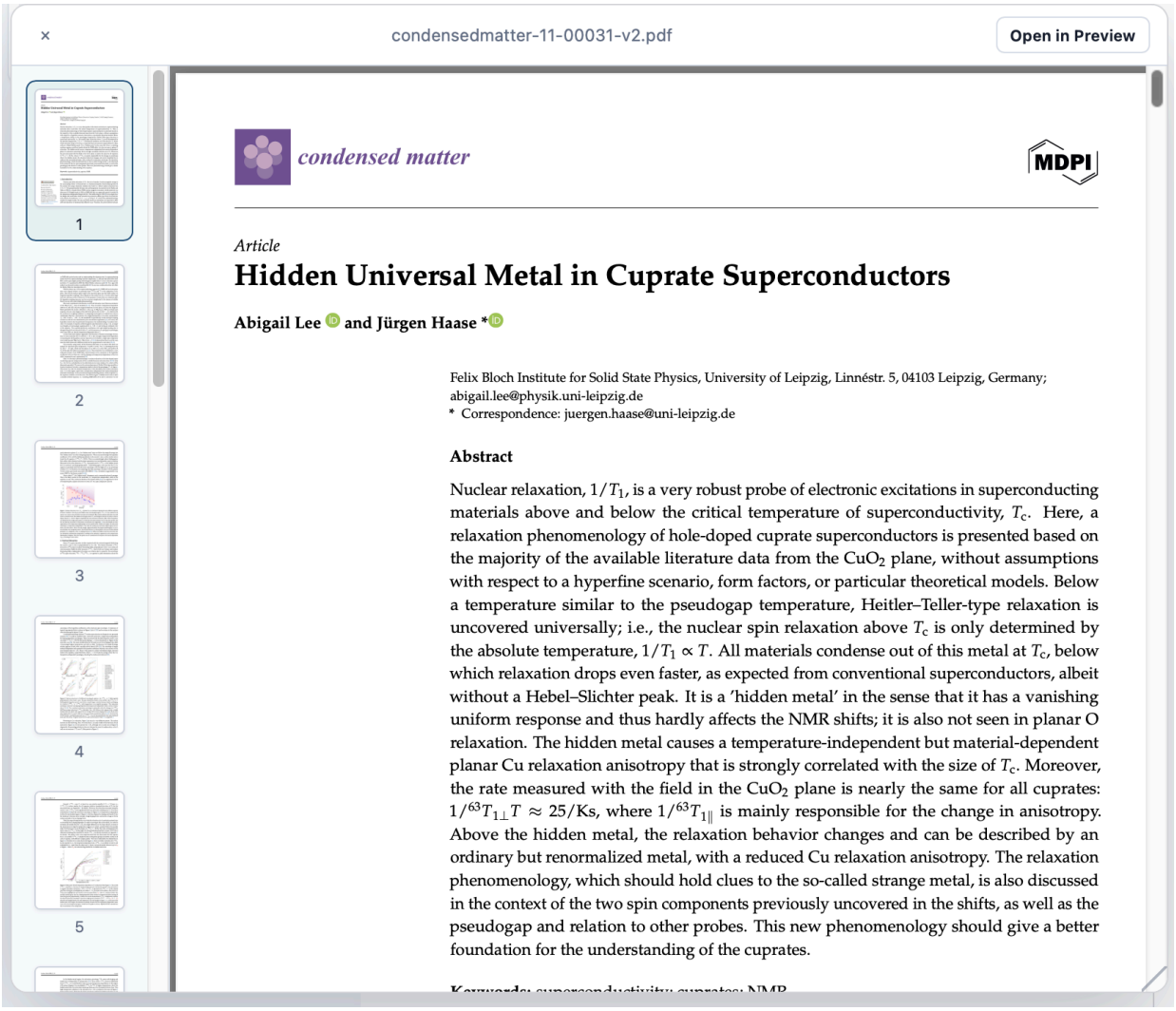
3-2.6 Preview Files with Quick Look

Press Space to inspect the selected resource without changing its paper record. Quick Look can stay inside Gewu Insight, use the macOS system panel, or combine both approaches.

Open and follow a preview

1. **Select a file in Library, a Space, or Paper Files.** Press `Space` to open the configured Quick Look.
2. **Move with ↑ or ↓.** When Quick Look is open, the preview follows the current Gewu Insight selection.
3. **Press Space again.** Close Quick Look and return keyboard focus to the same file.

With macOS Quick Look, Gewu Insight keeps keyboard focus so you can continue moving through Library or Paper Files while the system panel follows the selection. Selection changes are sent immediately, with the same file reused when possible. An older preview response cannot replace the latest selection. Document loading can still take time for large or unfamiliar files.



Built-in PDF Quick Look. The selected Library PDF appears with page thumbnails and Open in Preview. The document and thumbnail column scroll independently.

Read the built-in PDF loading sequence

The document viewer starts first and requests the bytes it needs from the local file instead of waiting for the whole PDF to be copied into memory. Page count and other metadata load in parallel. Page thumbnails are scheduled after the embedded document reports loading, so the document and thumbnail strip may appear at different times. That loading signal is not a guarantee that every PDF page has finished painting.

On the first open, the app may still need to inspect the PDF and render thumbnail pages. Later opens can reuse stable local-file validators, remembered page information, and a bounded thumbnail cache. The cache is automatically trimmed and can always be rebuilt; it is not part of the paper record or a substitute for the PDF.

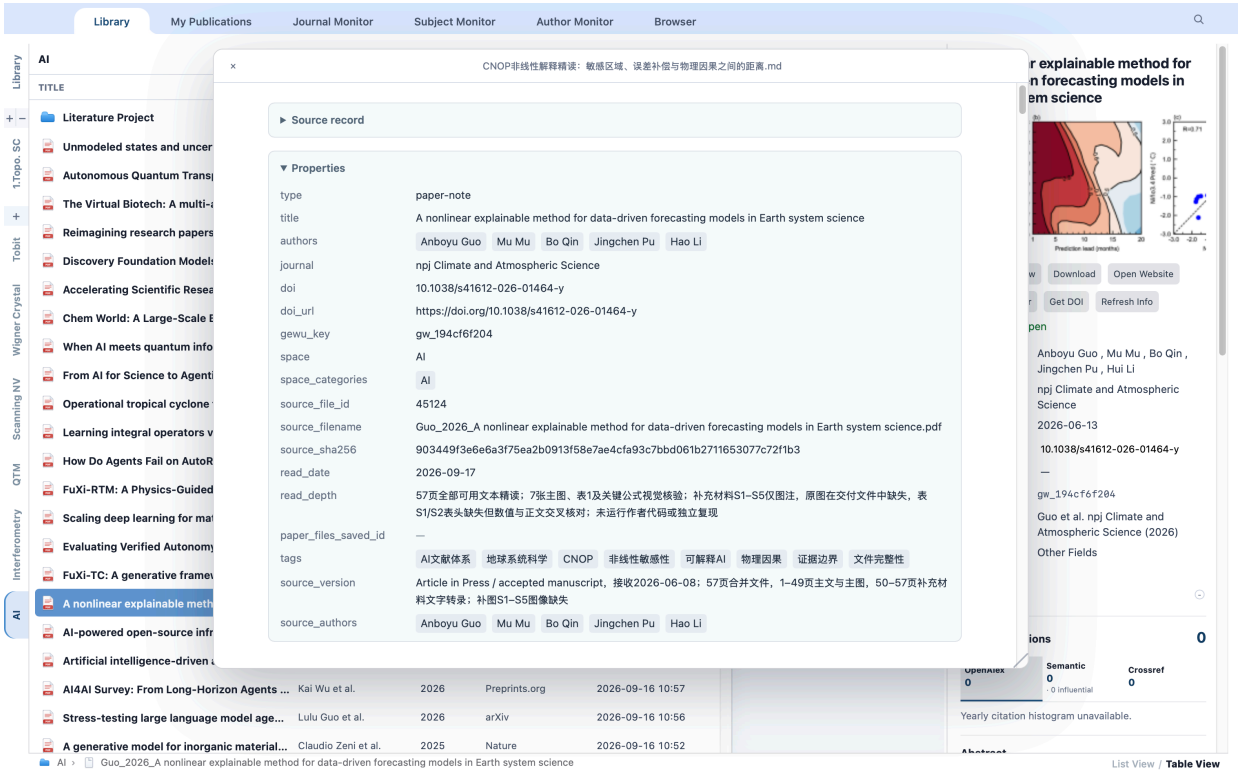
Read and navigate the built-in viewer

The PDF opens at page width. Scroll the document, click a page thumbnail to jump, or use a trackpad pinch to zoom the PDF. The document and thumbnail strip scroll independently. Drag the preview title bar to reposition it, or its resize corner to change the reading area. The strip provides page buttons for the first 200 pages; thumbnail images are limited to the first 160. The document remains scrollable beyond that range; use **Open in Preview** for the full reader's navigation and tools.

Markdown files (`.md`, `.markdown`, `.mdown`, and `.mkd`) use a formatted reading view in built-in Quick Look, including headings, emphasis, nested lists, quotations, code blocks, and tables. The view follows keyboard selection, including Markdown attachments in Paper Files. Nearby images in the same folder or its subfolders can appear inline; remote images are links that you open explicitly. Website links follow Website Links settings. Raw HTML is shown as text. Large files are limited to a 256 KiB preview and marked [Preview truncated]; the original file is unchanged. Complex notes may use a shorter preview to keep navigation responsive. The mixed mode keeps both PDFs and Markdown in these built-in readers and uses macOS for other formats. Only **macOS Quick Look for all files** sends Markdown to the system renderer, whose rendering depends on installed macOS support.

Supported images, audio, video, and text files can also be inspected inside the workspace. A long text preview may end with [Preview truncated]; use Open File to read the complete original. Other formats show a file card with **Open File**; that opens the file in its macOS application. Choose a native mode in Quick Look Settings for formats better handled by system Quick Look.

Switching between PDFs in the built-in viewer can briefly flash while the next document loads. It does not guarantee the same switching speed or continuity as macOS Quick Look. For rapid comparison, try **macOS Quick Look for all files**.



Built-in Markdown Quick Look. This existing paper note shows Source record collapsed and Properties expanded. Authors, tags and long values stay readable within the window; collapse Properties to reach the formatted body. The preview does not modify the note or Paper Details.

Choose Quick Look or Preview

Use	Best for
Built-in Quick Look	PDF inspection and keyboard comparison among versions while remaining inside the workspace.
macOS Quick Look	System-native preview support for presentations, media, images, PDFs, and other formats supported by macOS.
Open in Preview	Full macOS Preview tools, annotation, export, printing, and a separate reading window.

> **Note**
> Previewing is read-only from Gewu Insight's perspective. Opening Quick Look, loading thumbnails, or moving between pages does not change file type, annotation state, source, Default status, or the SQLite paper record.

Read Markdown properties and long notes

Leading YAML metadata is displayed as collapsible **Properties**. An AI note's separate provenance header appears under **Source record**. Expand either to inspect its values; collapse it to reach the body. These are values saved in the note, not an instruction to overwrite Paper Details. Invalid metadata falls back to readable source text.

The Markdown preview opens at a reading width that fits the window. Text and long property values wrap; wide tables and code blocks scroll within their own area. Markdown does not need the PDF thumbnail rail. Rendering is local, and an unchanged note can reuse a validated in-memory preview. If Loading Markdown does not finish, follow the retry guidance. Obsidian-specific plugins, diagrams and other extensions are not a guarantee of the built-in reader.

Read mathematical Markdown

The built-in Markdown reader supports inline formulas delimited by `$...$` or `\(...\)`, and displayed formulas delimited by `$$...$$` or `\[...\]`. Put displayed formulas on their own lines. The math renderer is bundled locally, so it does not require a separate internet connection.

Formulas inside inline code or fenced code blocks stay literal. Unsupported, malformed or exceptionally large formulas remain visible as escaped source instead of blocking the note. Wide equations scroll within their own area. This formats the preview only; the saved Markdown and any original PDF remain unchanged.

See also: Quick Look Settings · Navigate with Mouse and Keyboard · Understand Paper Files · Troubleshooting

3-2.7 Choose File Types and the Default File

The default file is the preferred local representation of the paper. Choose it manually or let Gewu Insight select it. Default appears first in the file list; Up and Down follow the visible order.

Automatic selection

The automatic order is **Corrected published version**, **Published version**, **Accepted manuscript**, then **Preprint**. Within a type, a later recognized version normally wins. If none is eligible, an available **Unknown** or **Other** file supplies a fallback: keep the existing main file where possible, otherwise choose the earliest PDF, otherwise the earliest remaining eligible file. This includes standalone RTF, image and other non-PDF imports; it does not relabel them as published papers.

Automatic and manual eligibility

The four publication types above take priority over the Unknown/Other fallback. Correction and retraction notices, expressions of concern, withdrawal notices, addenda, supplementary material, peer review files, Perspective, News and Views, Summary, Analysis, Source Data and Source Code are not automatic candidates. Missing, Invalid file and unresolved identity-conflict files are also excluded. You may manually choose an existing valid file, including a supporting file. A manual choice is sticky across rescans, metadata refreshes, and later downloads until you choose **Use Automatic Selection**. Newly arriving files remain peer rows and do not erase or silently reclassify older files.

Choose a file manually

1. **Enter Paper Files, or expand Files in My Publications.**
2. **Select the file you want.**
3. **Choose Set as Default.** This file will be used by **Open in Preview** and paper preview. You can also pin the automatically selected Default this way.

The selected file becomes the manual default. In a managed paper with an active placement, it becomes the main file and the former main file moves to Associated Files. This applies to non-PDF defaults too. The article and file identities stay unchanged. External originals and files shared with another article are protected; existing names are retained unless a collision requires a unique name.

Return to automatic selection

Choose **Use Automatic Selection** for a manually selected default. Gewu Insight evaluates the eligible file types again.

Classify files without collapsing annotation state

In Library, File type stays visible on each card. Click the size footer to reveal Annotation and Source, then change each control independently. In My Publications, expand Files to see all three controls directly. File type uses exactly these labels:

- **Preprint**, **Accepted manuscript**, **Published version**, and **Corrected published version**
- **Correction notice**, **Retraction notice**, **Expression of concern**, **Withdrawal notice**, **Addendum**, **Supplementary material**, and **Peer review file**
- **Perspective**, **News and Views**, **Summary**, and **Analysis**

- **Source Data** and **Source Code**
- **Other**, **Unknown**, and **Invalid file**

Use Perspective or News and Views for related commentary, Summary or Analysis for reading notes and interpretation, and Source Data or Source Code for the accompanying research material. These labels describe this file's role; they do not change the parent paper's publication type or Gewu Key. They are not automatically promoted to Default. File extension alone does not determine the role.

Annotation state is selected independently as **Annotated**, **Unannotated**, or **Unknown**. Source is independently recorded as **Publisher**, **Preprint server**, **Institutional repository**, **User import**, **Other**, or **Unknown**. Manual type and annotation choices survive later fingerprint rescans.

Publisher is used only when a recognized publisher host or a publisher-specific acquisition path provides that evidence. arXiv, bioRxiv, and medRxiv are shown as **Preprint server**; repository hosts remain **Institutional repository**; ordinary local additions are **User import**; and a generic web download with no stronger evidence remains **Other**. A real host can appear after the source label when useful. Internal migration or ingestion markers are never user-facing source details, and choosing a different source clears a stale provider detail rather than appending it to the new choice.

> **Note**
> Changing File type can change the automatic Default. The selected managed Default and its main file location are synchronized together. Annotation and Source remain independent properties; changing a label does not apply a filename template or rewrite the PDF.

See also: Download and Sync Versions · Add and Classify Attachments

3-2.8 Add and Classify Attachments

Import first, classify afterward. This keeps the add operation simple and lets you set type, annotation state, and source after seeing the file in context.

Add a file to the paper

1. **Control-click blank space in the Paper Files area and choose Add File.** In My Publications, expand Files and use Add File.
2. **Select one or more local files.** Confirm the file chooser.
3. **Wait for the new row.** A new item begins as **Unknown** unless its source proves a more specific type.
4. **Select its compact controls.** Choose File type; in Library, click the size footer to expand Annotation and Source. My Publications shows those controls directly in its Files panel.

Attachment types

The same file-type choices apply to every row and to Browser capture. Use **Summary** or **Analysis** for notes, **Source Data** for research data, and **Source Code** for code; use **Other** when none fits. File extension and MIME type remain separate from this classification. Existing files keep their saved labels until you change them.

Adding the same unchanged file to the same paper again reuses its existing resource, including when its source path or filename has changed. Matching uses exact file contents; an edited or annotated PDF with different bytes remains a separate version. A shared DOI alone does not make two files duplicates.

Add by drag and drop

Drag one or more Finder files anywhere in the visible Paper Files column: over its heading, an existing card, or the empty gray area. No card needs to be expanded. Wait for the whole-column drop highlight, confirm the selected destination paper, then release. The files are copied into that paper's associated bundle and appear in Paper Files. The same behavior applies in Library, shortcuts and Spaces.

> **Important**
> Invalid or unreadable PDFs are rejected at file ingress. Existing migrated records that can be proven invalid remain visible as **Invalid file** for audit and repair, but cannot become a default.

See also: Move a Standalone File into Paper Files · How Gewu Insight Stores Files

3-2.9 Download and Sync Versions

Downloads add a version to the paper bundle. They do not silently overwrite a different preserved PDF.

Download the best available paper

1. **Verify DOI and arXiv.** Make sure the identifiers belong to the selected work.
2. **Choose Download at the top of Paper Details.** Gewu Insight first tries the best available published file, then supported alternatives.
3. **Wait for the status message.** Existing files are preserved if the incoming file is a different version; a manual default stays selected.
4. **Inspect Paper Files.** Every preserved file appears in the same flat list. Automatic ranking may select the new PDF as Default and synchronize its main file location; a manual default remains unchanged. Earlier files stay in the bundle.

Keep downloading while you work

You can select other papers, switch workspaces or add another Monitor result to Library while a download runs. The running state belongs to each paper, and completion updates that paper without selecting it or returning to an earlier tab. Repeating Download for the same running paper shares its request; another paper has its own request. Ordinary Download reuses an existing valid local PDF. Downloads do not resume automatically after quitting the App.

Direct download and background webpage fallback

In the original macOS App, supported publisher downloads first try their direct routes. If those do not provide the main PDF, APS, ACS, Science and Nature-family downloads can load the article in an offscreen webpage and follow its PDF link. This background fallback shares the Gewu Browser website session and keeps your current workspace and selection in place. The App must remain running; a command-line or ordinary web-browser session does not provide this native fallback.

Publisher	Current download route
APS / Physical Review	The direct APS Harvest endpoint is tried before the background article/PDF route. Harvest remains the fast first route when the publisher provides access.
ACS / Nano Letters	Direct PDF retrieval, then the background article page when needed.
Science family	Direct retrieval, then the shared Browser session. The main article must be obtained before supplementary PDFs are appended.
Nature family	Direct retrieval, then the background article/PDF page. A publisher reference or accepted manuscript is retained as that version, not labeled as the published PDF.

Other publishers use their supported direct routes; this does not imply a webpage fallback for every website. Background browsing does not force Science or another publisher through a proxy. Your current network, system proxy/VPN rules, browser verification and article entitlement still determine access.

If the background attempt still cannot retrieve a PDF

1. Choose **Open Website** and open the article in Gewu's **Browser**. If Website Links is set to the system browser, open the address in Gewu Browser to use its shared session.
2. Complete the publisher's normal verification or institutional sign-in yourself, and check whether the main PDF actually opens. An abstract page alone does not establish full-text access.

3. Return to the paper and retry **Download**. Alternatively, with the PDF open in Browser, use **Save to Gewu**, verify the Key and loaded PDF in Details, then save.

A legitimate local PDF can also be attached with **Add File**. A timeout or authorization error leaves existing files intact; repeated retries do not create access rights. See Save an article and PDF to Library and Publisher and Network Access.

Download arXiv files

- Use the download icon in the **arXiv** row to retrieve the latest arXiv PDF.
- Control-click blank space in Paper Files and choose **Sync Versions** to retrieve all available arXiv revisions such as v1, v2, and v3. In My Publications, use Sync Versions inside the expanded Files panel. The command is available when the paper has an arXiv identifier.
- Each revision is labeled with its arXiv version when known.

Duplicate protection

Gewu Insight computes exact SHA-256 and PDF content fingerprints as each new resource is registered. An exact duplicate is reused instead of stored twice. A clean PDF and a genuinely annotated copy can share a content fingerprint while retaining different exact hashes, annotation counts, and labels.

Validation before replacement

A response is not accepted merely because it begins with `%PDF-`. Structural PDF validation rejects empty files, HTML responses, truncated downloads, missing valid endings, and content that cannot be parsed as a PDF. A download is written to a temporary file and replaces the destination only after validation succeeds. If the request or stream fails, an existing valid file remains untouched and a partial file is not registered as a resource.

Classification follows the file that actually arrived

A publisher download is classified from the file and source actually returned. A corrected published version outranks a published version automatically; a publisher reference or accepted preview remains an accepted manuscript. Science PDFs recognized as First Release are classified as Accepted manuscript in Gewu, even when the paper metadata already names Science. A correction notice remains a separate peer and never becomes automatic merely because it came from a publisher.

A PDF URL stored in bibliographic metadata is only a retrieval lead. It does not create a Paper Files row until a download actually succeeds, the returned bytes pass validation, and a local resource is registered.

If a download is skipped because the local bytes already exist, that skipped operation does not relabel the existing file from metadata or a newly discovered URL. Correct the file type explicitly if the saved classification itself is wrong.

> **Note**
> A Monitor result is not forced into a Library placement. Where download is available, a local resource can belong to the cataloged paper even before you add that paper to Library or My Publications.

Science and Nature main articles and supplementary PDFs

The Science and Nature-family workflows distinguish the main article from supplementary PDFs and can append retrieved PDF supplements to the combined reading file. A supplement, reporting checklist or supplementary-only response cannot stand in for the main article. If the main PDF cannot be obtained, the download reports failure instead of registering a supplement as a successful article download.

Inspect the combined PDF and any download message to check what was actually retrieved; access to the article webpage does not guarantee access to every PDF. Non-PDF material is not converted into PDF pages. The file type follows the main article's version, while the displayed filename uses the returned publisher name or a meaningful paper-based fallback. Re-downloading preserves other files and an explicitly chosen Default.

Non-PDF supplements do not make the article fail

Automatic supplement handling is PDF-only. MP4 videos, audio, archives, spreadsheets and other non-PDF attachments are skipped; they are not required for a successful main-PDF download. Nature review files, reporting summaries and source-data files are also excluded from the supplementary reading merge. You can obtain other useful files separately and attach them to Paper Files.

If the main PDF succeeds but an optional supplement is unavailable, the main file is kept and the message describes the missing supplement. A message such as `main PDF unavailable; downloaded 1 PDF supplement(s)` means the main article could not be retrieved, not that a video caused the failure. A `not a PDF` response may be an HTML access or verification page. Supplements alone are never presented as a successful main article.

See also: Choose File Types and the Default File · Publisher and Network Access · Edit DOI and arXiv Identifiers

3-2.10 Move a Standalone File into Paper Files

If a data, image, presentation, or other file was imported as its own Library item, drag it onto the correct paper's Paper Files area to make it an attachment.

Convert an independent item

1. **Select the destination paper.** Keep its Paper Files inspector visible.
2. **Begin dragging the standalone item.** The main paper remains the active destination while you drag.
3. **Move the pointer into the Paper Files drop zone.** Wait for the drop feedback that identifies the target paper.
4. **Release the pointer.** Gewu Insight validates the move, adopts the file into the paper bundle, and removes its independent Library position.
5. **Classify the attachment.** Choose **Supplementary material** for an actual paper supplement, **Source Data** for data, **Source Code** for code, **Summary** or **Analysis** for notes, or **Other** when none fits. Expand the size footer to set Annotation and Source separately. See the complete current file-type list.

> **Result.** The physical file and database link now belong to the destination paper; the former standalone entry is gone.

Make a Paper Files resource independent again

Drag a resource's icon or filename from Paper Files onto a Library folder or that folder's content area. The file leaves its former paper and moves from Associated Files into the destination's main folder. A PDF becomes an independent paper with a new Gewu Key; a non-PDF becomes an ordinary standalone file with only an internal storage key. Control-clicking the resource and choosing **Create Paper from File** performs the same separation in the current folder, with an editable title. The former paper keeps its remaining files.

To transfer just one resource between existing papers, use **Move to Another Paper** or drag it into the other paper's Paper Files area. A Library paper-row drag adopts the source paper's files as a group; it removes the former independent identity only when the conversion is safe. A single-file RTF therefore disappears from its former folder and becomes one resource of the destination paper.

> **Check the destination**
> Dragging within Gewu Insight changes ownership or location. Dragging a file to Finder or the Desktop makes an external copy and leaves Paper Files intact. Dragging from Finder into Paper Files imports a managed copy and keeps the external original.

If the conversion cannot finish

The file moves and database changes are coordinated as one recoverable operation. If any step fails, Gewu Insight restores the original standalone files and rolls back the paper, Library item, resource relationships, and fingerprint fields. The source should remain usable rather than becoming a half-converted attachment.

> **Important**
> If the source item is shared by another Library reference or protected relationship, Gewu Insight may refuse a destructive conversion. Add a copy with **Add File** instead.

See also: Add and Classify Attachments · Folders, Shortcuts, and Spaces

3-2.11 Delete and Recover Items

Deleting a paper reference either removes only that reference or moves the last active Library/Space reference to Recycle. Final deletion inside Recycle is reference-aware and uses the macOS Trash for managed files.

Delete a Library reference

1. **Control-click the item.**
2. **Choose Delete.**
3. **Check the result.** If another active Library or Space location, or My Publications, still refers to the same work, only the selected reference is removed. Otherwise the item moves to **Recycle** and remains recoverable.

What happens to the original and attachments

A paper's original can move into the local Recycle directory only when no normal Library, Space or My Publications placement still uses it. The same check applies when you delete or drag a whole folder: retained papers keep their originals in normal storage, or move to a surviving location. Recycle entries can remain shortcuts to those retained originals. Other paper records sharing the registered bytes are protected too.

Recovering, copying or adding a recycled paper back to an active location brings its original out of Recycle. This includes Monitor additions, Browser saves and Add to My Publications. If a recovery parent has also been recycled, an active fallback is used. The app synchronizes SQLite file paths, Associated Files and Finder shortcuts while retaining Gewu Keys, file IDs, classification, Default and file contents. These checks apply during file operations; they do not silently sweep historical disk remnants.

Deleting a single removable **Paper Files** attachment is a different action: its private managed copy goes directly to macOS Trash and its file registration is removed from the shared paper. Other files and the parent paper stay. All placements of the same Key see this attachment change. Shared bytes and external originals remain protected.

Delete from My Publications

Select one or more publications and choose **Delete**. If a Library, Space, Shortcut or Recycle location still refers to the work, its shared article record, Gewu Key and files remain. Otherwise, deleting this last placement also removes the article record and its identifier tasks from SQLite; private managed files move to macOS Trash. External originals and bytes shared with other articles are preserved. Import history does not keep an otherwise unreferenced article alive.

Removing every currently saved article of a running profile import pauses that import. Continue Import does not restore previously removed source entries. An explicit Add All Publications recheck after completion can add those articles again.

Recover an item

1. **Open Recycle.**
2. **Control-click the item and choose Recover.**
3. **Confirm the restored location.**

Delete permanently

Final deletion of Library items takes place inside Recycle. Choosing **Delete** on a folder moves that folder intact, including its subfolders and unregistered contents, to macOS Trash. The confirmation lists the folders and files to move. Original imported source files outside Gewu Insight are not touched. If another Library, Space, Recycle or My Publications location still refers to the paper, its shared record, Gewu Key, Paper Files and Default remain; any required local files are retained outside the folder being removed. Only papers with no remaining references are removed from SQLite.

> **Empty Recycle**
> Control-click the Recycle root folder or blank space directly inside it to choose **Empty Recycle**. This clears every Recycle entry and moves all contents of its local disk directory to macOS Trash, including unregistered files, empty folders and broken shortcuts. The Recycle root remains. This command is not offered on files or folders inside Recycle.

The deletion confirmation has a bounded height and an independently scrolling item list. Read the selected scope, then confirm. A failure remains visible for retry instead of reporting success. **Empty Recycle** clears the application's Recycle; it does not empty macOS Trash.

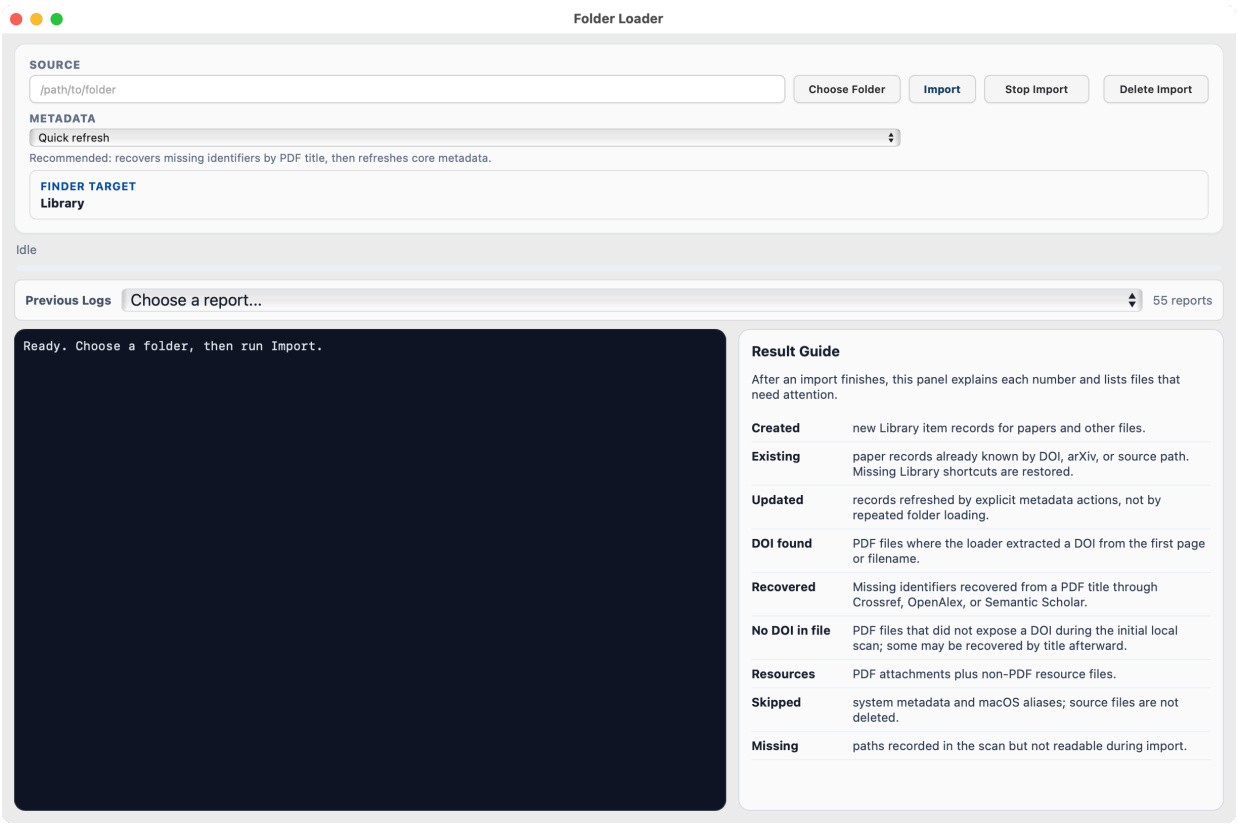
See also: Back Up or Move to Another Mac · How Gewu Insight Stores Files

3-3 Import and Transfer

3-3.1 Use Folder Loader

Folder Loader imports an existing directory tree into managed Library storage while preserving relative paths.

For a Zotero library, use Loader > Import from Zotero instead. It reads collections, attachment relationships and notes from Zotero; selecting Zotero's storage directory in Folder Loader does not preserve that library structure.



Folder Loader. Choose a source directory and metadata mode before Import. The live log and Result Guide fill the remaining window height and scroll independently.

Import a folder

1. **Choose Loader > Open Loader.** Outside Browser, you can also press ⌘L; inside Browser that shortcut focuses its address field.
2. **Choose the source folder.** Review the displayed path before starting.
3. **Select a metadata mode.**
   - **Skip metadata** — fastest; import files and structure only.
   - **Quick refresh** — recommended for most migrations.
   - **Full refresh** — more network work and a longer run.
4. **Choose Import.** Follow progress in the live log. Use **Stop Import** if necessary.
5. **Review the result.** Open previous logs or the result guide when items need attention.

Follow a folder drop without keeping Loader open

Dropping an external folder into Library starts Folder Loader in the background; it does not need to open the Loader window first. The Library footer shows import progress. Choose **Details** beside the activity indicators to open the live Loader window, inspect progress and read the log. Closing it leaves the import running, and Details opens it again. When the import has finished, Details still opens the retained result until you choose **Dismiss**; previous reports remain available in Loader.

Click a log entry linked to a paper to **Show in Library**. The window closes after the item is revealed, without interrupting the job. General progress messages cannot identify a Library item. Use the footer's Details to return, or **Loader > Open Loader** to review saved reports. See log navigation and task controls.

Stop Import finishes at the next safe checkpoint, preserves completed files and ends any linked metadata queue. It does not undo the import. The separate **Delete Import** command has a different, destructive purpose.

What Folder Loader preserves

- The source folder's relative hierarchy.
- Existing managed targets and files on repeated imports.
- An existing main file rather than overwriting it, plus edited versions with different contents.
- A verified exported Paper Files relationship when a Gewu Insight manifest is present.

Folder Loader copies into Gewu Insight's managed storage. It does not move, rename, or delete the external source files and folders you selected. Loader, Paper Details, folder and all-Library refresh use the same service and refresh history. **Quick refresh** updates core information (title, authors, journal and date) and verified DOI/arXiv links; **Full refresh** adds detailed authors, citations and other source data. Recovered identifiers use the selected mode. Partial source failures remain visible in the log and are not recorded as a fully successful refresh.

Identity mismatches and partial imports

Loader and Finder folder drops share the same matching and file-preservation policy. When a DOI, title or existing identifier disagrees with the matched paper, the incoming file is still saved in its Paper Files and the paper is placed in the target folder. Existing metadata and Default are preserved. PDFs with different bytes but the same DOI are kept as separate files; an exact unchanged copy already owned by that paper reuses its existing resource even after a source rename or move; the import continues without a mandatory review prompt.

The live log and **Result Guide** identify each mismatch by filename, detected DOI, existing DOI and Gewu Key. Correct a wrong assignment later from the file card's **Move to Another Paper** or **Create Paper from File** command. This preserves the PDF; it does not certify that the identified DOI is correct.

A disk or other import error can still stop a batch. Completed items remain available, and the result shows the failing file and actual progress. For a background drop, use the bottom bar's **Details** to reopen its current result and **Dismiss** when finished. Previously saved results remain available in Loader logs.

Delete an import batch

Delete Import is a rollback for the selected Loader batch, not a way to forget a log. Stop an active import first. The command removes database records created by that batch, eligible empty folders, its Loader reports, and managed local copies associated with the batch; eligible paths are sent to the macOS Trash. It never deletes the original external source directory.

> **Important**
> Review the selected batch and make a backup before using Delete Import, especially if you have already reorganized or reused imported items.

> **Note**
> For an ordinary folder, Loader does not guess paper-to-attachment relationships from similar filenames. A verified Gewu Insight export manifest can restore those relationships. After paper identity and ownership are known, the normal Default and Associated Files rules determine where that paper's managed files belong.

See also: Make a Portable Desktop Copy · Refresh Info: One Paper, a Folder, or the Library · Back Up or Move to Another Mac

3-3.2 Import a Zotero Library

Copy a Zotero library or selected collections into your current Gewu Database, with collection hierarchy, Paper Files and readable notes. Zotero's records, original attachments and annotations stay unchanged.

> **Choose the workflow you mean**
> Opening the Gewu tab in Zotero displays your existing Gewu workspace. It does not import Zotero's own library. **Import from Zotero** makes an explicit copy from that separate library into Gewu; it is not continuous synchronization.

Before you start

Use a Gewu App with companion **0.4.0 or later** and keep Zotero running during preview and import. This preview has been exercised with Zotero 10.0.6 on macOS; older Zotero versions, Windows and every third-party plugin configuration are not covered by that validation. In the standalone App, **Plugin > Zotero Integration** shows the bundled and installed versions. Follow its installation guide or use **Update Plugin** when offered.

Download cloud-only attachments in Zotero first, check that linked files open on this Mac, and allow enough free space for independent copies and any exported annotations or notes. You do not need to export a Zotero database or locate its internal storage folders.

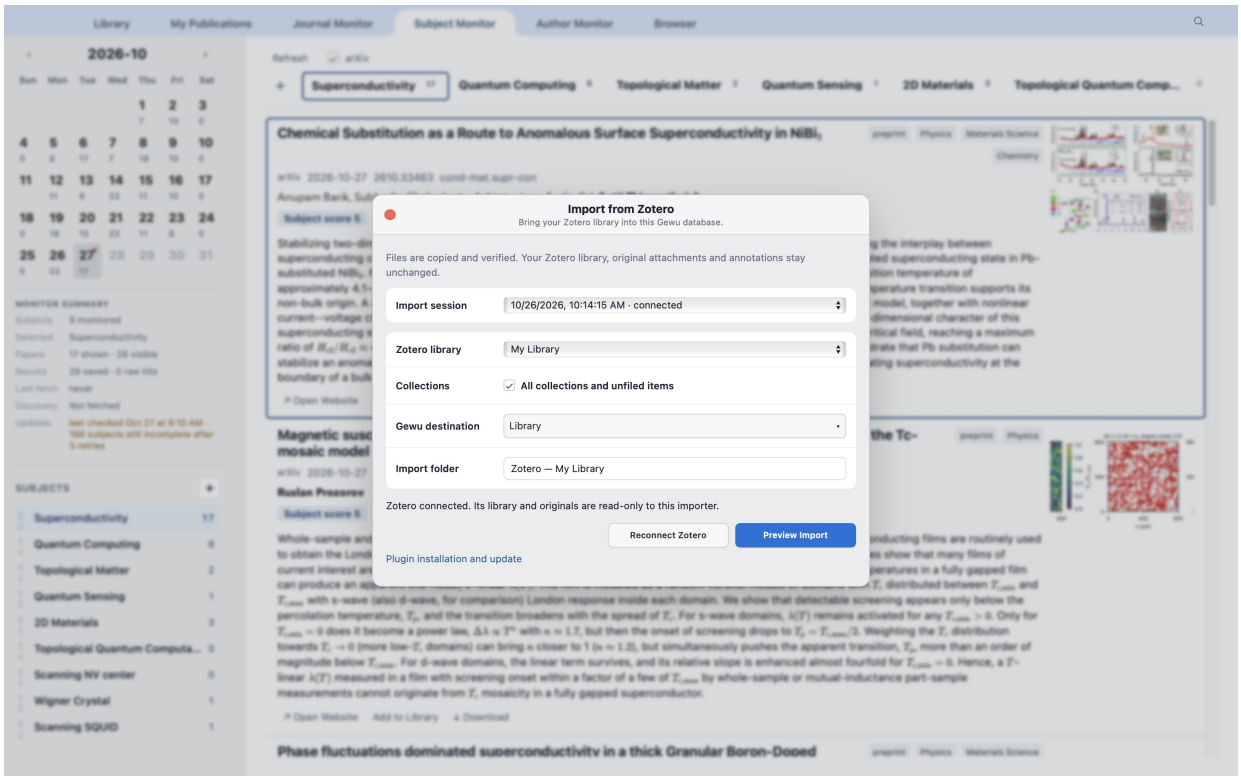
Open the importer in either interface

Where you work	Entry point
Standalone Gewu Insight	macOS menu: **Loader > Import from Zotero**.
Gewu Insight tab inside Zotero	Upper-right **… > Loader > Import from Zotero**.

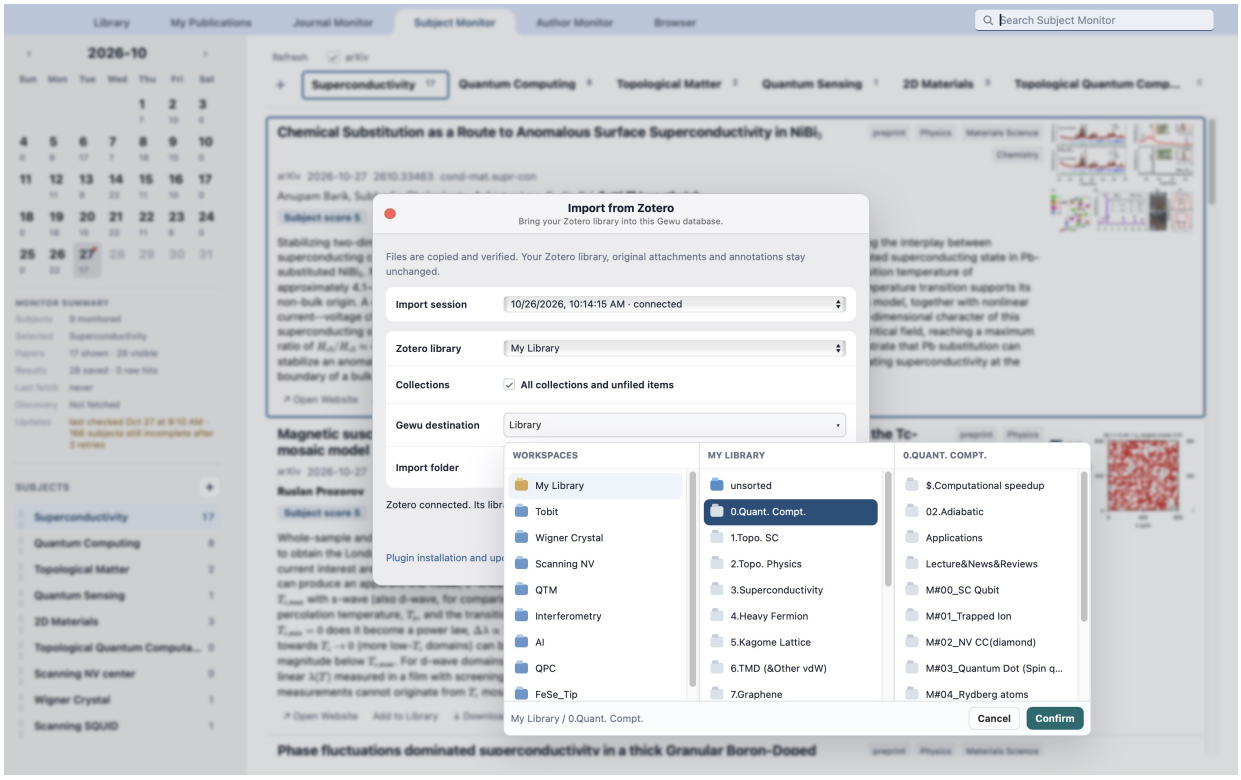
Both open the same importer for the same selected Gewu Database. You do not import once for the App and again for the plugin. **Connect Zotero** or **Reconnect Zotero** connects this import session to Zotero's loaded library; it is separate from the workspace's Connection Settings.

Choose scope and destination

1. Open the importer, choose **New import** if needed, and click **Connect Zotero**. Wait for the library choices to appear.
2. Choose **Zotero library**. Group libraries appear only when available in your local Zotero profile.
3. Keep **All collections and unfiled items** selected to copy the whole library, or clear it and select collections. A selected collection includes its subcollections; unselected ancestors are not added above the selected subtree.
4. Click **Gewu destination**. Browse the columns from Library or a Space into the required folder. Read the full path at the bottom and click **Confirm**. This selects the import target; it does not change your default Save to Gewu destination.
5. Set **Import folder**, for example **Zotero — My Library**. This folder contains the imported hierarchy beneath the selected destination.



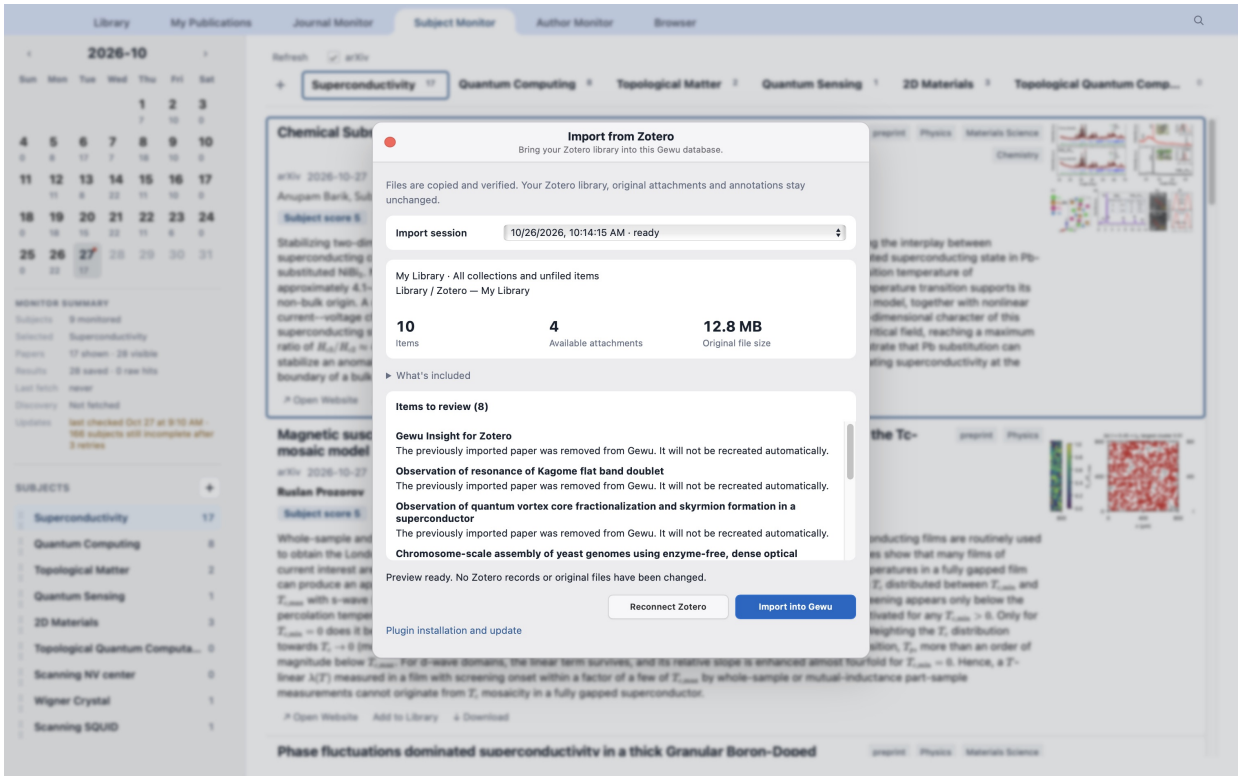
Select the source and target. This connected session is ready for Preview Import. The source is Zotero's My Library; the target is a new import folder under the Gewu Library. Connecting alone does not copy papers.



Choose a destination by hierarchy. Each column opens the next level. Confirm accepts the path shown at the bottom; Cancel keeps the previous destination. The folder names shown are examples from this workspace.

Preview, then start the copy

1. Click **Preview Import**. This reads the source and prepares a saved preview. It does not create Library papers, collection folders or their disk directories.
2. Check the source, destination, item count, available attachments and original file size. Expand **What's included** and read **Items to review**. Available attachments are local source files, not a promise that every item has a PDF; original file size is not a complete estimate of exported notes and annotated copies.
3. Repair missing source files or uncertain identities before proceeding. Files or papers deliberately removed from Gewu are not silently recreated. A changed Zotero item or repaired file needs **New import** and a fresh preview.
4. When the scope and review results are correct, click **Import into Gewu**. The importer creates the selected hierarchy and verified copies, then reports completed and unfinished entries.
5. Inspect the result in the chosen Gewu folder. Check a nested collection and its Paper Files; use Finder to inspect the corresponding managed directories if needed.



Review before importing. This real preview includes previously imported records that were later removed from Gewu. They are flagged instead of silently restored. Counts and warnings depend on your library. No import was started for this capture.

How collections and files are stored

In Zotero	In Gewu
Collections and subcollections	Folders under the import folder, in the same selected hierarchy. Empty collections also become real directories, with matching Associated Files directories, when import starts.
One item in several collections	One paper and Gewu Key with several folder memberships. Secondary placements follow Gewu's existing shortcut rules.
Stored or linked attachments	Independent, verified copies in Paper Files. The source remains in its original location.
Multiple PDFs and supplements	Separate Paper Files. The Default file occupies the canonical Library or Space location; other files belong in its mirrored Associated Files bundle. Existing manual Default choices and classifications are retained.
Item without a local PDF	A metadata record. Import does not fabricate a DOI or automatically download a missing PDF.

In Zotero	In Gewu
Notes and PDF annotations	Readable notes plus original archives; native PDF annotations can produce a separate annotated PDF and annotation text. Annotations already embedded in a PDF remain in its copied bytes.
Tags, citation keys and extra fields	Preserved in Zotero Metadata and the migration record. This does not add a new tag editor or reproduce every plugin's interface.

For example, choosing **Library / Research** as the destination and **Zotero — My Library** as the import folder places **Materials / Empty collection** beneath that root both in Gewu and under **data / Library / Research / Zotero — My Library / Materials / Empty collection**. The matching hierarchy exists under **data / Associated Files**. A Space target uses its existing Space storage rules. The importer does not clone Zotero's internal `storage` directory layout. See How Gewu Insight Stores Files for canonical files and secondary aliases.

Original note archives retain richer formatting and embedded assets. Zotero-specific interactive annotation links, saved searches, Trash, sync credentials and plugin settings are not recreated as Gewu controls.

If you use ZotFile, ZotMoov or Attanger

The importer follows the attachment registered in Zotero, including stored files, absolute linked paths and paths relative to its Linked Attachment Base Directory. The arrangement used by a file-management plugin can therefore be imported when the file is registered and locally accessible. It does not scan an unrelated plugin folder, reproduce that folder's arbitrary hierarchy, or claim compatibility with every plugin version.

External tablet or backup copies that are not registered as Zotero attachments are not included. Better Notes content stored as ordinary Zotero notes follows the note mapping; external synchronized Markdown files and custom plugin state are not reconstructed. Fix missing links or legacy Mac aliases in Zotero, then prepare a new preview.

Pause, resume and repeat safely

Pause Import keeps completed copies. **Resume Import** continues the saved session; **Retry Unfinished** retries incomplete entries. You can close the window and reopen it through Loader, then choose the session from **Import session**. After either app or its service restarts, use **Reconnect Zotero** before resuming.

Repeated imports recognize previously mapped source records and unchanged files; they do not make a second full copy. Mapped folders are reused, while changed file versions may coexist with earlier copies. Later Zotero collection renames or deletions do not rename or delete Gewu folders. Folder Loader's **Delete Import** belongs to that separate workflow, not this importer.

> **Source safety**
> Import copies and verifies files rather than moving Zotero's originals. It does not edit Zotero's library records, transfer away PDF annotations or rewrite linked paths. A blocking file, shortcut or conflicting destination is reported without being overwritten. Manage the resulting Gewu copies through Gewu's normal file controls.

See also: One Workspace, Two Interfaces · Install, Update, and Remove the Plugin · How Gewu Insight Stores Files · Edit DOI and arXiv Identifiers

3-3.3 Make a Portable Desktop Copy

Make Copy Desktop creates a self-contained copy for transfer, inspection, or re-import.

Create a copy

1. **Control-click the non-root Library folder you want to export.**
2. **Choose Make Copy Desktop.**
3. **Wait for the completion message.** Open the resulting Desktop folder in Finder.
4. **Confirm representative files.** Check a paper with versions or attachments as well as a normal folder.

What is included

The export replaces managed filesystem shortcuts with real files, includes relevant Associated Files, and writes a verified manifest describing paper-file relationships. This lets Folder Loader restore the bundle on another installation without guessing from filenames.

> **Important**
> Do not edit or discard the export manifest before migration. Without it, Loader correctly treats the files as an ordinary folder and does not infer hidden associations.

See also: Use Folder Loader · Back Up or Move to Another Mac · How Gewu Insight Stores Files

3-3.4 Back Up or Move to Another Mac

A reliable transfer preserves both the SQLite database and the managed file roots. The app bundle alone is not your library.

Know what you are backing up

Location	What it contains	What an update does
`/Applications/Gewu Insight.app`	The daily application and its packaged resources, including Print Guide.	Replace or update the App while it is closed; the selected research folder stays in place.
Your selected `Gewu Database`	`data`, `reports`, `config` and `backups`. This includes SQLite and managed research files.	Continue using the same database, Gewu Keys, classifications, placements and analyses.
A source checkout, when used	Development code, assets and the local Python environment.	Preserve local edits and update code separately. A source-backed App still needs this checkout and its configured environment.

Find the actual research folder with **Gewu Insight > Database Location…**. A desktop folder named `Gewu Database` is one possible choice, not a required path for everyone. Removing a Dock shortcut or replacing the App does not delete this folder. A GitHub checkout or App-only copy is not a research-data backup.

Create a full backup

1. **Finish or stop active work, then quit with ⌘Q.** Finish unsaved webpage forms first. Closing a window is not the same as quitting; any separately started Gewu service must also stop before its database is copied.
2. **Copy the complete selected `Gewu Database`.** Include its hidden files, `data`, `reports`, `config` and existing `backups`. SQLite and the files it describes must come from the same closed state.
3. **Include any external files still used by links.** A Finder alias does not contain its target's bytes. Preserve those originals and record their paths. Historical import-source paths are provenance, not proof that every old source is still an active dependency.
4. **Preserve macOS file information.** Keep filenames, permissions, symbolic links, Finder aliases and resource forks. Use a backup method that preserves them. If a USB filesystem cannot hold large files or these attributes, use a suitable Mac backup container on the disk rather than reformatting it or splitting individual research files.
5. **Keep software and machine configuration recoverable.** Before an upgrade, also retain the installed App, any source checkout and local edits, database-location preference and relevant AI-client configuration. They are separate from the research folder.
6. **Verify and store separately.** Check the copied SQLite, file list and representative PDF/Markdown files before relying on the backup. Use another disk or a versioned backup location, outside the live research folder. Keep full backups private because they can include service credentials and connection tokens.

A ZIP or filesystem copy that loses alias information is not an equivalent replacement for a verified Mac backup. A `.sparsebundle` backup container can be mounted in Finder to inspect its contents; keep the container intact and eject its mounted volume after use.

Back Up and Upgrade

1. **Note the active database location, then quit and back up.** Preserve the research folder and recoverable software together.
2. **Update the existing installation.** A standalone App includes its own runtime. A source-backed installation uses its configured checkout and Python environment; preserve local edits when updating GitHub code and retain its local startup configuration. Downloading source alone does not update a separately packaged App.
3. **Install matching native resources while the App is stopped.** Native-code or bundled-resource changes require a matching App build, including the fixed-layout guide. A source-backed Python-only update needs a restart to load the changed code. Keep the daily entry in Applications; do not launch an archived App as a second service.
4. **Start once and confirm Database Location….** Continue with the same research folder. The App checks database identity, version compatibility and whether another process already has it open. Before a required schema upgrade, it saves a SQLite snapshot under the workspace's `backups`; this snapshot is not a full research-file backup.
5. **Verify before retiring the old installation.** Check Library, a Space, a default PDF, another Paper Files version, an analysis, My Publications and a monitor. Check **Help > User Guide** and **Print Guide**. Quit and reopen once to confirm the saved database location and any existing MCP connection.

The local build uses ad-hoc signing and does not require an Apple developer certificate, Team ID or keychain password. This is a local integrity signature, not Apple notarization. Build and packaging details are in `docs/DEVELOPMENT.md` and `DISTRIBUTION.md`. Retain any source-backed startup adapter when rebuilding; a generic build does not automatically reproduce a machine's private configuration.

Schema migration updates database structure; it does not rebuild your collection from exports or require another Library import. Later ordinary App/source updates keep using the same research folder without repeating the directory migration.

Move an older source-folder installation once

Older installations kept `data` and `reports` inside the source project. Use the supplied `docs/SOURCE_DATA_MIGRATION.md` procedure to separate them. A maintainer or local AI coding assistant can perform the backup, protect local edits, update the code, copy the workspace to your chosen new folder, verify it and configure the daily App. Confirm the destination before migration; no new DMG or fresh Library is required.

The migration must update managed absolute paths and Finder aliases as well as copy SQLite and files. Check paper and resource counts, Gewu Keys, classifications, Default choices, placements, analyses and permissions. Record any missing files that predate migration separately from new failures. Keep the original recoverable until the new App, real Paper Files, MCP and a normal restart have been checked.

After verification, the old research folders can remain in a clearly labelled archive outside the source checkout. Keep them as recovery material; do not open the old App against the upgraded database or use the archive as another daily collection. Update the Dock shortcut when the App's location changes.

Move to another Mac or another folder

1. **Prepare a verified, closed backup and a compatible App.** Keep both the source and destination App closed during the copy.
2. **Move the complete workspace through the migration procedure.** A different username, disk or parent folder can change absolute paths. Merely dragging the folder or choosing it in Open Existing Database does not rewrite those references.
3. **Open the verified destination workspace.** In **Open Existing Database**, select the folder containing data/paper_monitor.sqlite. A remembered location is a local machine preference; configure the new machine for its actual path rather than copying another Mac's path blindly.
4. **Check real records and files, then restart.** Verify a Space, versions, an attachment, saved analysis, My Publications, monitor state and the database location. Reconnect AI clients with their actual runtime paths if needed.

Restore without losing newer work

Quit the current App and first preserve the current workspace, especially if you have added papers or notes since the backup. Restore a matching software/database/files set; do not replace only SQLite or combine unrelated Associated Files trees. An old App must not open a database upgraded by a newer version. If the restored location differs, migrate its file references before using it. Verify the result before deleting any retained copy.

Transfer only part of a Library

Use **Make Copy Desktop** on the source Mac, move the complete exported folder, then use Folder Loader on the destination Mac. Keep its verified manifest intact.

Browser and credential backup boundaries

The selected database includes Browser bookmarks, folders, history, and saved sidebar preferences. A bookmark HTML export transfers only bookmarks, not history or research papers. Open webpage tabs and unsubmitted forms are memory-only and are not restored after quitting.

WebKit cookies, sign-ins and cache live in a macOS-managed profile outside Gewu Database. Research-folder copies and bookmark exports do not transfer it; expect to sign in again on another Mac. Apple Passwords and passkeys remain under macOS/iCloud management. Metadata API keys saved in SQLite travel with the database: keep full backups private.

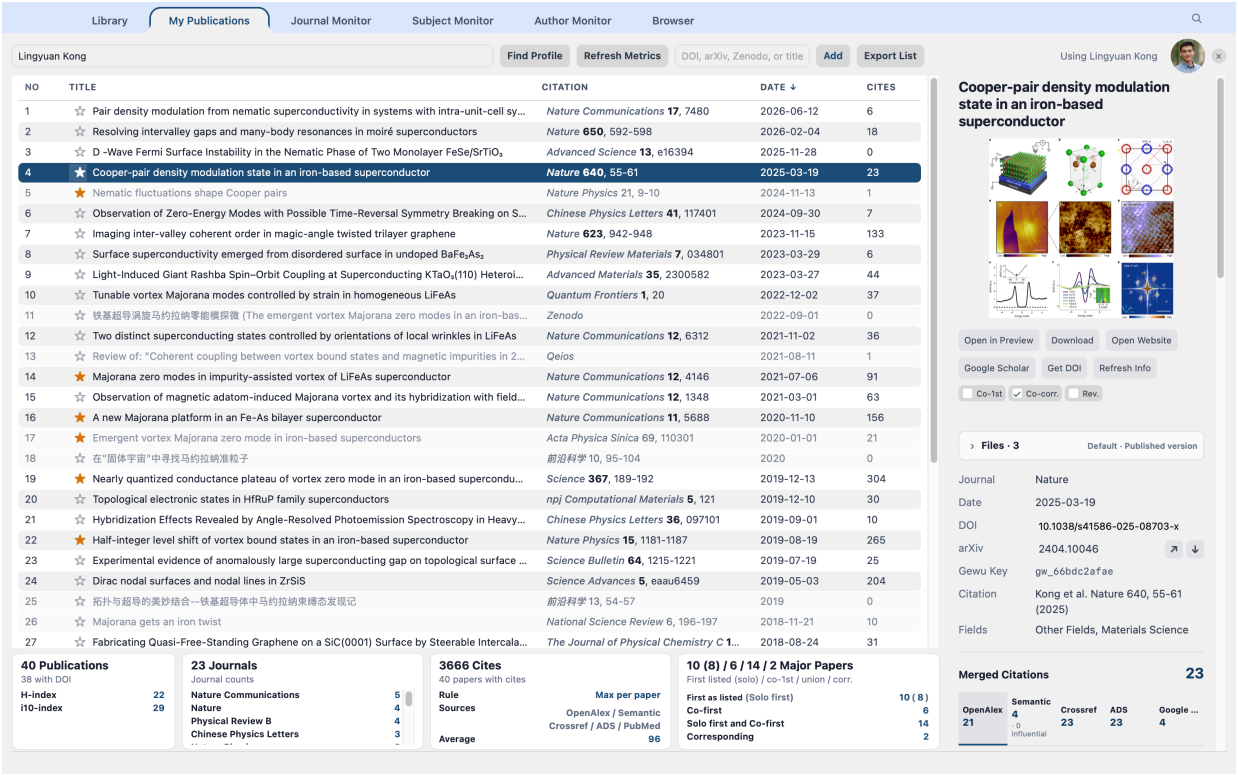
> **Important**
> Never migrate by copying only `paper_monitor.sqlite` or only the visible Library folder. Either choice leaves half of the managed state behind.

See also: Make a Portable Desktop Copy · Storage Reference · Troubleshooting

3-4 My Publications

3-4.1 My Publications

My Publications keeps a managed publication list, citation sources, author-role statistics, and export tools for your own work.



My Publications. The active profile, DOI-resolved publication list, citation evidence, article figure, and authorship summary share one view.

Choose your profile

1. **Enter a name, optional affiliations, or an exact Google Scholar profile URL.** For a name search, use `Name; affiliation A; affiliation B`.
2. **Choose Find Profile.** Review candidate profiles carefully.
3. **Choose Use Profile to review candidates individually, or Add All Publications to import the complete profile.** The saved strip shows the active name and profile photo. Select the photo to open the profile using Website Links; use the remove control to forget it.

Already have the paper in Library? Control-click its row and choose **Add to My Publications**. Multiple selected papers can be added together. The publication list and Library then refer to the same work, files, and Gewu Key.

Add all publications or choose individual papers

Use Profile opens a candidate list where each article retains its own **Add** button. To import the author's entire profile, choose **Add All Publications** beside the correct search profile or at the top of its article candidate list. These controls appear only inside the search/add area; the normal publication table keeps its original layout. The complete import reads successive source pages until the profile ends; it is independent of the 200-title candidate preview and has no 300- or 5,000-paper cutoff. Existing canonical articles and My Publications entries are reused.

The bottom status bar continuously shows active import counts and DOI/arXiv progress, including running, processed, paused, and retry-needed states. Completed imports move into **Import history**, which

distinguishes original import counts from articles currently saved in My Publications. When only history remains, the footer entry reads **History**. Old imports do not appear in a new author search; its own status or error stays visible in the search panel. Choose **Details** in that bar to view full progress and import controls without searching again. The search/add panel opens when searching or choosing a profile and stays open until you choose its **Close** button; clicking a publication or completing a task does not dismiss it. Closing the panel keeps background work running and preserves the table position. **Pause Import** preserves completed work; **Continue Import** resumes it. An interrupted source page is retried without adding its successful articles twice. A blocked, unreadable, repeated, or interrupted source page is reported as incomplete with a short reason and guidance to try again later. Technical connection details stay in the diagnostic log. Individual identity conflicts do not stop later articles and remain reported for review and retry. Adding all again after completion explicitly checks the profile from its first page for additional articles.

Both individual Add and Add All automatically queue missing DOI/arXiv identifiers. Articles are saved first, and identifier completion runs separately in the background. Its progress distinguishes matched, unmatched, failed, and pending work. Switching tabs does not stop either task; unfinished work resumes after app restart. Identifier updates preserve the article's Gewu Key and file relationships. The existing per-candidate **Retry ID** remains available. When saved articles have unmatched or failed identifiers, **Retry Unresolved IDs** appears with progress inside the search/add area. Pausing the profile import does not pause identifier work already queued. Articles without a confident match remain saved. This operation does not automatically download PDFs.

After Scholar reports an access restriction, Gewu Insight applies a shared 24-hour pause to Scholar requests. The prompt shows the local retry time; this is an application pause, not a Google-provided IP-ban deadline. Changing authors or restarting does not clear it.

Add and update publications

- Use the identifier/title field and **Add** to add a DOI, arXiv record, supported identifier, or title.
- Choose **Refresh Metrics** to update citation services and PubMed data.
- **First as listed** is calculated automatically from the active profile name and the resolved author order, including the profile's English and Chinese name forms.
- Use **Co-1st**, **Co-corr.**, and **Rev.** in Paper Details to mark co-first authorship, corresponding authorship, and review/perspective papers. Leave Co-1st unchecked for a solo first-author paper. These manual roles are preserved during metric refresh.
- Use the star to highlight a paper. Star state is independent from citation refresh and export inclusion.
- Drag rows into a manual order when a curated sequence matters more than date or citation sorting.
- Choose **Comments** to build or open a citation-comments report from citing-paper metadata. This is not a personal note field.
- Use **Hide** or **Show** in the context menu to control export inclusion. Use **Open in Finder** or **Delete** for the corresponding local action.

Edit manual records and author marks

A publication without either a DOI or arXiv identifier exposes **Manual Metadata**. Edit Title, Authors, Journal, Date, Volume, Issue, and Pages, then choose **Save Metadata**. Separate author names with semicolons or new lines. Once an identifier is present, use identifier-based metadata controls instead.

Select an author name in the publication's author list to inspect affiliations and set **First author**, **Co-first author**, or **Corresponding author** for that named author. Checkbox changes save immediately. These individual author marks are separate from the publication's Co-1st, Co-corr., and Rev. summary controls.

Build a citation-comments report

Comments gathers citing-paper metadata and available Semantic Scholar citation-context sentences, then writes a local report that groups strong praise, positive impact statements, cautions, and unmatched citing records. It helps you review how later literature describes a publication; it does not modify the publication metadata or replace your own notes.

The report is a generated local reading artifact. Network services may return fewer contexts than citation records, and a temporary rate limit can leave some entries without quoted context. Reopen or rebuild the

report later to retry the available sources; an existing report remains on disk until you deliberately replace or remove it.

Work with files

My Publications uses the same Paper Details and flat Paper Files model as Library. Expand **Files · n** within the details to inspect its resources; Library instead has an adjacent Paper Files column. The publication Files panel shows File type, Annotation, and Source controls directly; the size-footer expansion belongs to Library cards. A publication can keep a default PDF, earlier versions, supplements, notes, presentations, data, and other attachments under the same Gewu Key.

Read the summary

The lower summary includes publication counts, identifier coverage, h-index and i10 information, journal distribution, merged citation maxima, and major-paper role counts. Select a role statistic to filter the table where supported.

Export a list

Choose **Export List**, then select LaTeX PDF, LaTeX source, Overleaf project ZIP, or Word document. Papers excluded from export remain visible in My Publications but are omitted from the generated list. The export follows the current visible table, including its search, role/journal filters, and row order, and excludes rows marked Hide. Clear filters and search first if you want the full included list. Exporting never replaces the SQLite source.

The Overleaf option creates a project ZIP, copies it to the Desktop, and opens Overleaf using Website Links. The status message reports the saved path. It does not upload your files or create an Overleaf project automatically; import the ZIP yourself on the website.

See where a reordered row will land

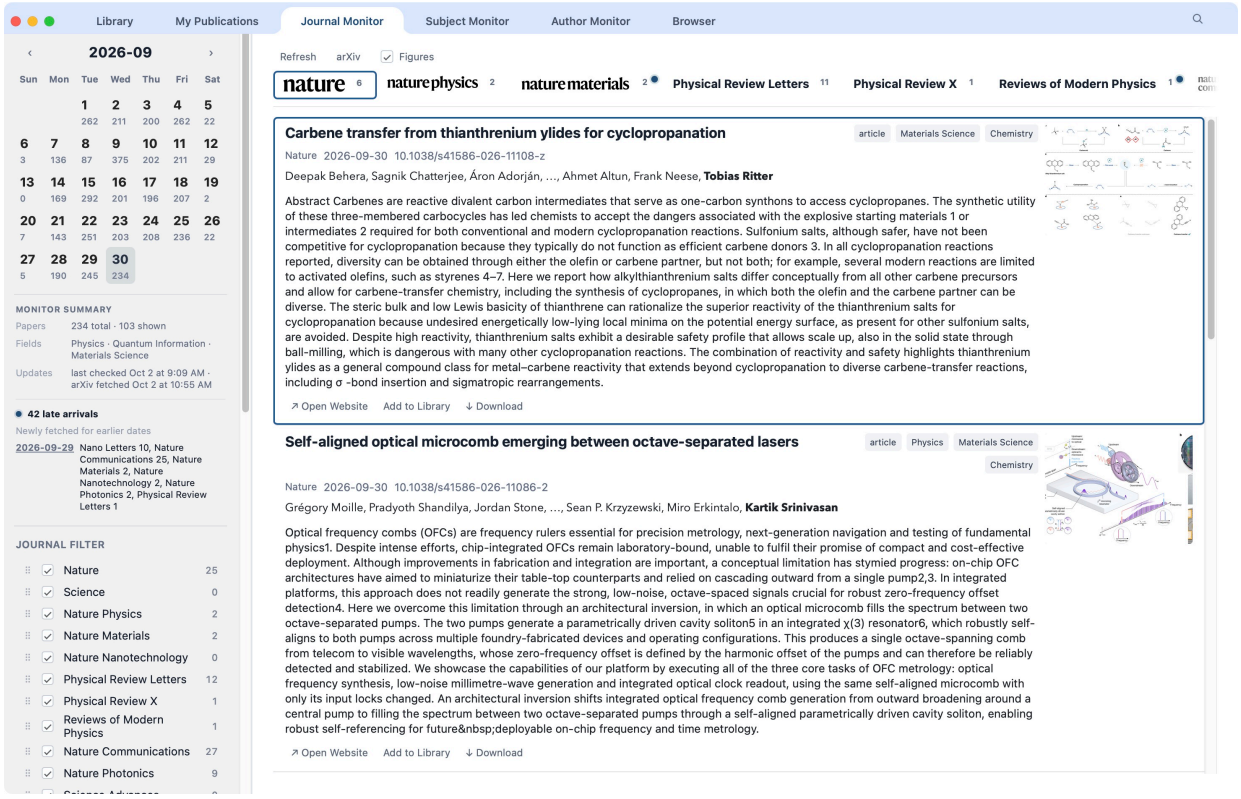
When dragging a publication to change its manual order, a translucent copy follows the pointer. A blue circle and horizontal line mark the insertion position. The other rows stay in place during the drag; release to apply the order. This changes the publication list order, not the paper's file location.

See also: Literature Data Source · Use Paper Details · Local Data and Privacy

3-5 Journal Monitor

3-5.1 Journal Monitor

Journal Monitor collects papers for a selected publication date and lets you narrow the result by journal, preprint source, and field.



Journal Monitor. Nature is selected for September 30, 2026, with scientific figures visible beside the paper cards. Sunday starts the calendar and Saturday ends it. The independent sidebar retains stored counts, Monitor Summary, late arrivals and filters. The top Nature tab shows six visible papers; Journal Filter retains the full count of 25.

Refresh a date

1. **Select a day in the calendar.** Small numbers below a day show stored result counts.
2. **Choose Refresh.** The app requests all configured journal sources for that date.
3. **Control-click the journal in Journal Filter and use Refresh This Journal when only that journal needs another attempt.** This avoids re-requesting journals that are already complete.
4. **Choose arXiv separately if needed.** This keeps journal and preprint work explicit.
5. **Watch the bottom status area.** It reports the current source and completion without changing the page height.

Manage monitored journals

Open **Journal Settings** to add or manage a journal by name, ISSN, or known publication history. When an ISSN is missing, the app can look up a matching journal record before saving it. Review the match rather than relying on a similar title alone.

Drag journals vertically in **Journal Filter** or **Journal Settings** to set their order. The saved order controls the filter list and the journal groups in the report. Removing a journal disables the subscription without erasing its previously collected papers or its saved history.

Your enabled list, disabled history, ISSNs, source identifier, section policy, logo source, and display order are saved in SQLite. Product updates can add better built-in source definitions, but those defaults do not replace this personal state. Adding a previously removed journal restores it at the end of the active order.

Understand counts and filters

The count beside a journal in the left **Journal Filter** is its stored total for the selected day, independent of **Field Filter**. The journal tabs above the cards count the currently visible papers after filtering. **Monitor Summary** distinguishes total from shown. A left-side journal count can therefore be nonzero while the current fields hide all its cards. Counts describe results; blue dots describe unread state.

Calendar counts summarize stored day results. A zero count is meaningful only after the relevant source completed successfully. Read the source status before interpreting zero as "the journal published nothing."

Blue dots follow the papers you select

Late-arriving discoveries can have an unread blue dot. Click an individual paper or use Up/Down to select it with the keyboard to mark that paper read. Opening a date, journal or subject, changing fields, refreshing, or automatically highlighting the first card does not mark the whole list read.

A journal or subject tab's blue dot counts only unread papers currently visible under its filters. It disappears after all those visible papers are read. Hidden papers keep their own unread state; if you change fields or another filter to reveal them, the tab's dot can return. Read state is shared for the same paper between Journal and Subject Monitor. Author Monitor retains its separate New and Read All workflow.

The calendar runs **Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday**. Small numbers show stored results, not unread totals. The late-arrivals section links to earlier publication dates whose results arrived later; it is separate from publication notices.

Daily scheduled updates

Journal Monitor is designed to keep a daily cached record through its scheduled update workflow. A scheduled run and a manual refresh use the same stored day results, so opening a date does not require downloading it again. Use a manual refresh when a scheduled source was missed, partial, or needs verification.

The scheduler belongs to the running local UI service; it does not wake a sleeping Mac or run while that service is stopped. See When automatic updates run for the current timing and restart boundaries.

Journal logos and mathematical text

Configured journal logos are downloaded into a local cache and reused from disk. The first successful request may require network access; changing dates and later launches should not repeatedly download the same logo. The cache is presentation data and can be rebuilt without changing subscriptions or papers.

Supported inline LaTeX in titles and abstracts is typeset after the math renderer becomes ready. On a cold launch, raw source may be visible briefly, then the current day is rendered again automatically; changing to another date and back should not be necessary. Title normalization also preserves recognized proper names and source capitalization such as *Su–Schrieffer–Heeger* and *Monte Carlo* while still cleaning all-uppercase metadata and formula spacing.

How discovery provenance is preserved

Each stored Journal result is related to the canonical paper, the fetch run, and the source that reported it. The run preserves the journal, requested date window, status, and timing. Seeing the same paper from Crossref and OpenAlex records both source observations against the same paper identity; seeing it again within one run does not create a duplicate relationship. Later runs remain separately traceable.

The same canonical paper can also acquire Subject Monitor and Author Monitor relationships. These discovery paths explain why the work is visible; they do not automatically add a Library placement or create a local Paper Files resource.

Filter and inspect

Use **Journal Filter**, **Preprint Filter**, and **Field Filter**. Journal and field choices combine to determine the visible cards, while the left Journal Filter counts continue to describe the full selected day. Enable **Figures** to show a representative image when available. Selecting a result opens its report and actions.

Act on a result

- **Open Website** opens the publisher or best known record using Website Links: a new built-in Browser tab by default, or the macOS default browser when selected.
- **Add to Library** begins the placement workflow described below; after that, **Show in Library** returns to the managed paper.
- **Download** retrieves the paper file. **PDF** opens an existing local PDF in Preview, and **Delete PDF** removes that managed download and resets its status.
- When a reading note exists, **Note** opens it in the right-side drawer. The list compresses but keeps its own scroll position. **Delete Note** removes the note and resets analysis status.

Read an existing Note

Note appears when a reading note has already been generated. It opens the saved note on demand in a right-side drawer; the report area compresses and each side scrolls independently. Drag the divider to resize the note. **Hide list** gives the note more space; **Show list** restores the report. Closing the drawer keeps the note. **Delete Note** is a separate action that removes it and resets its analysis status. Links in the note follow Website Links.

Add a Journal result to Library

1. **Choose Add to Library on the result.** The Library destination chooser opens without leaving the selected date.
2. **Choose the target Library folder.** This is the visible folder placement for the managed paper.
3. **Complete the add.** The card changes to **Show in Library**; use it to reveal the existing placement later.

The operation links the Journal discovery to the canonical paper, preserves its source and run provenance, and adds the chosen Library placement. It does not automatically download a PDF. Choose **Download** separately when you also want a local file. If the paper is already managed, **Show in Library** reveals it instead of creating a duplicate identity.

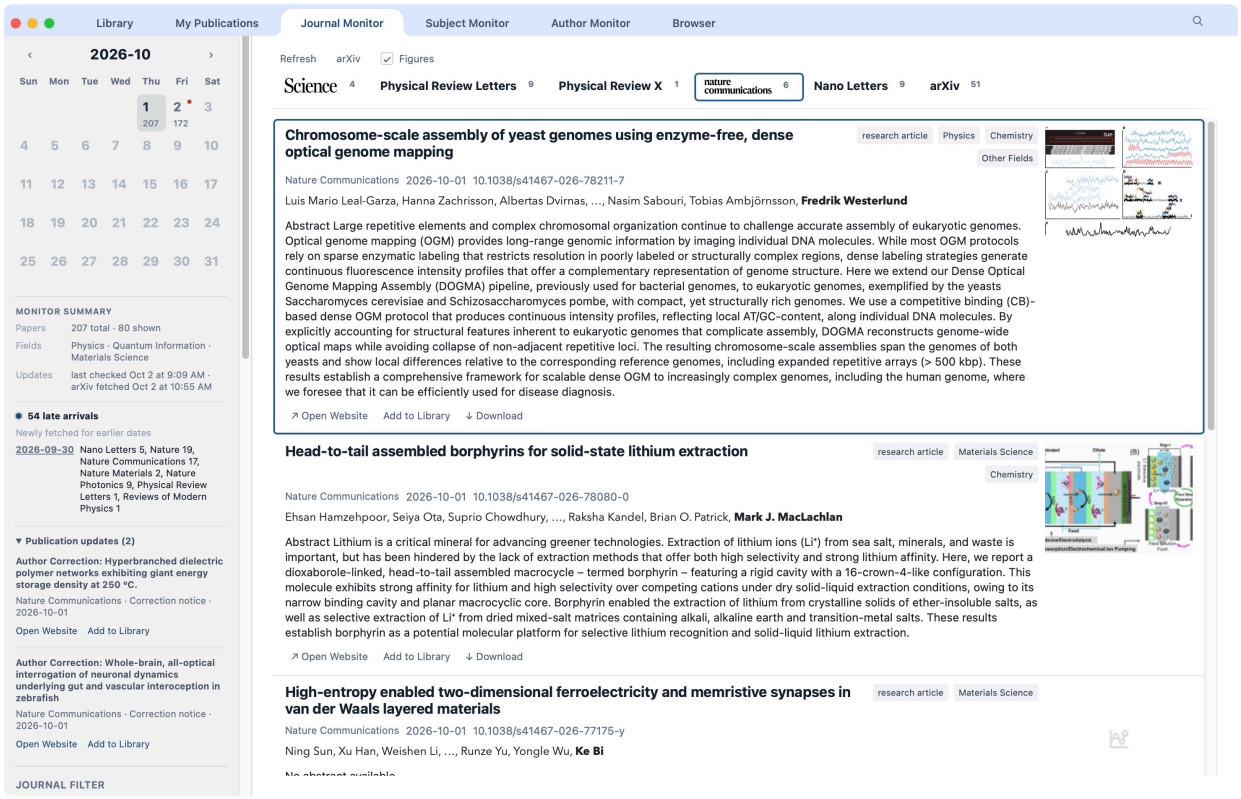
Per-paper status text and the bottom progress area make completed downloads and background work visible, which helps avoid repeating the same action.

> **Note**
> "No matching papers" and "source fetch failed" are different outcomes. Check the source status before concluding that a date is empty.

The destination chooser initially selects your Setting > Add to Library default, initially **Library / unsorted**. You can confirm another active Library or Space folder for this addition. The choice does not change the global default.

Journal tabs identify the journal by a supported wordmark or by text. APS titles, including Physical Review Letters, Physical Review X and Reviews of Modern Physics, use readable journal names rather than a publisher icon or a page image.

Review corrections and retractions



Publication updates. On October 1, two notices match the current Physics, Quantum Information and Materials Science selection. The expanded section sits below Monitor Summary and late arrivals in the left sidebar; its Open Website and Add to Library actions belong to each notice. Selecting the Nature Communications tab changes the main report, while notice visibility follows the left Journal and Field filters.

Publication updates is a collapsed section in the left sidebar below **Monitor Summary** and any late-arrivals section. Expand it to see notices for the selected publication date. Both its count and list follow **Journal Filter** and **Field Filter**. A linked notice uses the original article's fields; an unresolved notice uses its retained classification. With no matching notices, the section is hidden.

1. Choose **Open Website** to inspect the notice and its original-article relation.
2. Choose **Add to Library** on the notice to associate it with the original work. An existing original retains its Gewu Key. If absent, Gewu can retrieve the explicitly identified original's metadata and create its paper record without a PDF.
3. Read the result below the action, then inspect the original's Paper Files. The notice keeps its DOI, title, date and kind; it does not receive an independent research-paper Key. This action does not automatically download its PDF.

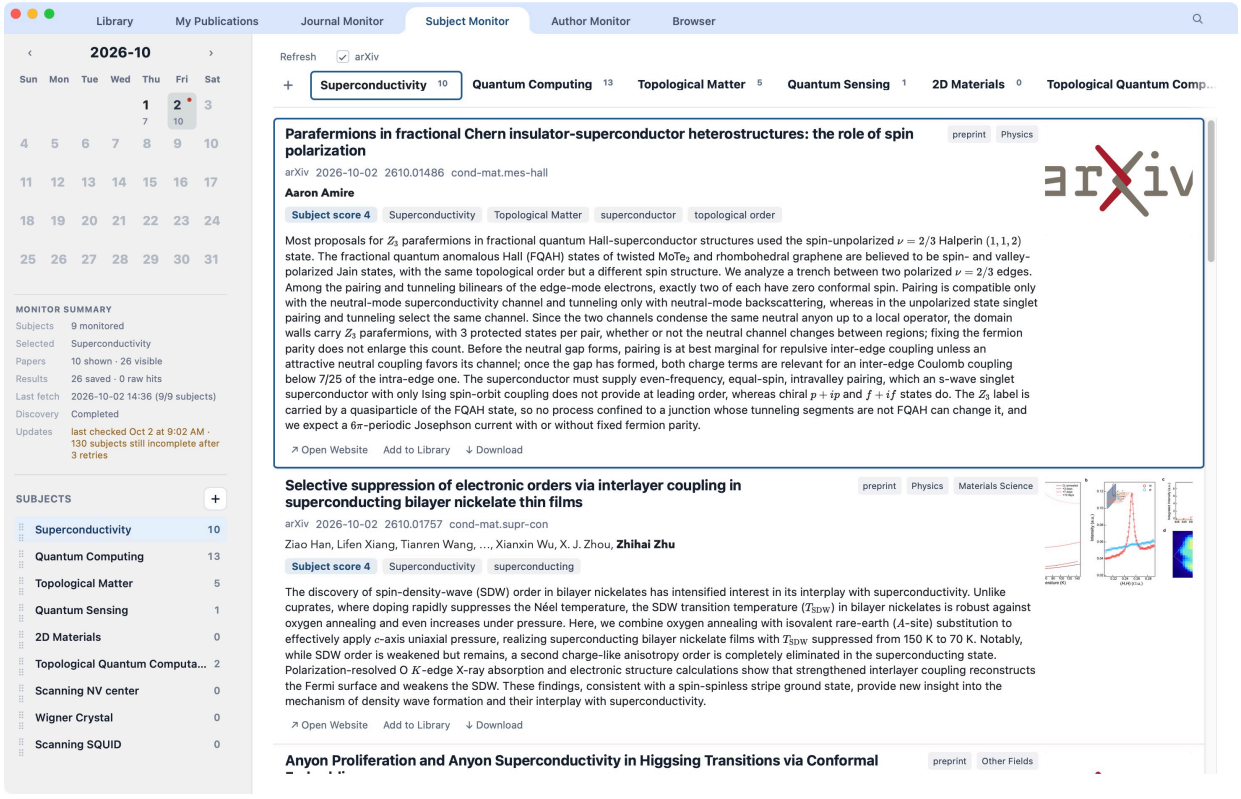
Missing or conflicting relations stop the association for review; a similar title alone is insufficient. The notice does not replace the original bibliography or Default. Encountering a notice during discovery does not stop processing the journal's later research articles. To check an older saved paper directly, use Refresh Info.

See also: Subject Monitor · Publisher and Network Access · Add Papers and Files

3-6 Subject Monitor

3-6.1 Subject Monitor

Subject Monitor discovers papers for reusable topics. Each subject stores a name and literal discovery terms; an optional copied prompt can help you draft those terms outside the app.

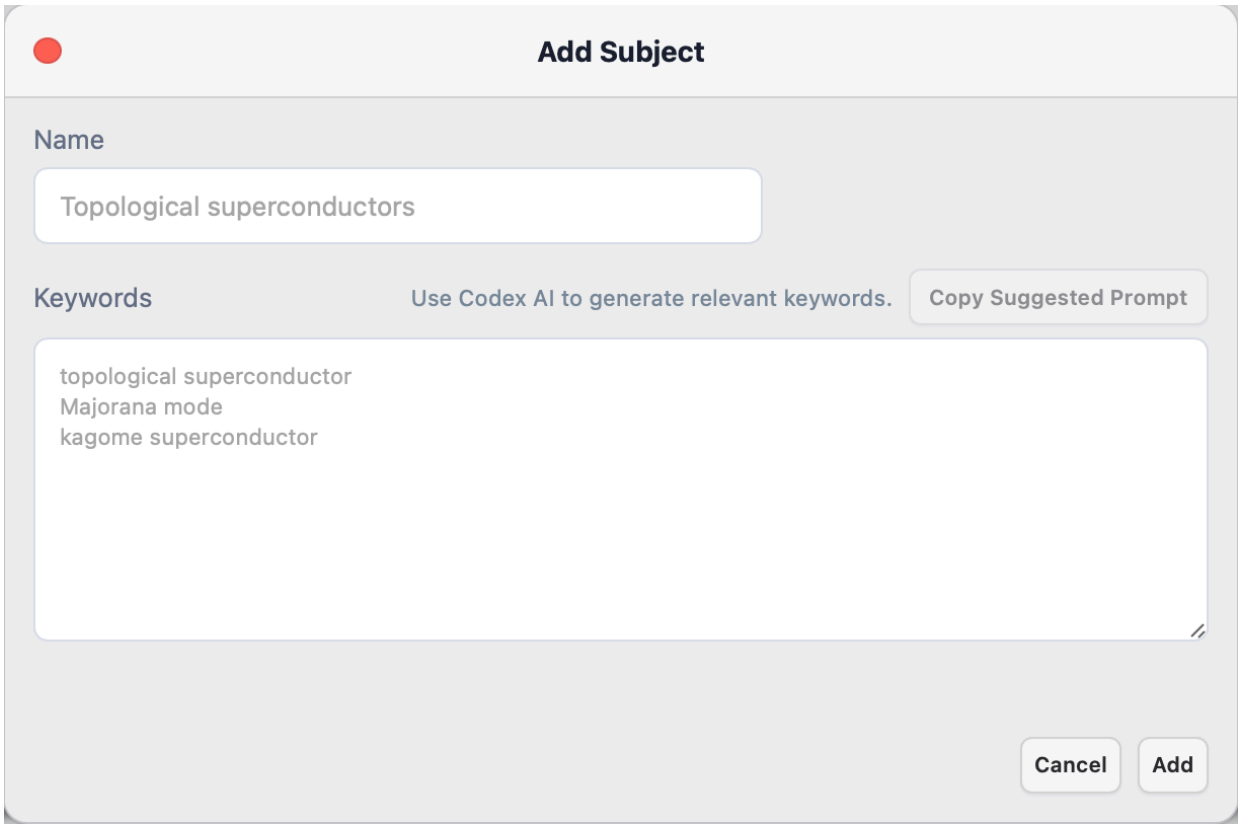


Subject Monitor. Superconductivity is selected on October 2, 2026. The Sunday-first calendar shows numeric counts for that subject; future or unstored dates remain blank. The current date has completed results, while Updates can separately report outstanding retries from other stored windows.

Create a subject

1. **Choose Add Subject.**
2. **Enter a clear name and discovery keywords.** Describe the field broadly enough to find variants, but avoid unrelated general terms.
3. **Use Copy Suggested Prompt if you want Codex-assisted wording.** Review the prompt before using it.
4. **Save and arrange the subject.** Subjects can be edited, deleted, and reordered.

Keywords use case-insensitive literal matching. Enter the words or phrases you expect to appear; they are not regular expressions and are not silently expanded into unrelated concepts.



Add Subject. Enter a name and one keyword or phrase per line. The same form is used when editing an existing subject; Cancel leaves its saved keywords unchanged.

Edit keywords without losing history

Control-click a subject and choose **Edit**. Saving a new name or keywords keeps its existing paper list, calendar counts, selection, and reading position. New keywords apply to future retrieval. Existing date windows, including empty and incomplete results, remain preserved after a keyword change; background retries and later local arXiv matching do not silently reclassify those windows.

Choose **Refresh This Subject** explicitly to rebuild that subject's selected date with the current keywords. A successful refresh replaces only that result window; other dates and subjects remain unchanged. An incomplete refresh preserves previous results. Editing during an active retrieval discards the obsolete result instead of replacing history. Deleting a subject still clears its Subject Monitor matches, without deleting the underlying shared papers or Library files. Previously cleared matches from an older software version are not automatically reconstructed by this fix.

Keep one subject selected across dates

Selecting a subject makes it the active topic until you choose another one. Moving to a date with no result does not switch the tab back to the first subject, and a subject with zero papers remains selectable. The calendar's day numbers describe the active subject, so you can see which dates contain papers for that topic rather than a total across every subject. The count beside each subject tab describes that subject on the selected date.

The calendar shows the stored count as a number, including **0**, for the selected subject and date. Counts have no completeness marker. A date with no stored results and no fetch record remains blank. While a fetch is running, the progress indicator remains visible.

The **Discovery** row describes the recorded retrieval status for the selected subject. A completed retrieval is not a guarantee that every relevant paper has been found. Switching subjects changes the counts and recorded status; reopening the app preserves the status stored in SQLite.

Refresh subjects

1. **Select a calendar date.** Selecting a date only loads the result already stored for that day; it does not start a network request.
2. **Choose Refresh.** The command refreshes every configured subject for the currently selected date. The arXiv display option is enabled by default; clear it when you want to hide arXiv cards without changing what is stored.
3. **Read source status.** A source can be complete, partial, empty, or failed. Source-level retries are managed by the background scheduler.
4. **Cancel if necessary.** Completed results remain distinguishable from unfinished work.

Refresh one subject

Control-click a subject and choose **Refresh This Subject** when only that topic needs another pass. Use the main **Refresh** command when every configured subject should run for the selected date.

The current candidate layer can draw from local arXiv data, local APS RSS data, OpenAlex, Crossref, Semantic Scholar, and PubMed. It does not depend on a fixed number of feeds. Results still pass topic filtering and deduplication before display.

Inspect and keep a result

Select a paper to read its title, authors, abstract, source, date, identifier, topic score, and matched subject terms. Use **Open Website**, **Download**, **PDF**, and the note controls in the same way as Journal Monitor. Choose **Add to Library** to create a managed reference, or **Show in Library** when one already exists.

Add a Subject result to Library

1. **Choose Add to Library on the matching paper.** The subject and selected date remain in place while the destination chooser opens.
2. **Select the target Library folder.** The same canonical paper can be placed in a durable topic hierarchy without losing the Subject relationship.
3. **Complete the add.** Use **Show in Library** on later visits to reveal the managed placement.

Adding the result preserves the matched subject, score, terms, and discovery provenance. The paper remains visible in Subject Monitor immediately after adding it; its action becomes **Show in Library**, while the selected subject, date, and reading position stay in place. It does not automatically download a PDF; use **Download** separately. Repeated discovery of the same identity updates relationships rather than creating unrelated paper records.

> **Important**
> Subject Monitor rejects Zenodo records during ingestion and when cached results are displayed. There is no Zenodo show/hide option.

The destination chooser initially selects your Setting > Add to Library default, initially **Library / unsorted**. You can confirm another active Library or Space folder for this addition. The choice does not change the global default.

Read state, calendar and publication notices

The calendar runs Sunday first through Saturday last. Its counts remain plain numbers for the selected subject. Selecting an individual paper by mouse or Up/Down acknowledges that paper only; opening the topic or date does not acknowledge all its papers. A subject or journal tab's blue dot reflects only its currently visible unread papers. Hidden results remain unread and can restore the dot when shown. See the shared Journal/Subject rules.

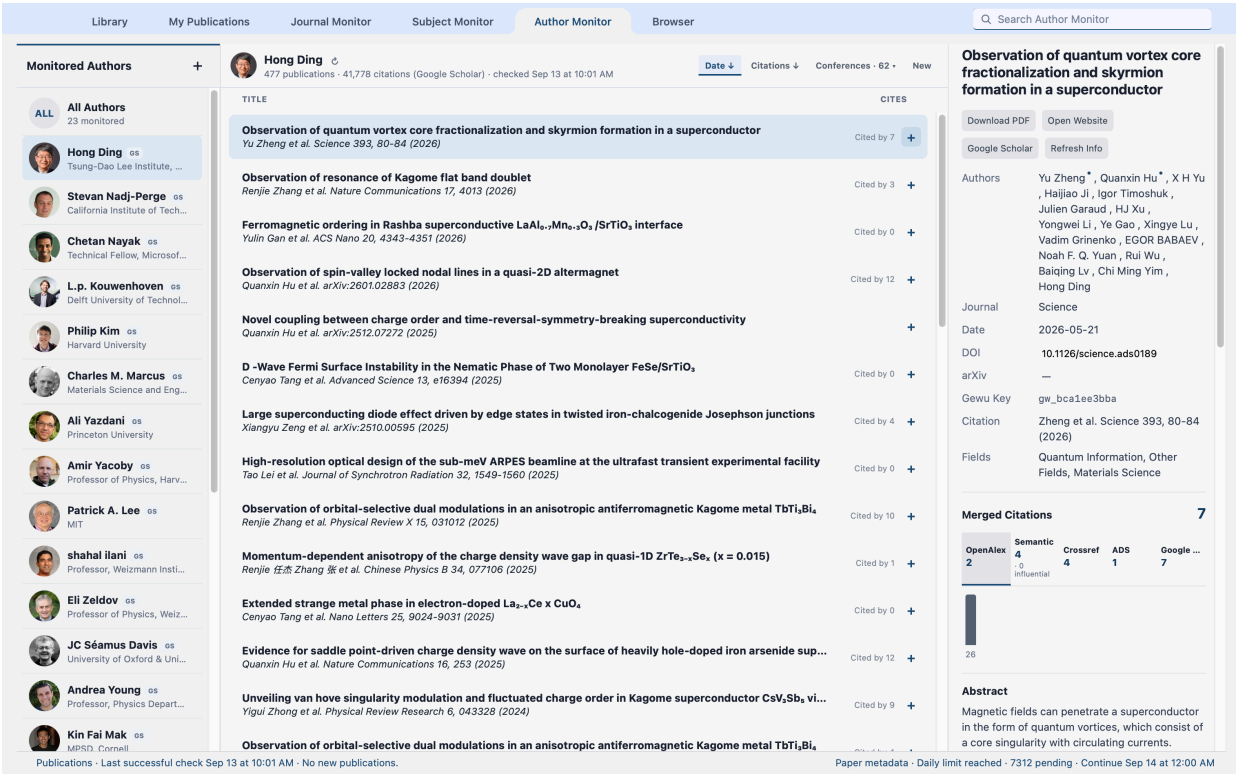
Subject ingestion separates recognized correction and retraction notices from ordinary research-paper candidates and continues processing later results. Notices do not acquire independent research-paper Keys. The expandable **Publication updates** panel described in this guide is in Journal Monitor; for a saved original use **Refresh Info** and Paper Files.

See also: Journal Monitor · Publisher and Network Access

3-7 Author Monitor

3-7.1 Author Monitor

Author Monitor follows researchers and separates a first-import baseline from newly discovered work.



Author Monitor. Hong Ding's profile is selected. The paper title leads the details pane, followed by compact actions and a visible Gewu Key; local portraits keep the author list stable.

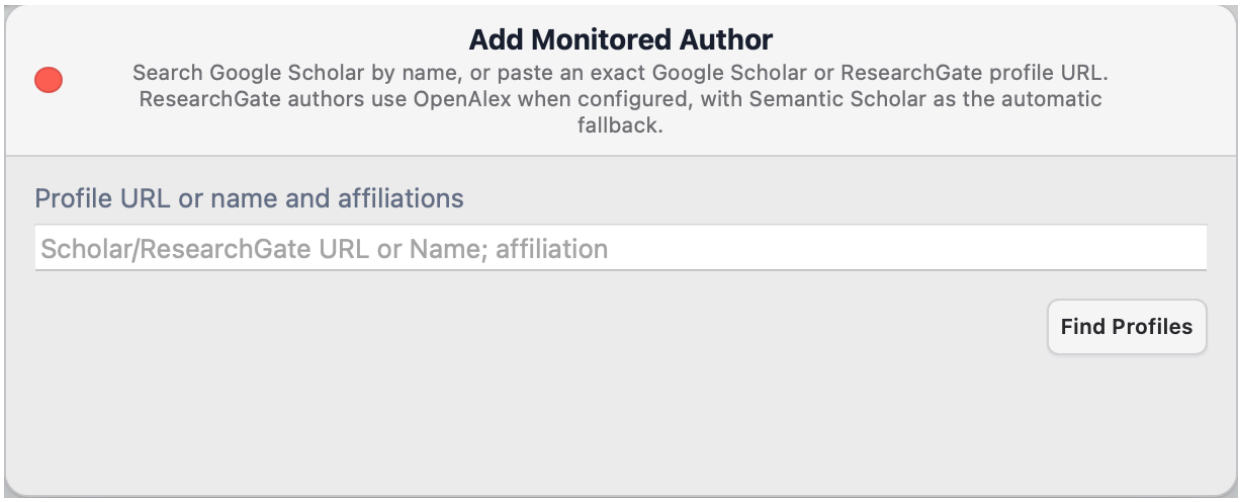
Add an author

1. **Choose the add-author control.**
2. **Enter a name and optional affiliations, or an exact profile URL.** Google Scholar and ResearchGate profile URLs are supported where the profile can be resolved.
3. **Confirm the correct person.** The initial result set becomes the baseline rather than a batch of unread "new" papers.

Understand profile and publication sources

Google Scholar profiles refresh through Scholar with shared pacing and cooldown. A ResearchGate URL identifies the public profile; its publications are matched separately through OpenAlex when configured. If unavailable, Gewu Insight automatically falls back to Semantic Scholar. Confirm the proposed person, especially for similar names.

Without a verified publication-data match, the ResearchGate profile and homepage can be saved, but automatic publication refresh remains unavailable. Profile, publication, and citation sources remain distinct evidence; a fallback does not change the author identity.



Add Monitored Author. Enter an author name or a profile URL. The empty form is ready for a name or profile URL; finding a profile does not add it until you confirm the result.

Reorder monitored authors

Drag an author in the left list to change its order. A translucent preview follows the pointer, and a blue circle with a horizontal line shows where it will be inserted. Other authors stay in place until you release. The order does not change the author's profile source or publication records.

Review updates

Later refreshes can mark newly discovered publications as unread. Use the **New** filter and other author/result controls to focus the list. Select **Date** or **Citations** to sort; selecting the active sort reverses its direction. **Conferences · n** opens conference publications in a separate, independently scrolling popover, so they are not missing from the ordinary article list. Close the popover with its close control or Escape.

Control-click **New** and choose **Read All**. For one selected author, it marks every unread publication as read, including offscreen papers; other authors keep their unread state, even for a shared paper. With **All Authors** selected, it marks all authors' new publications as read. Papers and Library placements are preserved. The New filter remains active; click New again to see read papers. Read state persists after reopening.

Background work and status

Opening this tab reuses available local cards, including older cached lists, while the background checks for changes. It does not start another Scholar fetch. Background list updates retain your selection, reading position, open controls and details.

The bottom bar keeps publication checks separate from **Author metadata · All authors**. **Checking status...** means that these summary counts are still loading; it does not mean all papers are being repaired. Metadata counts cover all monitored authors, not just the selected author, and are recalculated from current missing information even during a daily pause. Ready work, retry waits and papers needing manual lookup are distinct; hover over shortened text for its full message.

Automatic DOI and abstract completion uses other metadata services, independently of Scholar profile checks. It rotates between authors and reuses shared paper identities: up to 20 papers per batch and 80 per local day, yielding between papers after about two minutes and resting at least five minutes between batches. Recent attempts wait seven days before automatic retry, with at most three automatic attempts per paper. Papers still unresolved then need manual lookup; completed papers are not periodically refreshed. Progress reservations and rest times survive restarting; explicit identifier commands remain available separately.

Scholar requests remain serialized, with 4–8 seconds between requests, 12–24 seconds between profile pages and 45–90 seconds between automatic author checks. Every request checks the shared workspace cooldown again before sending, including My Publications imports. Access restrictions pause Scholar for 24 hours; manual actions and restarts do not bypass or reset that pause. This is Gewu's protective waiting period, not a promised unblock time from Google.

A red **Google Scholar paused** message means the shared Scholar access cooldown is active. New restriction messages may identify HTTP 403, HTTP 429 or a verification page; an older generic message may lack that detail. A VPN exit can affect access, but the pause text alone cannot identify the cause. Changing networks or restarting does not clear the saved waiting period. Cached papers and non-Scholar metadata sources remain available.

A refresh reports identity review

If one publication conflicts with an existing identity, the refresh preserves existing papers and files and continues with the remaining publications. The result reports how many need review and identifies the first affected record, Gewu Key and reason. Read the full message before editing: incomplete identity evidence, contradictory metadata and a protected merge are different cases. Compare the title, authors and identifiers; do not delete files or create another copy to silence the warning. A review message can persist after the refresh finishes and is not itself a running repair task.

Recover after a network interruption

A temporary connection failure shows **Google Scholar temporarily unavailable** in the bottom bar. Hover over it for the author, error and retry details. Cached publications remain available. After the connection returns, keep the App running: failed transport requests have a paced automatic retry queue, including before the normal 10:00 daily check. The message clears after a successful retry, not immediately when Wi-Fi reconnects.

An unreadable Scholar publication list is a failed check, not evidence of no publications. Saved papers and the last successful check are retained. Automatic checks stop probing other Scholar authors in that pass, wait an hour, and allow at most three unsuccessful page probes per day before waiting for the next daily check. An access restriction has its separate cooldown. You can use **Refresh This Author** for a deliberate retry when permitted.

Refresh one author or find missing identifiers

Control-click the author and choose **Refresh This Author** to update that person's publications. **Find Missing Identifiers** checks eligible papers for that author. To check only one unidentified paper, Control-click the paper and choose **Find Missing Identifier**; it is disabled when an identifier already exists or a lookup is running. These online lookups differ from **Get DOI**, which uses an available local PDF.

Update citation profiles

For a Google Scholar author, use the circular arrow beside the name or Control-click the author and choose **Update All Scholar Citations**. Review the request estimate, then confirm **Update All Citations**. This rereads only that author's Scholar profile, updates citation counts and discovers new publications. It does not start a Library-wide metadata refresh. Follow the bottom progress status until it finishes.

Manage the profile image

Control-click an author and choose **Edit Homepage** to edit the website, or **Change Photo** / **Add Photo** to choose a local portrait. Downloaded Scholar avatars are saved locally. Opening the author list does not need to reopen SQLite or contact the image host for every row; a manual photo takes precedence. Without a valid local photo, the list can show initials until you refresh or replace the image.

Discovery entries and managed papers

An Author Monitor result is a cataloged shared record, with a Gewu Key shown in its details even before it has Library placements or local files. A missing key displays `(undefined)` until record data is available. Add to Library preserves the existing key and updates the view; downloading or attaching a file creates a resource relationship.

When repeated discoveries identify the same work

Stable Scholar entries, normalized DOI and version-independent arXiv identities are reused across discovery and later metadata updates. When two existing records are safely confirmed as one work, Author Monitor keeps one card for that author. The same paper can still appear under several different authors. Unread state is retained if either repeated entry was unread.

Reconciliation preserves Library and publication relationships, source history, registered files, classification and manual Default. Old Gewu Keys continue to resolve to the surviving paper. Contradictory manual identifiers, competing manual Default choices or insufficient identity evidence keep records separate for review. A shared DOI is not permission to indiscriminately merge every historical duplicate; startup performs no historical merge sweep.

Add an author result to Library

1. **Select the publication and choose Add to Library.** This opens the Library destination chooser without removing the author from view.
2. **Choose the target Library folder.** The selected placement becomes the durable organizational path for the shared paper record.
3. **Complete the add.** The action becomes **Show in Library**, which reveals the managed paper later.

The author-publication relationship and canonical identity remain intact. Adding a placement does not automatically download a PDF, and a paper already managed elsewhere is reused rather than duplicated.

Remove a monitored author

Control-click the author and choose **Remove from Monitor**. Removing an author deletes the monitoring relationship and its monitor state. It does not delete papers already added to Library, their managed PDFs, or their Paper Files bundles.

> **Note**
> Google Scholar requests are paced and protected by a cooldown. Repeated manual refreshes during a block do not improve the result.

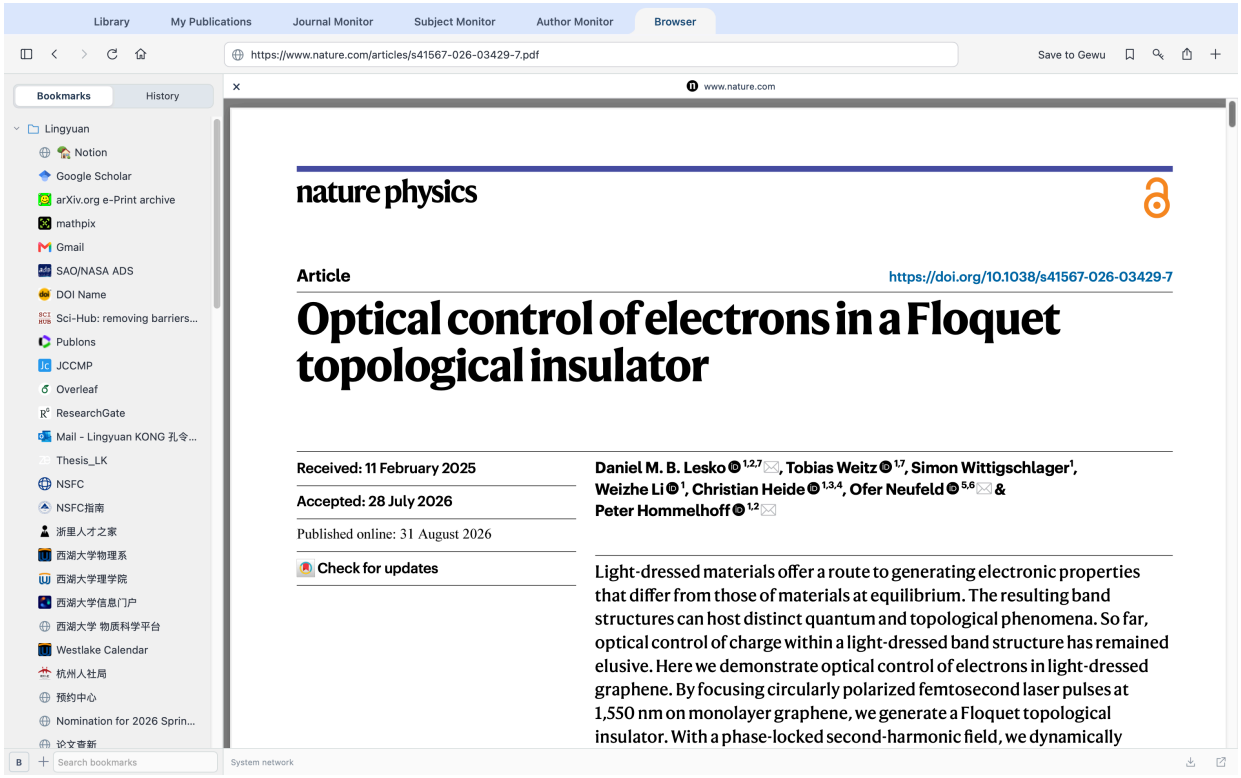
The destination chooser initially selects your Setting > Add to Library default, initially **Library / unsorted**. You can confirm another active Library or Space folder for this addition. The choice does not change the global default.

See also: My Publications · Publisher and Network Access · Local Data and Privacy

3-8 Browser

3-8.1 Browse, Bookmark, and Revisit Websites

Browser is the sixth main tab, after Author Monitor. By default, website links across Library, My Publications and all three Monitors switch here, opening a new webpage tab without replacing existing pages. This includes Open Website and Google Scholar links. Choose Setting > Website Links to use the system default browser instead.



Browser in use. Captured from the original App displaying a fully loaded Nature Physics PDF. The left bookmark sidebar, centered webpage icon and title, left-side close button, and Save to Gewu are visible. This is a live interface capture, not a generated demonstration.

Search and navigate

1. **Open Browser.** On first use, Home displays your local Favorites folder. Later switches return to your selected webpage; use Home to return to Favorites. A paper link opens its destination directly in the unused first tab. No Google webpage is loaded for Home.
2. **Use the address field.** Enter a website address or search terms. Plain search terms go to Google; an address opens directly. Use ⌘L to focus the field.
3. **Navigate normally.** Back, Forward, Reload/Stop, and Home act on the selected tab. Home uses the content area for Favorites; websites, search results, sign-in pages and PDFs use it for browsing.
4. **Switch between workspaces.** Returning from Library or a Monitor preserves the current page and browsing position without reloading merely because the main tab changed.

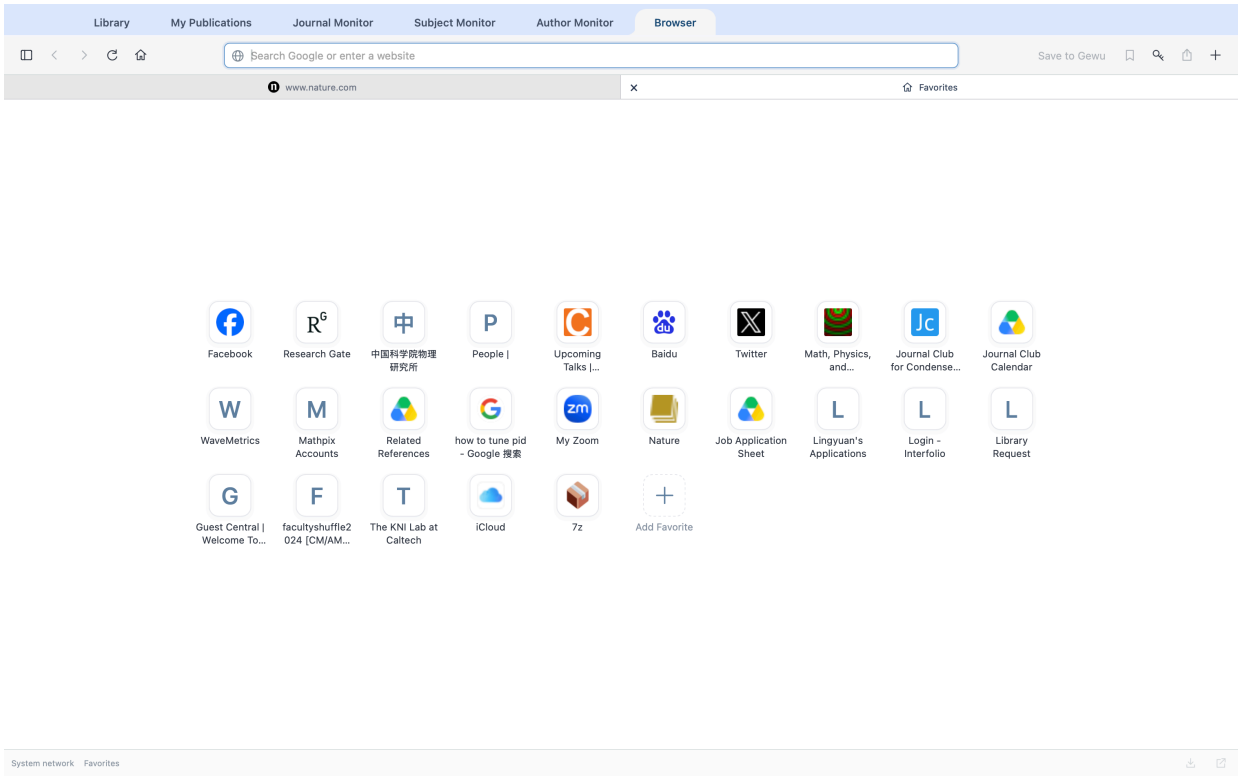
Return to the source workspace: ⌘[. After a website link in Library, My Publications, Journal Monitor, Subject Monitor, or Author Monitor brings you here, press Command-left bracket to return to that workspace's retained selection and scroll position. The physical left-bracket key also works with a Chinese input method selected. It works from the webpage or address field and leaves Browser tabs open; it is not webpage Back. Each linked webpage tab retains its own source for this App session, including related PDF tabs. Switching tabs selects the corresponding return destination. Library links retain the original paper and occurrence; an unrelated address entered in the toolbar clears that tab's association. Entering Browser directly does not create a source. Close any open dialog first.

Local Home, online search. Favorites is available without loading Google, even after restarting the App. Home has no Google logo, search box or account controls; use the toolbar address field to search. Reload on Home refreshes local bookmarks, not a website. Site icons may need a connection; missing icons show initials. Webpages and sign-in remain online and share Browser's existing profile.

When you search or follow a link from Home, Home remains visible while the destination begins loading, then gives way to that destination. It should not flash the page you visited before Home. A cancelled or failed pending navigation leaves Home available rather than presenting that older page as the result.

⌘F opens Find on Page. Enter moves to the next match; Shift-Enter moves to the previous match. Browser is available in the native **Gewu Insight.app**; opening the local UI in a separate web browser does not embed another browser engine.

Favorites on Home. Favorites/Favourites (including localized names) supplies the tiles; use **Create Favorites Folder** if missing. Open nested folders and return through breadcrumbs; use **Add Favorite** or right-click to edit. The grid uses the whole main area, without a Google section or divider. Excess tiles use Previous/Next, not vertical scrolling. Bottom bars align; hiding the sidebar keeps Favorites visible. Opening google.com explicitly shows an ordinary webpage, without Home-specific styling or zoom. Bookmarks are not sent to Google.



Browser Home. The original App displays the user's real Favorites with the bookmark sidebar collapsed. Home and Nature occupy separate tabs with each icon beside its centered title. Home is local; use the address field or a bookmark to open a website.

Use multiple webpage tabs

Below the address toolbar, adjoining rectangular tabs show × at the far left of each tab, with its icon and title centered together, above the webpage column only. The sidebar begins directly below the toolbar; it is not pushed down by the webpage tabs. The top-right + (**New Tab**, ⌘T) opens Home. Switching preserves each page, position and Back/Forward history without reloading or stopping background loads.

Right-click links for **Open Link in New Tab**, or Command-click. New-window links also create tabs, including inline PDFs. Bookmarks/Favorites/History offer **Open in New Tab**. New tabs are selected immediately; ordinary clicks use the current tab.

× or ⌘W closes a tab; closing the last opens Home. Left/Right switches focused tabs. Up to 32 tabs stay in memory, not restored after App quit. Save drafts before closing; bookmarks, history, downloads and sign-ins remain. AutoFill only triggers in the visible page.

Save and organize bookmarks

The sidebar's top tabs switch between **Bookmarks** and **History**, without a repeated heading. The compact **B** (Bookmarks menu), **+** (New Folder), and Search sit along the bottom, even at the minimum sidebar width. History shows Clear History and Search there instead. Drag the sidebar's right edge to resize it or use the toolbar to collapse it. Width, tab, and expanded folders are saved locally.

1. **Add a page.** Use the bookmark toolbar button, **Share > Add Bookmark**, or ⌘D. The native sheet starts with the page title and address.
2. **Choose a location.** Edit the name or address and choose a folder under **Add to**. The previous destination is remembered. Saving the same address in the same folder reuses the existing bookmark.
3. **Create nested folders.** Use **New Folder**, or right-click a folder to create a child folder. The editor can also rename a folder or choose another parent.
4. **Arrange the tree.** Drag an item into the middle of a folder row to move it inside, or to a row edge to reorder it. Drop on the top **Bookmarks** tab to move to the root. A folder cannot move inside itself or its descendants.
5. **Save by dragging.** Drag the website icon beside the address field, or a webpage link, into a bookmark folder. Confirm the name and destination in the same native sheet.

Search filters bookmark names and addresses. Right-click a bookmark for **Open**, **Edit Bookmark**, or **Delete**. Deleting a folder requires confirmation and removes its contained bookmarks, not websites, browsing history, downloaded files, or papers. A bookmark is a saved link, not an offline webpage copy.

Import and export bookmarks

Choose **B > Import** at the bottom of Bookmarks and select a Safari or Chrome bookmark HTML file. Safari versions with **File > Export Bookmarks** produce HTML; newer Safari versions may use **File > Export Browsing Data to File** and produce ZIP. Select bookmarks only when exporting. Gewu accepts a ZIP containing exactly one bookmark HTML file and ignores all other members. In Chrome, use **Bookmark Manager > menu > Export Bookmarks**.

Imports go into **Imported Bookmarks**, retaining folders, order, names, and supported links. Repeating an import merges matching folder names and reuses the same URL in the same folder; existing bookmarks are not overwritten. The result reports added bookmarks, new folders, duplicates, and skipped links. Local addresses, application links, and recognized authorization links are skipped. Limits are 10 MB, 20,000 source entries, and 64 folder levels; invalid files do not partially import.

B > Export saves the full tree as standard bookmark HTML; right-click a folder for **Export Folder**. Safari and Chrome can import this format. Exports contain no history, cookies, or passwords, but their bookmark names and URLs are readable by anyone with the file. Keep personal exports private.

Sync one source folder into Gewu

1. **Choose B > Sync from Safari or Sync from Chrome.** This is a manual, one-way operation, not a background watcher or a write-back to either browser.
2. **Select the source bookmark file.** For Safari, choose `~/Library/Safari/Bookmarks.plist`. For Chrome, choose `Bookmarks` in the intended profile under `~/Library/Application Support/Google/Chrome/`, such as Default or Profile 1. The picker remembers the last chosen location; select the same file for later refreshes. Only this explicitly selected bookmark file is read.
3. **Choose one folder.** The next sheet lists source folder paths and bookmark counts. **Sync Selected Folder** includes that folder and its descendants, not its siblings or the whole profile. Cancel leaves Gewu bookmarks unchanged.
4. **Review the result.** New links and folders are added. Stable source identifiers allow later source title/address changes to update entries created by sync. Pre-existing matches and items edited or moved in Gewu are kept. Source deletions do not delete local bookmarks. The result reports additions, updates, local changes kept, and skipped links.

The selected folder is placed at the Bookmarks root; matching folder names and same-folder URLs are reused. Sync tracking lives in SQLite, independently of papers and website sessions. Legacy source entries without stable identifiers merge by folder/name/URL rather than overwriting local edits. Switching source

files or profiles is a separate source. Limits match import (10 MB, 20,000 entries, 64 levels), with at most 2,000 folders in the chooser. Invalid input or a failed transaction leaves no partial sync.

If macOS denies Safari file access, use a Safari bookmark HTML export with **B > Import**. No privacy settings are changed, and no password, cookie, history, or cloud-account databases are read. Only bookmarks already saved in the chosen local source file are available.

Navigate Browser with arrows

From the top **Browser** workspace tab, press Down to enter the toolbar. Left/Right moves among available toolbar controls; Enter activates one. At the address field, Enter or typing starts editing; its arrows then edit text or browse address suggestions. Escape returns to the toolbar. Up from the toolbar returns to the top Browser tab, where Left/Right changes workspaces. The Library exception is unchanged: Up within Library content does not ascend to its top workspace tab.

Down from the toolbar selects the current webpage tab. Left/Right selects adjacent webpage tabs; Down enters Favorites or the native webpage. Favorites uses directional tile selection and Enter to open. At the left edge, Left enters the visible sidebar on its currently active Bookmarks/History tab; Down traverses its list and footer controls. Right returns to the main area. A hidden sidebar is never selected. In the sidebar header, Left focuses the other collection tab and Enter activates it.

Inside an ordinary webpage, arrows remain WebKit scrolling controls. Once Up reaches the document top, it can hand focus back to webpage tabs; two more Up presses reach the top Browser tab. Left at a document's left boundary can enter the visible sidebar. Inputs, editable text, framed/PDF content, and recognized or uncertain custom widgets keep their own keys; no form values are read or filled. A boundary check runs only after an eligible arrow press, without background polling. Use ⌘L, then Escape, to return to the toolbar from a document that retains its own arrows.

Website icons

Bookmarks, Favorites and webpage tabs use a disposable local icon cache, separate from SQLite. Icons come directly from sites (at most two concurrent requests), without cookies, credentials, whole-article fetching or third-party icon services. Visits may supply a nonstandard icon URL. Missing icons use globes or Home initials; bookmark lists are never sent to an icon service.

Revisit and clear history

History groups loaded main pages by Today, Yesterday, and date, with names, sites, visit times, search, and pagination. Subresources, main-tab switches, and cached Home returns add no visits; Reload updates the existing record. Click to revisit; right-click to bookmark or delete a visit.

Clear History confirms Last hour, Last 24 hours, Last 7 days, or All history. It preserves bookmarks, downloads, cookies, and sign-ins. History remains until deleted; authorization callbacks and recognized signed-download URLs are excluded.

Address suggestions from your history

As you type in the toolbar address field, Gewu searches saved local visit titles and URLs. Address prefixes rank first, followed by matching titles; visit frequency and recency help order the results. A matching address can complete the typed prefix with its remaining text selected. These are local history suggestions, not live Google suggestions, and typing does not send a suggestion request to a search engine.

Use ↑ / ↓, then Enter to open a result, or click it to open directly. Escape dismisses suggestions and restores your typed query. Without a chosen completion, Enter opens an address or searches Google normally. Clearing history also removes those visits from suggestions; bookmark search remains a separate sidebar control.

Read PDFs and save downloads

Viewable PDFs stay inline, including attachment/download links, with article/PDF Back/Forward navigation. In a publisher's HTML viewer, click its PDF download link: Browser receives the PDF using its current website session and opens it in that tab, including supported page-generated files. **Opening PDF…** shows progress. Switching tabs does not interrupt it or let completion take focus; leaving the source page or closing its tab cancels that pending preview. Website sign-in and consent pages remain visible.

Preview uses temporary browser storage, not your Downloads folder or Library. PDF attachment previews accept up to 128 MB and two minutes per transfer. Save to Gewu and the bottom-right **Save Download** arrow reuse a complete loaded PDF; they do not silently download it again. Both the bottom-right arrow and the PDF viewer's own download arrow open a native save dialog, initially in Downloads. Confirm the filename and destination to write the file; Cancel leaves it unsaved. Original-file copies are temporary and bounded; navigating to another document or closing its tab releases them. To import a PDF, use **Save to Gewu** below or import a saved file. Library PDFs still open in macOS Preview.

Choose the right save action

Action	Result
Save Download	Save the browser file to a location such as Downloads. This alone adds no Library placement or Paper Files record.
Save to Gewu	Review the article, optional PDF and destination, then save them to Library. An article can be saved without a PDF and receive or reuse its Gewu Key.
Download on a research result	Run Gewu's managed download workflow for that paper, with supported direct routes and background webpage fallback. **Add to Library** remains a separate placement action.

System networking and Clash rules

WebKit follows macOS networking without changing proxies/Clash. When system proxy or TUN routes traffic to Clash, its rules choose DIRECT/proxy. **System network** is policy, not proof of routing. DIRECT needs a school-recognized exit or institutional login. PDF/login/CDN domains match separately; browser-specific extensions/process rules are not inherited. Verify Clash connections; see Publisher and Network Access.

Save an article and PDF to Library

Save to Gewu opens a compact dialog with **File type**, **Add to Library**, Cancel and Save to Library. Click the double-arrow **Details** to expand article identity and PDF controls. Set its default under **Setting > Website Links**.

1. **Open the article or PDF.** Follow **Open Website** from a research workspace, or find the article directly in Browser. Wait for the page to finish loading and click **Save to Gewu**.
2. **Review identity in Details.** Check the title, DOI and Gewu Key. An existing DOI or known arXiv identity reuses its paper record, including a record discovered only in a Monitor. A new article receives its own Key when saved. If needed, enter a DOI/arXiv ID and click **Identify**. Without an identifier, enter the article title and choose **Use this title**; metadata can be completed later.
3. **Choose the PDF in Details.** For a directly opened PDF or a detected embedded PDF (such as an IEEE full-text viewer), Gewu reads the loaded original and parses its information again. It does not request the PDF from the publisher again. If several loaded PDFs are present, select the intended one first. **Use Loaded PDF** retries this local read. If WebKit cannot supply the complete original, an error is shown; **Download PDF Again** explicitly requests another download. On an article page, select a detected link and click **Get PDF**, or choose a local PDF. Each file must be readable and no larger than 128 MB. Check the filename and leave **Include this PDF in Paper Files** selected to attach it.
4. **Choose the destination.** Click the folder button under **Add to Library**. The same column picker used by Journal and Subject Monitor shows **My Library** and Spaces on the left, then subfolders in successive columns. Check the full path and click **Confirm** to select the destination. Choose the file type, then click **Save to Library** to import. Recognized publisher article PDFs suggest **Published version**, while recognized Science First Release PDFs suggest **Accepted manuscript**. Recognized arXiv, bioRxiv, medRxiv, ChemRxiv and Research Square preprints suggest **Preprint**, regardless of the article's journal metadata. Review the suggestion; uncertain files remain **Unknown**. The first article PDF becomes Default in the main folder, even if Unknown. Supplements stay separate, and manual Default is preserved. Source follows the PDF's actual website: Publisher, Preprint server, Institutional repository or Other. Local files use User import without a webpage source.

Opening Details, identifying or cancelling imports nothing. Save reuses an existing active Library or Space placement, preferring the selected occurrence of that same paper. Attaching a PDF does not add a root-level entry. To add another location, explicitly choose and confirm its folder. Papers with no active placement use the folder saved in Setting > Add to Library, initially **Library / unsorted**, unless you confirm another folder. An unavailable saved default falls back to unsorted. Review creates no placement; the destination is checked again on Save. Paper Files refreshes immediately. Exact duplicates retain their existing source and settings; distinct versions stay separate, and manual Default is preserved. **Show in Library** selects the saved occurrence. Browser keeps the webpage.

Save a correction or retraction under its original article

A normal article PDF that resolves to an existing work can be attached to that work, including one with both DOI and arXiv identifiers or placements in several folders. Check the title and Gewu Key in Details, keep **Include this PDF in Paper Files** selected and save. The selected placement and existing identity are reused.

When Browser recognizes a correction, retraction, expression of concern, withdrawal or addendum, it resolves the explicitly related original article during review. Details shows the original that will receive the file. If that original is not yet cataloged and its DOI can be resolved, Save can create the original's record and attach the notice there. The notice retains its own metadata and file type; it does not receive a separate research-paper Gewu Key or replace the article's automatic Default.

If the original cannot be determined, use an existing original's **Gewu Key** and click **Use Key**. Review the target and confirm the explicit attachment when prompted. **Automatic** returns to automatic identification. Resolve missing or conflicting evidence before saving to avoid attaching a notice to the wrong paper. Opening Details or reviewing a target does not itself save anything. MCP and Journal Monitor retain their own notice-import workflows.

Save a Science ePDF already open in Browser

Science's online ePDF reader displays the PDF inside a webpage. Once the document has loaded, **Save to Gewu** can read that reader's original PDF, including pages whose download control is an icon. This is the article file, not a screenshot or a printout of the webpage.

1. **Wait for the actual Science PDF to load.** An abstract page or access screen is not a loaded article PDF.
2. **Open Save to Gewu > Details.** Check the article title, DOI, Gewu Key and PDF status. If needed, choose **Use Loaded PDF**. Each original must be complete and no larger than 128 MB.
3. **Check File type and the destination, then save.** A byte-identical file already attached to that paper reports **Existing PDF reused**; it keeps the existing resource and your manual Default.

Keep the source document open during review. If it changes, reopen Save to Gewu for the new document. After Save or Cancel, the Science reader resumes displaying its pages. If the original cannot be read, wait for loading to finish and retry, or choose a PDF you have saved locally. **Download PDF Again** requests a fresh transfer and can still fail even when the browser has a readable copy. This workflow shares your existing article access; it cannot supply a missing subscription.

Use the original paper or correct the target

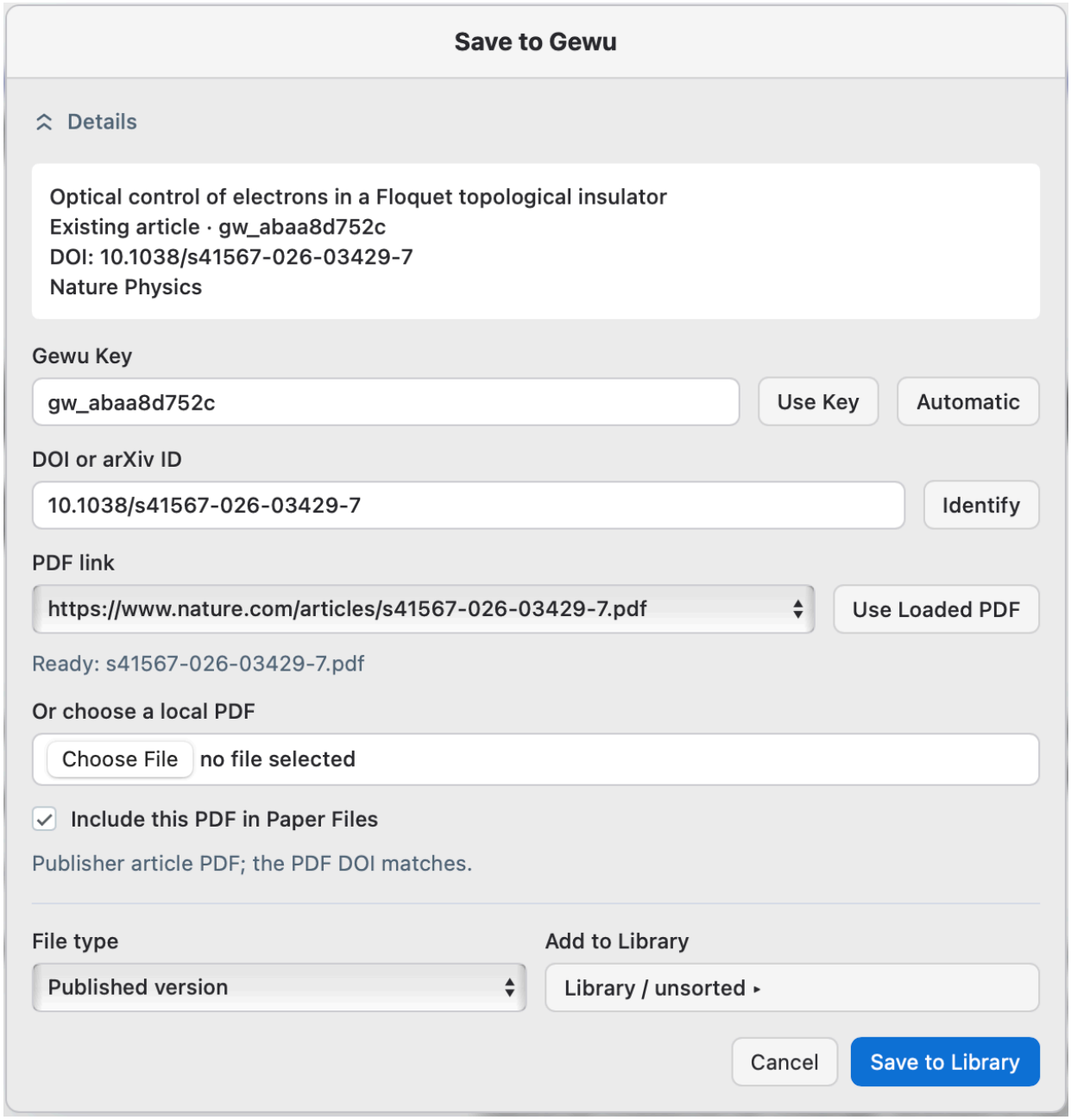
A tab opened with Open Website or the arXiv arrow remembers its source paper and Library occurrence, including related PDF tabs. Save to Gewu uses that existing paper when the current evidence agrees or provides no conflicting identifier. Manually corrected DOI/arXiv values take precedence over old automatic matches. If the page identifies another work, the source is not forced onto it.

1. **Enter the correct existing Gewu Key in Details.** Press Return or choose **Use Key**.
2. **Wait for Target updated.** Check the displayed title, Key, existing-article status and **Add to Library** path. The review switches to that paper's active location when available. An unknown Key produces an error and creates nothing.
3. **Review any conflicting evidence.** If the page or PDF identifies another work, explicitly confirm that this file belongs to the selected target before saving. Changing the identifier afterward requires another review.
4. **Check the PDF and File type, then save.** Selecting a target attaches the file without replacing that paper's DOI, metadata or manual Default. Choose **Automatic** to return to automatic identity detection.

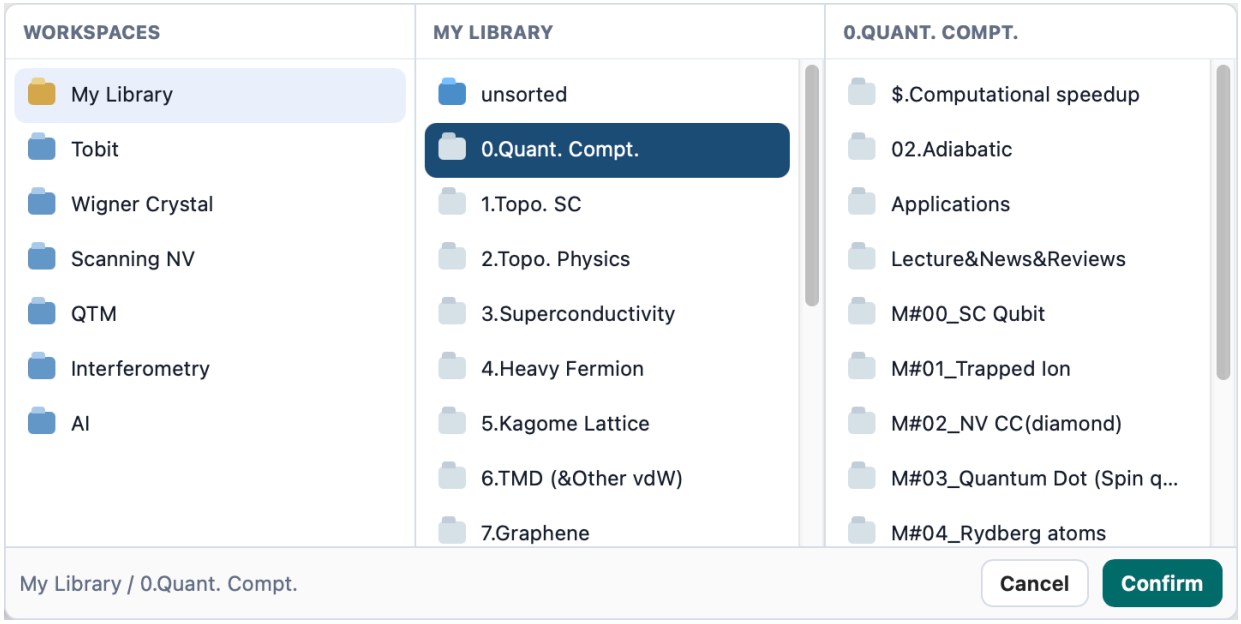
What Article only means. No PDF is currently included in the review. Article identification and PDF attachment are separate: identifying a DOI can find the existing paper without attaching a file. Expand Details and inspect the PDF controls. A loaded complete PDF can be reused through **Use Loaded PDF**; otherwise open the site's actual PDF or choose a local file. This status alone does not mean the publisher has no PDF.

If PDF retrieval fails: retry after opening the site's PDF, choose a downloaded local PDF, or save the article alone. Errors stay visible; missing or conflicting identity opens Details. If storage fails after saving the article, retry reuses its record and placement. Browser capture uses its current WebKit session without exporting sign-in cookies. It cannot bypass a site's login, access rights or embedded-browser restrictions. The collector currently belongs to Gewu's built-in Browser; a Safari extension is not included.

Save to Gewu: compact review. After the article and PDF have been identified, the compact panel keeps File type, Add to Library and the final save action visible.



Save to Gewu: Details. Expanded Details identifies this existing Nature Physics article by its Gewu Key and DOI. Use Key and Automatic control the target; Use Loaded PDF reuses the original file. The PDF is Ready and classified as Published version before saving.



Choose a Library destination. Read the columns from workspace to folder to subfolder. The bottom path follows the selection. Confirm selects a destination; Save to Library performs the import.

Privacy and compatibility boundaries

Bookmarks/history use local SQLite; cookies/cache use Browser's persistent profile, not Safari synchronization. Browsing/search reaches websites, but external pages have no Library/credential bridge. Local/file/application addresses are blocked. Standard internal child documents use WebKit's origin and sandbox rules; they cannot become address-bar destinations. HTTP recovery retries once as same-host HTTPS GET, without replaying forms or bypassing certificates; errors include reason/code. The local server rejects cross-site API requests and unrecognized hosts.

Use macOS Passwords

Open Passwords opens Apple's app (or use Passwords in System Settings on older Macs). Unlock and choose credentials there yourself. Gewu has no password vault and never imports, exports, lists or relays credentials.

Conservative AutoFill trigger. A single, unmodified click on an explicitly marked HTTPS login username or current-password input can open **AutoFill > Passwords**. Search, ordinary email, registration, codes, embedded frames and uncertain fields are skipped, as are keyboard focus, double-clicks and dragging. Each field kind is offered once per site per App session, including cancellation; right-click AutoFill retries manually. Navigation or intervening input cancels a pending trigger. macOS handles authentication, selection and filling; Gewu never reads credentials or submits login. Site matching, Safari-style suggestions and successful filling are not guaranteed.

Password versus passkey. A Password contains text; a Passkey may show only a creation date and cannot fill a password field. Use the site's passkey sign-in flow or select a genuine Password. See AutoFill troubleshooting; triggering AutoFill does not guarantee successful filling.

Website keys and macOS authorization

The current Browser declines persistent storage and retrieval of webpage cryptographic keys (WebCrypto keys). The same policy applies to background publisher views. It prevents those webpage-key operations from requesting the old **Gewu Insight WebCrypto Master Key** through Keychain. Ordinary webpage cryptography, cookies and local website storage remain enabled; this is separate from Apple Passwords and AutoFill.

A site that depends on saved WebCrypto keys may not work in this Browser, and keys saved by an earlier Gewu version cannot be read through the current Browser. Use the site's supported recovery/sign-in flow or

Open in Safari when needed; Safari has its own session. Updating Gewu does not delete the old keychain item or reset the Browser profile. See Repeated keychain prompts if that older prompt still appears.

> **Website links**
> Research-view links follow Website Links: Gewu Browser by default, or the macOS default browser. Existing Browser navigation stays here; local PDF/Preview and Finder actions stay native. For incompatible sites, choose **Open in Safari**; cookies are not transferred. The browser-hosted UI uses ordinary tabs. Access controls still apply.

Share article access with background downloads

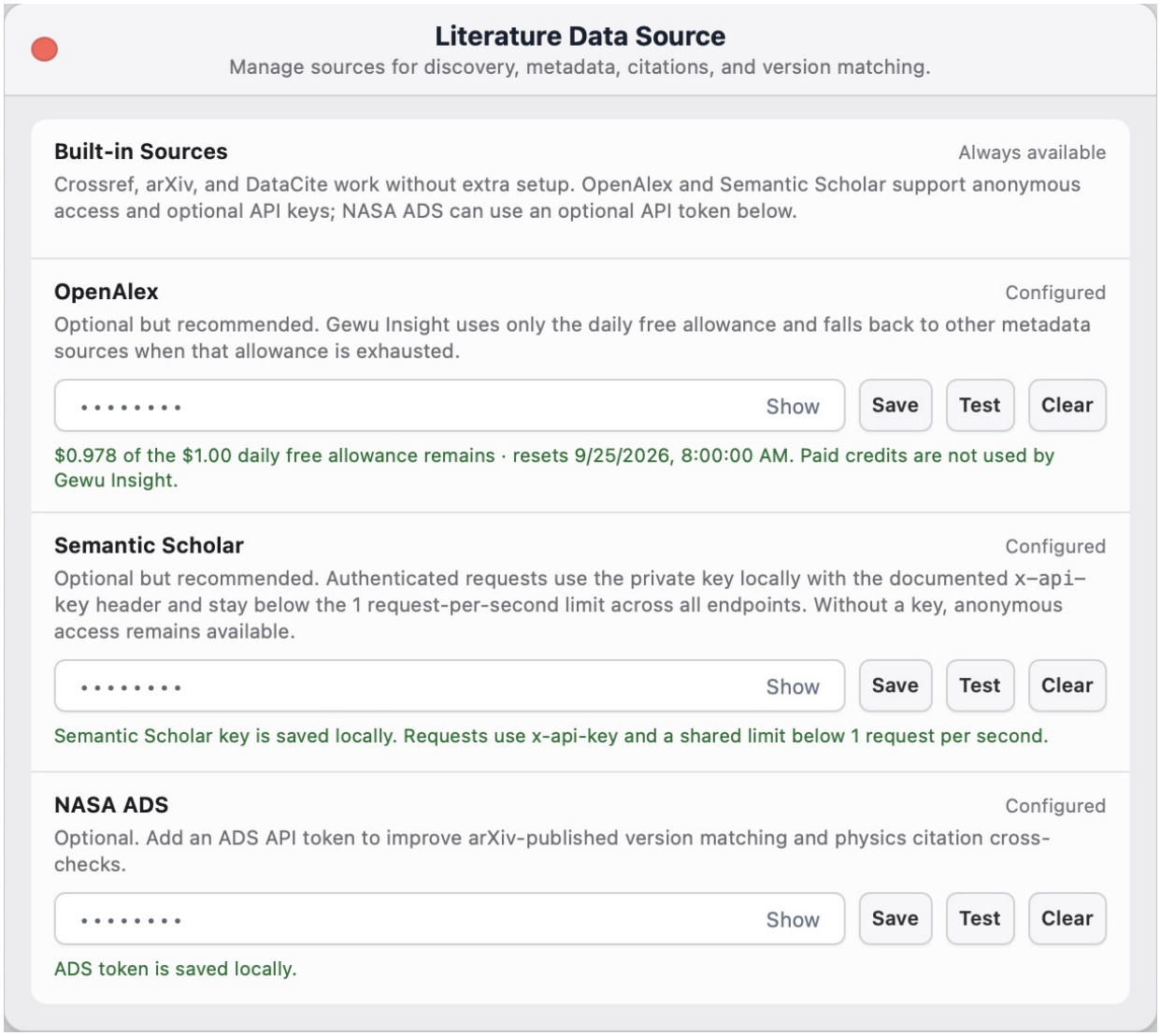
The native background fallback for APS, ACS, Science and Nature uses the Gewu Browser website session. Complete a required website interaction in this Browser, check the actual PDF, and retry the paper's Download action. A login in Safari or another browser does not automatically transfer its cookies here. Background downloads keep the workspace you are using visible. See Download and Sync Versions for supported routes and PDF-only supplement handling.

3-9 Settings and Safety

3-9.1 Literature Data Source

Use **Literature Data Source** to configure services for literature discovery, metadata, citations, and version matching.

Built-in sources work without setup. Optional credentials improve access or coverage for supported services.



Literature Data Source. Configure discovery and metadata services in one panel. Saved credentials remain masked; each service has its own Save, Test and Clear controls.

Open settings

Choose **Setting > Literature Data Source**, or press ⌘,.

Available services

Service	Configuration
Crossref, arXiv, DataCite	Built in; no key field is required.
OpenAlex	Optional OpenAlex API key.

Service	Configuration
Semantic Scholar	Optional API key. Anonymous access remains available.
NASA ADS	Optional API token for astronomy and physics coverage.

Save a credential

1. **Paste the credential into its service field.**
2. **Choose Save beside that service.** Each service has its own Save, Test, and Clear controls.
3. **Choose Test and read the status underneath.** This checks service access; it does not refresh every paper. Then refresh one paper to inspect the available citation evidence.

A configured field shows fixed round dots by default, while the detailed status below continues to report saving, connection, or quota information. The dots do not expose the key or its length. Click **Show** inside the field to reveal the saved key, or inspect a newly typed key; click **Hide** to mask it again. A saved key is retrieved only on Show, and its displayed value is cleared on Hide or when settings closes. Type a replacement and choose **Save** to change the key; **Clear** removes the locally saved key. A configured environment key can remain available. Avoid screenshots or screen sharing while a key is visible.

Semantic Scholar sends its key in the official `x-api-key` header. Requests share a low-rate queue and use exponential backoff after rate limits or temporary failures.

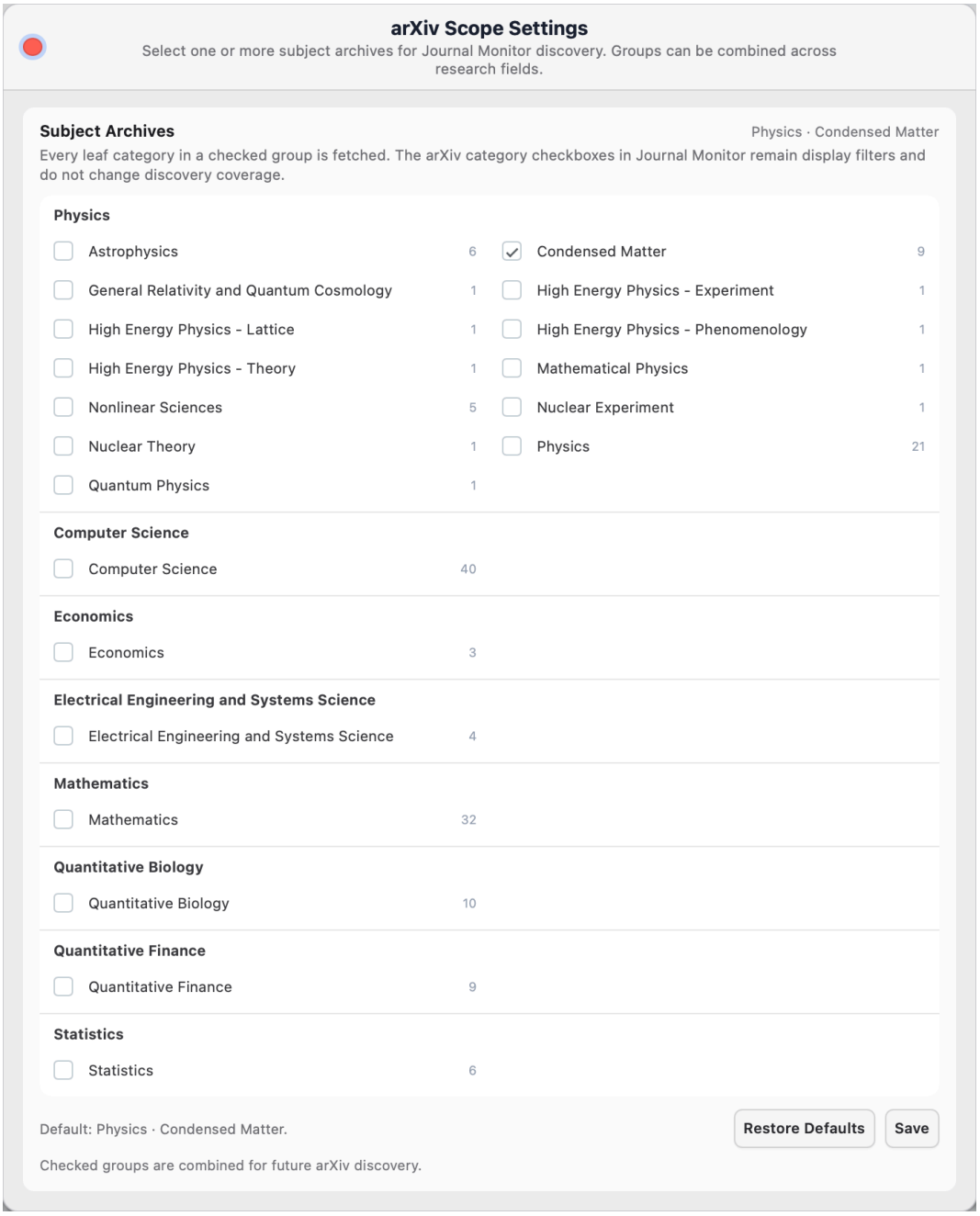
OpenAlex status reports the remaining daily free allowance and its reset time when available. Gewu uses only that free allowance and falls back to other metadata sources when it is exhausted; it does not spend paid OpenAlex credits. A saved key is not proof that every remote request will succeed.

> **Important**
> Credentials are local-only: store them in local SQLite or approved environment variables. Do not place keys in source files, logs, exported reports, tests, or Git.

See also: arXiv Scope Settings · Field Settings · Local Data and Privacy · Publisher and Network Access

3-9.2 arXiv Scope Settings

Choose one or more arXiv subject archives for discovery. Selections can span several research fields or combine several groups within one field.



arXiv Scope Settings. Choose subject archives for future Journal Monitor discovery. Each group shows its category count; the displayed Condensed Matter selection is an example of a saved preference.

Open arXiv Scope Settings

Choose **Setting > arXiv Scope Settings**. The window lists every supported primary subject and its archive groups. Select at least one group, then choose **Save**.

Understand the hierarchy

The first level identifies a broad arXiv subject such as Physics, Computer Science, Mathematics, Economics, Electrical Engineering and Systems Science, Quantitative Biology, Quantitative Finance, or Statistics. Each checkbox represents an archive group beneath that subject. For example, **Physics > Condensed Matter** includes all `cond-mat.*` leaf categories.

You may select several groups under Physics, or combine groups across primary subjects. Categories shared by more than one scope are fetched only once.

Default and saved behavior

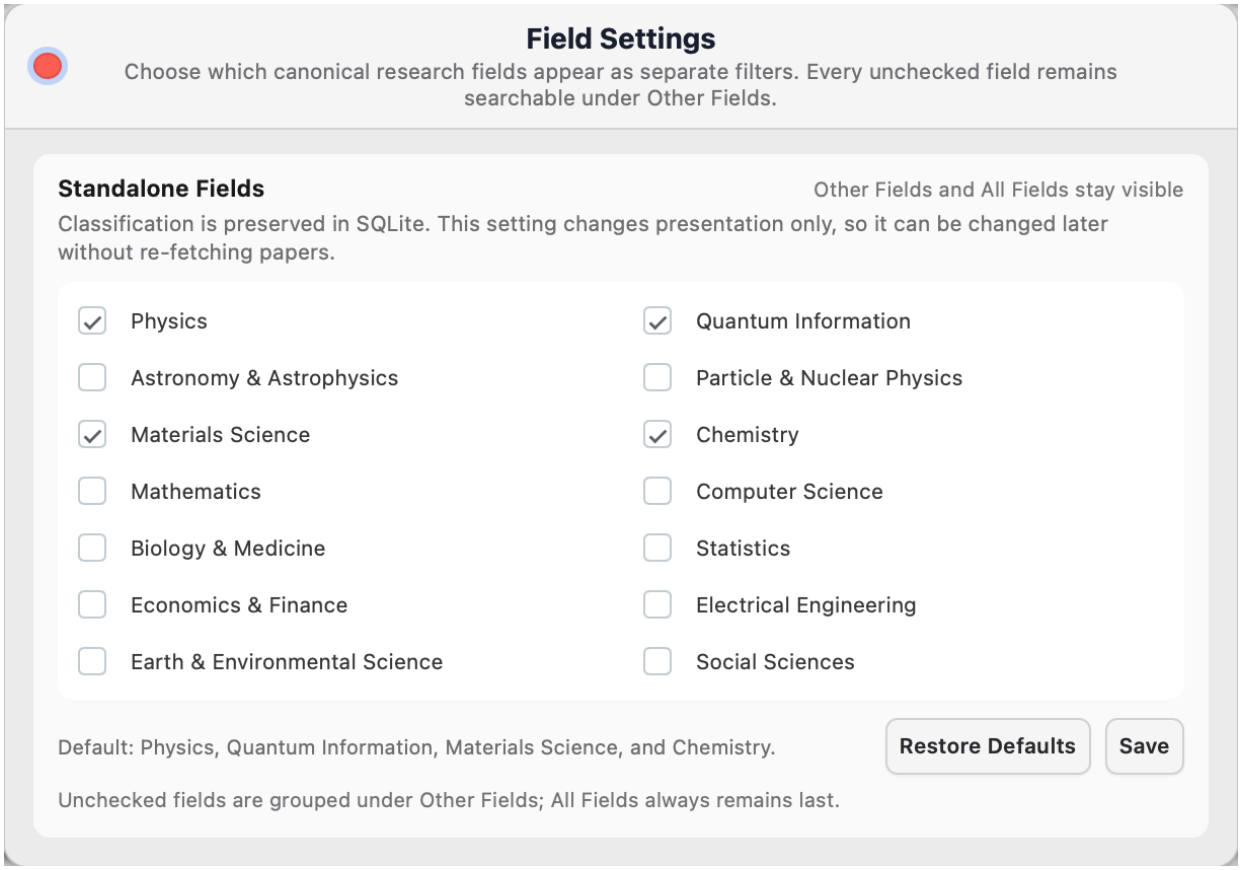
- The default selection is only **Physics > Condensed Matter**.
- **Restore Defaults** changes the selection in the window; choose **Save** to apply it.
- Saving changes future arXiv refresh coverage and rebuilds the available **Preprint Filter** categories.
- Previously saved papers remain in SQLite. Changing scopes never deletes earlier discoveries.

> **Note**
> The category checkboxes in Journal Monitor are display filters. They do not narrow the discovery scope selected here.

See also: Journal Monitor · Field Settings · Literature Data Source · Publisher and Network Access

3-9.3 Field Settings

Choose which research fields appear as separate filters across the workspaces.



Field Settings. Choose which research fields appear as separate filters. Unchecked fields remain searchable under Other Fields.

Open Field Settings

Choose **Setting > Field Settings**. Select a field to show it as a separate filter and label. Clear it to collect that field under **Other Fields**, then choose **Save**. Closing without saving leaves the earlier presentation in effect.

Default organization

Shown separately by default	Grouped under Other Fields by default
Physics	Astronomy & Astrophysics
Quantum Information	Particle & Nuclear Physics
Materials Science	Mathematics
Chemistry	Computer Science
	Biology & Medicine
	Statistics
	Economics & Finance
	Electrical Engineering
	Earth & Environmental Science
	Social Sciences
	Unclassified records

How the visible filters behave

- **Other Fields** appears immediately above **All Fields**.
- **All Fields** is always the final choice and includes every classified paper.
- Changing this setting updates field filters, counts, badges, and Paper Details across Journal Monitor, Subject Monitor, Library, My Publications, and Author Monitor.
- Canonical field assignments stay in SQLite and reappear when a field is selected again.

> **Note**
> Use **Restore Defaults** to save the four-field default immediately. Field Settings do not change arXiv discovery coverage; configure that separately in **arXiv Scope Settings**.

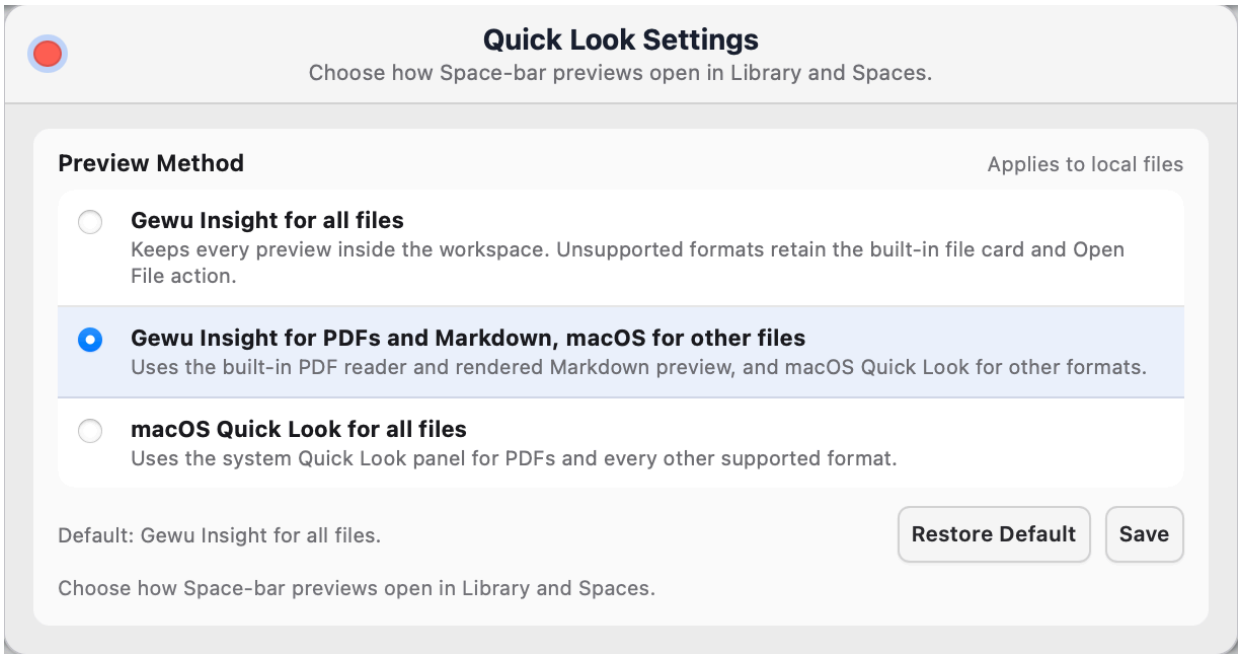
Filter visible papers and their indicators

This settings panel chooses which field names appear separately. The Monitor's **Field Filter** checkboxes then choose which papers to show. Journal and Subject tab blue dots count only unread papers visible under the current filters; hiding papers never marks them read. Showing another field can bring its unread dots back. Journal Monitor's Publication updates count and list also follow the selected fields and journals. Left-side Journal Filter totals remain the complete stored counts.

See also: arXiv Scope Settings · Literature Data Source · Journal Monitor · Subject Monitor

3-9.4 Quick Look Settings

Choose whether Space-bar previews stay inside Gewu Insight, use macOS Quick Look, or divide the work by file format.



Quick Look Settings. Choose built-in previews for all files, a mixed PDF-and-Markdown/native workflow, or macOS Quick Look for all supported files.

Open Quick Look Settings

Choose **Setting > Quick Look Settings**. Select a mode and choose **Save**. The setting applies to Space-bar previews in Library, Spaces, Paper Files, and the other research views when a local file is available. Closing without saving leaves the previous mode in effect. **Restore Default** immediately saves Gewu Insight for all files. The saved mode persists after restarting the App; it does not change Open in Preview. If a preview is already open, saving a different mode refreshes that preview with the newly selected behavior.

Mode	Behavior
Gewu Insight for all files	The default. Every preview stays inside the workspace. PDFs use the built-in reader; unsupported formats keep the built-in file card and **Open File** action.
Gewu Insight for PDFs and Markdown, macOS for other files	PDFs and Markdown use the built-in readers, so Markdown retains formatted rendering. Presentations, media, images, and other formats use the macOS Quick Look panel.
macOS Quick Look for all files	PDFs and every other format supported by macOS use the system Quick Look panel.

Choose a mode for your workflow

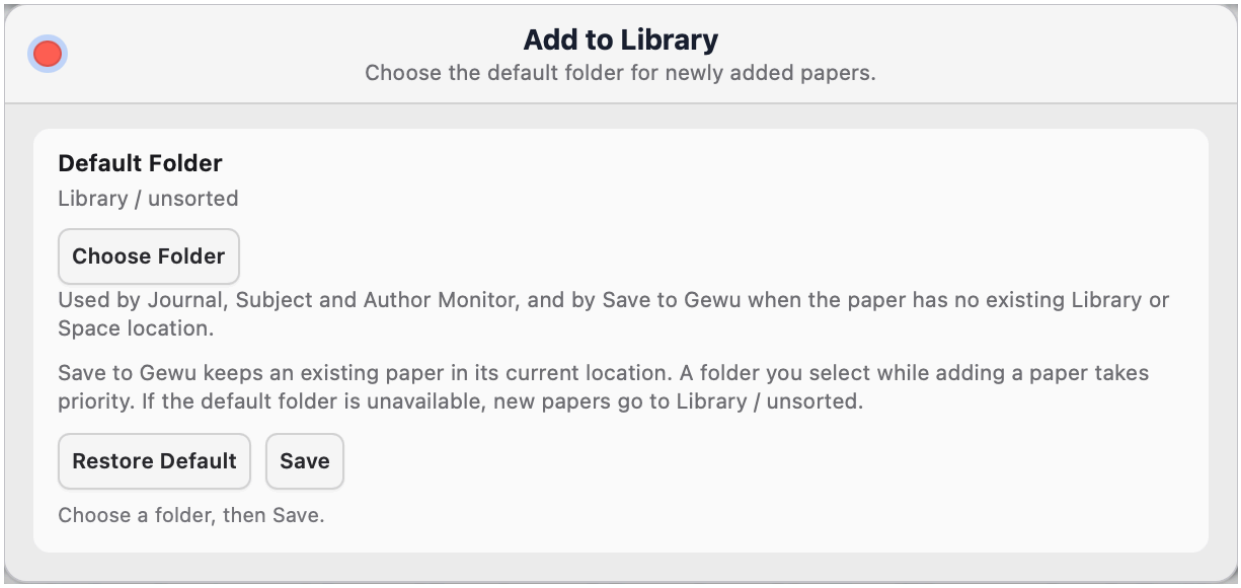
- Use the default when you want one consistent, contained workspace with an embedded PDF reader.
- Use the mixed mode to read both PDFs and formatted Markdown inside Gewu while presentations, video, audio, and images use native macOS preview support.
- Use the all-native mode for the macOS Quick Look panel, including when you prefer its behavior during rapid PDF switching. In the native App, this is separate from opening the full Preview application.

> **Note**
> When the macOS panel is used, keyboard focus remains in Gewu Insight so ↑ and ↓ continue changing the selected Library or Paper Files item instead of turning into page-navigation commands in the system panel.

See also: Preview Files with Quick Look · Understand Paper Files · Navigate with Mouse and Keyboard

3-9.5 Default Add to Library Folder

Choose where a newly added paper starts. The initial default is Library / unsorted ; existing Browser captures keep their active paper location.



Add to Library settings. Choose an existing Library or Space folder, then Save. The shown path is the current saved preference.

Set or restore the default

1. Open **Setting > Add to Library**.
2. Choose **Choose Folder**, select an active Library or Space folder in the column picker, and confirm it.
3. Choose **Save**. Closing the settings window without saving discards the new choice. **Restore Default** immediately saves Library / unsorted again.

How each entry point uses it

Entry point	Destination
Journal, Subject or Author Monitor > Add to Library	The picker preselects the saved default. Confirm another folder to override it for this addition.
Browser > Save to Gewu, existing active paper	Reuse that paper's active Library/Space placement, preferring the originating selected occurrence of the same Key. The new default does not create another location.
Browser > Save to Gewu, no active placement	Use the saved default, unless you explicitly choose and confirm another folder.
Folder Add Paper, Finder drop or Folder Loader	Keep the destination of that explicit operation. This preference does not redirect an import already aimed at a folder.

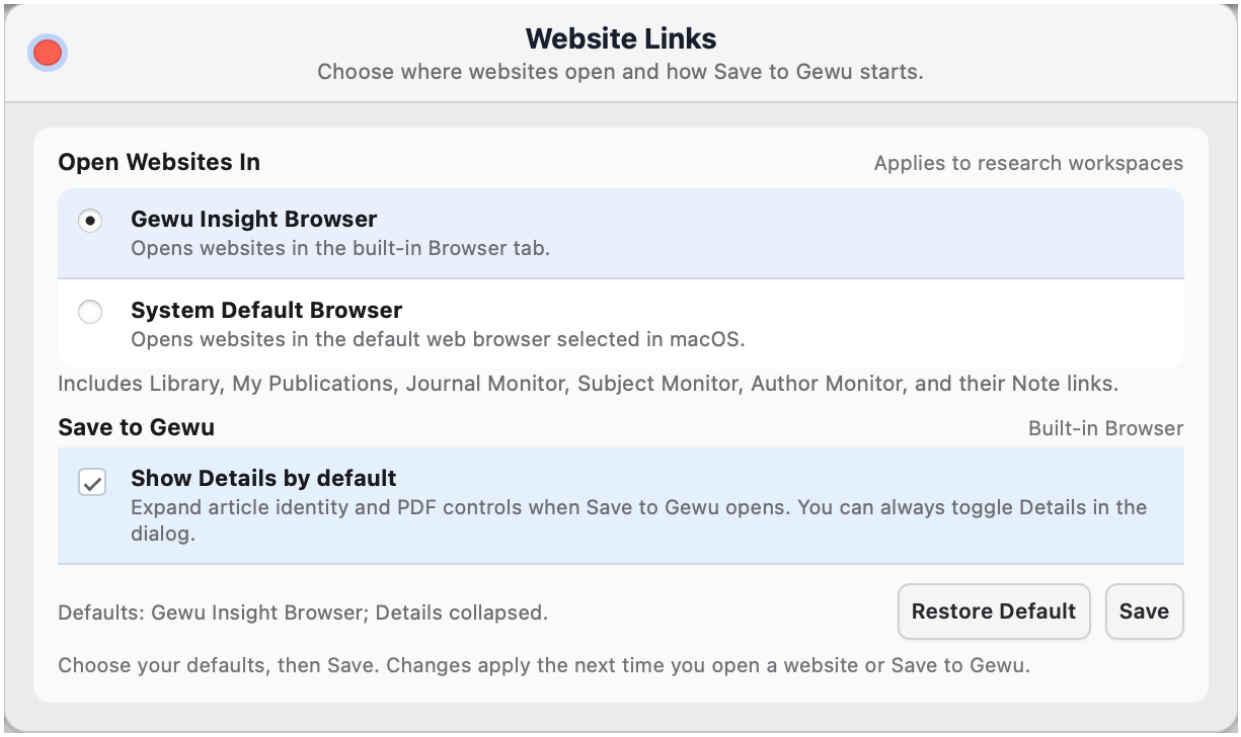
The preference follows the selected folder when it is renamed or moved. If it is deleted, recycled or unavailable, a later default addition falls back to Library / unsorted. Recycle cannot be chosen. If an explicitly selected destination becomes unavailable before Save, choose a valid destination again.

This affects future additions only. It does not relocate existing papers, change Gewu Keys, copy PDFs, alter file classification or change Default. Choosing a local disk path is not the same as choosing a managed Library folder.

See also: Save an article and PDF to Library · Folders, Shortcuts, and Spaces · Use Folder Loader

3-9.6 Website Links

Choose one browser preference for website actions across your research workspaces.



Website Links. Choose the browser used by Open Website and the initial Details state of Save to Gewu. The checkbox shown here is a saved user preference; the shipped Details default is collapsed.

Open **Setting > Website Links**, select an option, and click **Save**. The choice applies to the next website action and is remembered after restarting Gewu Insight. Closing the dialog without saving leaves the previous preference in effect.

Option	Behavior
Gewu Insight Browser (default)	Opens a new built-in Browser tab, preserving existing pages. Press ⌘[to return to the source workspace.
System Default Browser	Opens the website in the default web browser selected in macOS. Your current Gewu workspace stays in place.

This applies to Library (including Spaces), My Publications, Journal Monitor, Subject Monitor, and Author Monitor: **Open Website**, publisher arrows, DOI/arXiv links, Google Scholar, author profiles and homepages, citation sources, remote PDF links, Overleaf, and links inside Note. **Restore Default** saves Gewu Insight Browser and collapses Save to Gewu Details by default.

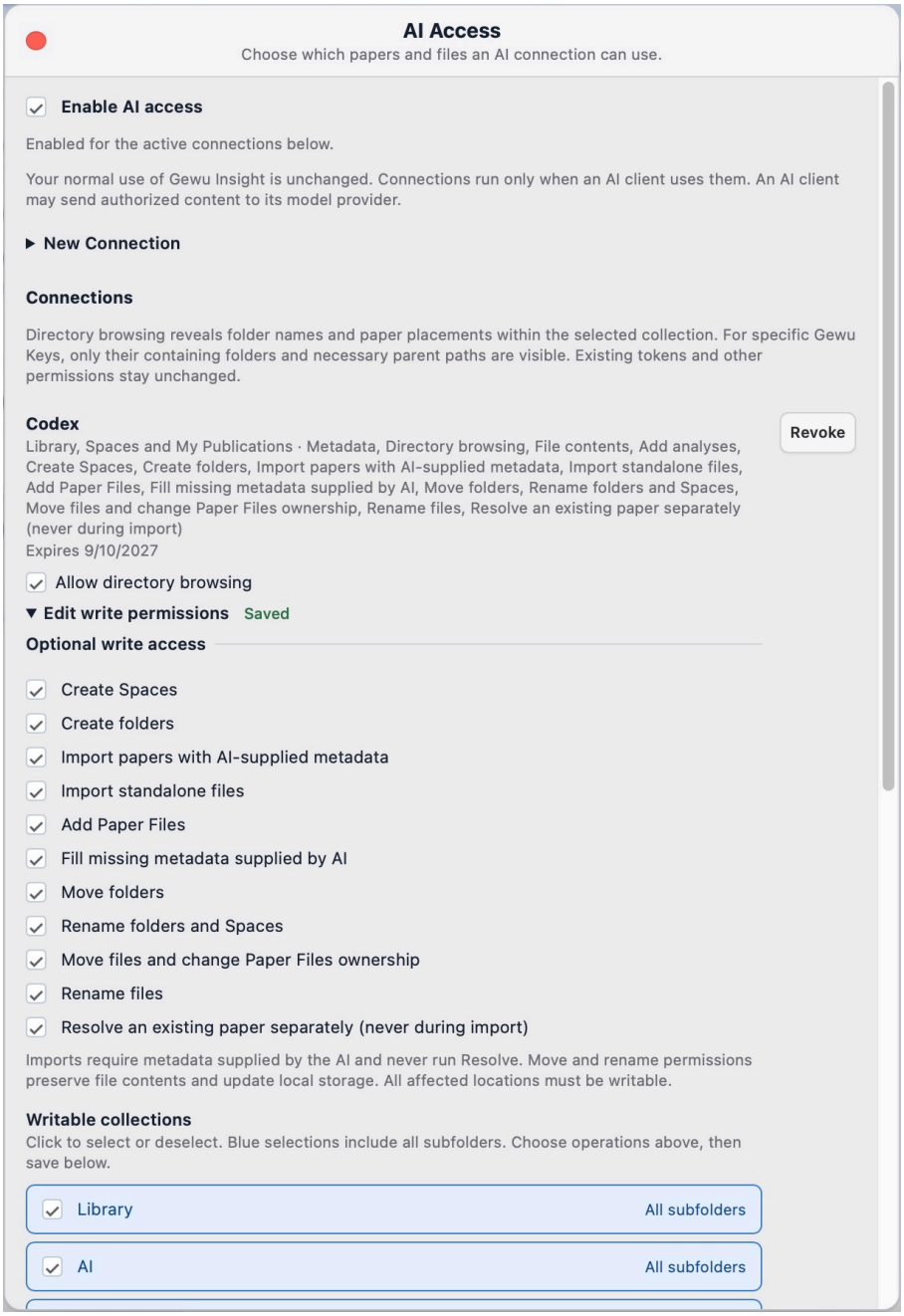
Show Details by default controls how Save to Gewu opens. It is off initially; Save remembers it after restart. Toggling Details inside one dialog preserves its fields and does not change this preference. Unsaved settings changes are discarded.

Local PDFs still open in Preview; Finder, Quick Look, and file downloads retain their own actions. Browsing directly inside the Browser tab stays there. Its explicit **Open in Safari** command still opens Safari. Selecting a different browser does not transfer website sign-ins, cookies, or existing tabs. When using the local UI in a separate web browser, the built-in option opens an ordinary tab in that browser.

See also: Browse, Bookmark, and Revisit Websites · Use Paper Details · Quick Look Settings

3-9.7 AI Access

Give an AI client access to selected papers through their stable Gewu Keys, without navigating the interface.



AI Access. This existing connection shows its permission summary and all eleven optional write operations under Edit write permissions. Scroll down for the remaining writable collections and Save Write Access; the detailed saved-permissions figure below shows them. These saved selections are examples, not new-connection defaults. Directory browsing is controlled separately.

Setting > AI Access is off by default. Normal App use needs no AI token. Enabling it does not start a model, analyze papers or upload your library. Requests happen only when a connected AI client calls the local service.

One MCP interface, multiple AI clients

Gewu Insight uses the standard MCP interface to work with compatible AI clients. **Codex and DeepSeek Harness are two configuration examples.** Claude Code and other clients that support local **stdio MCP servers** can use the same adapter, connection settings and permission model. The client provides the AI; Gewu provides access to the authorized papers and operations.

Choose a client that can start a local process on the same computer as Gewu. Copy the generated `command`, `args` and `env` into that client's configuration format. The local API address belongs in `GEWU_API_URL`; it is not a remote MCP server URL. Create a separate named connection for each client so that permissions, expiry, activity and revocation can be managed independently.

API, MCP and your token

Name	Meaning in this connection
API	**Application Programming Interface**: the operations another program can request from Gewu Insight.
MCP	**Model Context Protocol**: a shared way for an AI client to discover and call tools. Gewu's local MCP adapter translates those calls into its API requests.
Access token	A private credential issued by Gewu Insight. It authorizes a connection's paper collection and permissions; it is separate from your AI account login and from text tokens counted by models.
Gewu Key	The stable identifier of a paper. It selects the article; the access token authorizes access to it.

Methodology: a stateful information infrastructure

Gewu Insight gives people and AI clients a shared, persistent research workspace. **Stateful** means that cataloged papers, their identities, registered files and saved analyses remain available between tasks. SQLite is the source of truth for records and relationships; the documents remain local files. The Gewu Key lets a person or an authorized AI return to the same paper across conversations and clients.

The API exposes supported operations with defined inputs and structured results. An AI can retrieve a paper by key without navigating the interface or depending on the internal database schema. MCP makes those operations discoverable as tools; a script can use the HTTP API directly. Both use the same underlying records and authorization rules. Gewu supplies the material and stores approved results; the connected AI performs the analysis.

Design choice	How it helps research work
Stable paper identity	Use one Gewu Key to connect the article, its files and analyses across tasks. A title or filename is not the identity, and knowing a key does not grant access.
Explicit, bounded operations	Search with pagination and read selected pages or text chunks. Programs can compose these steps and limit unnecessary retrieval without relying on screen coordinates.
Persistent, traceable results	Save a Markdown analysis under the same key with source file references. People can inspect it in Paper Files, and another authorized session can retrieve it later.

This provides a foundation for continuing literature analysis and reusing saved knowledge. It does not automatically preserve an AI's conversation or reasoning, build a semantic memory index, or synchronize model memory. A later session must explicitly retrieve the records and notes it needs; access still depends on its current grant and the continued availability of those records and files.

Two API routes, one authorization boundary

Route	Where permission is checked
AI client → MCP adapter → Gewu API	The adapter sends the connection token to the API. MCP does not grant additional access.

Route	Where permission is checked
AI client or script → Gewu API	The caller supplies the same kind of token directly. Skipping MCP does not skip API authorization.

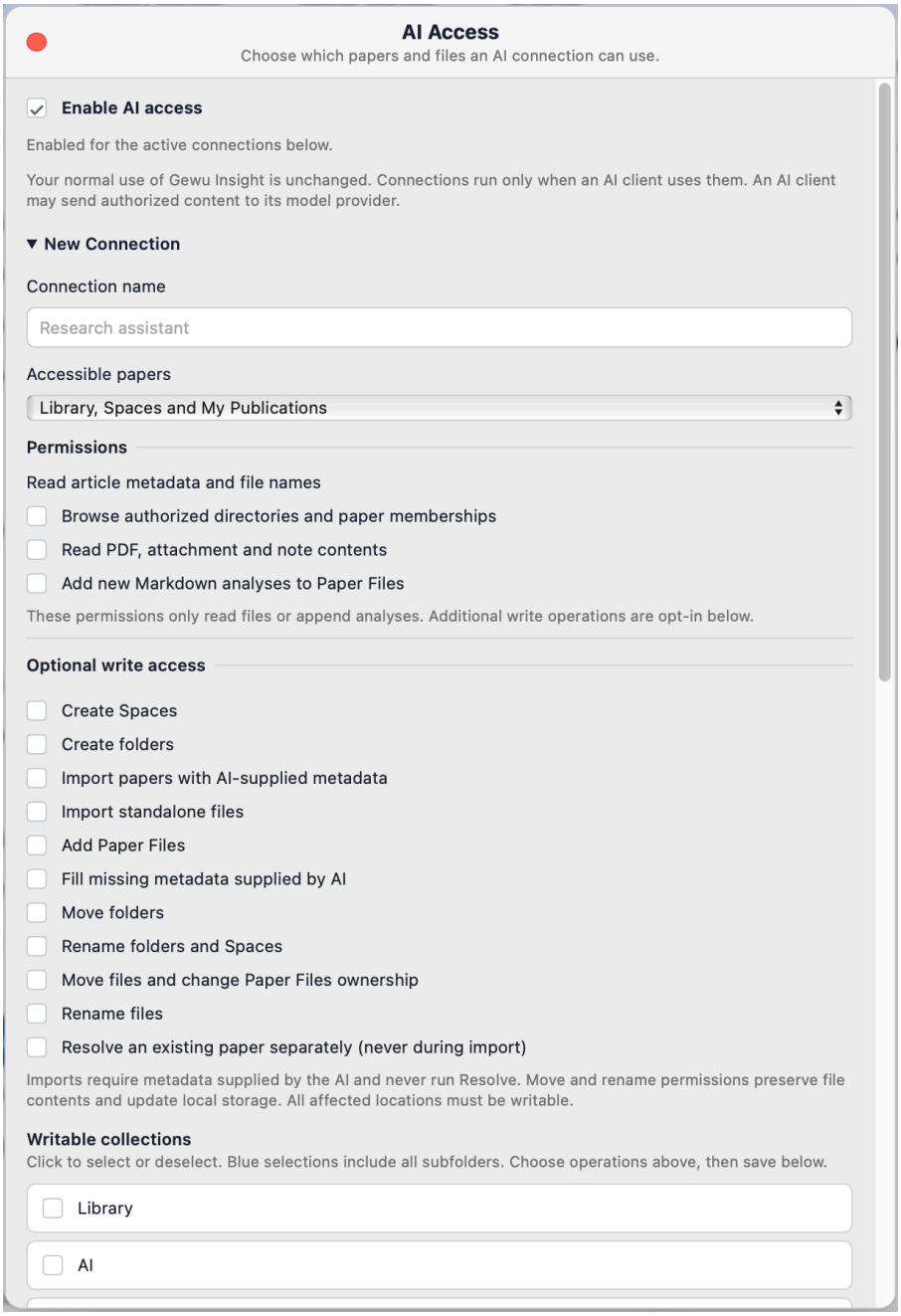
For either route, the API checks that AI access is enabled, the token is valid, unexpired and unrevoked, the requested paper is in the connection's collection, and the operation is permitted. Metadata access alone does not authorize directory browsing, file reading or analysis writing. The native App uses a separate private workspace session for normal human use; an AI token cannot call those internal UI endpoints.

Interface authorization and local file permissions
AI Access controls requests through Gewu's interfaces. It does not encrypt or isolate SQLite and document files from software that already has operating-system permission to read them. Such software can read the disk without contacting MCP or the API; administrator privileges are not necessarily required. Turning off AI Access or revoking a token stops access through that grant, but does not revoke independent filesystem permissions. This broader local access is outside the AI Access permission boundary.

Create a connection

1. Keep Gewu Insight running and enable **AI access**.
2. Expand **New Connection** and enter a name you will recognize.
3. Choose Library, Spaces and My Publications; one folder and its subfolders using the usual folder picker; or specific Gewu Keys. A folder connection follows current membership.
4. Article metadata and file names are readable by default. Separately allow directory browsing, file contents and/or adding new Markdown analyses when your task needs them.
5. Choose an expiry, then **Create Connection**. The default is 30 days.
6. Use **Copy MCP Configuration** to obtain the local adapter path, Python executable, port and token. Enter these in your AI client's MCP settings using its required format. The Codex and DeepSeek Harness examples below show two ways to use these same values.

Keep the connection private
The token is shown only after creation. Store it in client configuration, not in a chat message or shared document. If it is lost, revoke that connection and create another. An AI client may send authorized content to its model provider.



New AI connection. The current blank form separates basic permissions from optional write operations. All optional choices are unchecked here. Scroll further down to choose writable collections and expiry, then use Create Connection when ready.

What an authorized AI can do

Tool	Result
`search_papers`	Search authorized titles, DOI, Gewu Keys and abstracts, with pagination.
`list_folders`	List authorized Library and Space roots, or a folder's direct children, with stable IDs, logical paths and filtered counts. Requires directory-browsing permission.
`list_folder_papers`	List each authorized paper placement and its Gewu Key in a directory, directly or recursively. Includes pagination and minimal metadata. Requires directory-browsing permission.
`get_paper`	Get the article metadata, file inventory and Default file ID for a Gewu Key.

Tool	Result
`get_paper_file`	Get an original PDF or registered attachment as a read-only temporary copy on the local AI client's machine, with paper metadata, file ID and actual SHA-256. Requires file-reading permission.
`get_file_delivery_status`	Show used and remaining temporary capacity, file slots and active delivery IDs for this MCP connection.
`release_paper_file`	Release a temporary copy by its delivery ID after reading finishes, and show the restored quota. Original Paper Files and saved analyses are preserved.
`read_paper`	Extract PDF text by page or read text/Markdown chunks. Use `get_paper_file` for the original file when text extraction is insufficient.
`save_analysis`	Add a new Markdown analysis to the same paper's Paper Files when writing analyses is allowed.

For example, give the AI a Gewu Key, ask it to read the Default PDF, and save a Chinese analysis with source page references. The saved note records its connection, time, key, request ID and supplied source file hash/pages. It remains an ordinary associated Markdown file, available in built-in Quick Look. Returning to Library refreshes the selected paper's file panel.

Saving an analysis preserves the article identity, existing files, metadata and Default choice. A retry using the same request ID and identical content reuses the saved file. A new analysis needs a new request ID. Existing file contents, including earlier AI analyses, cannot be edited or overwritten. Creating folders, importing files, moving or renaming existing items and filling missing metadata require separate permissions.

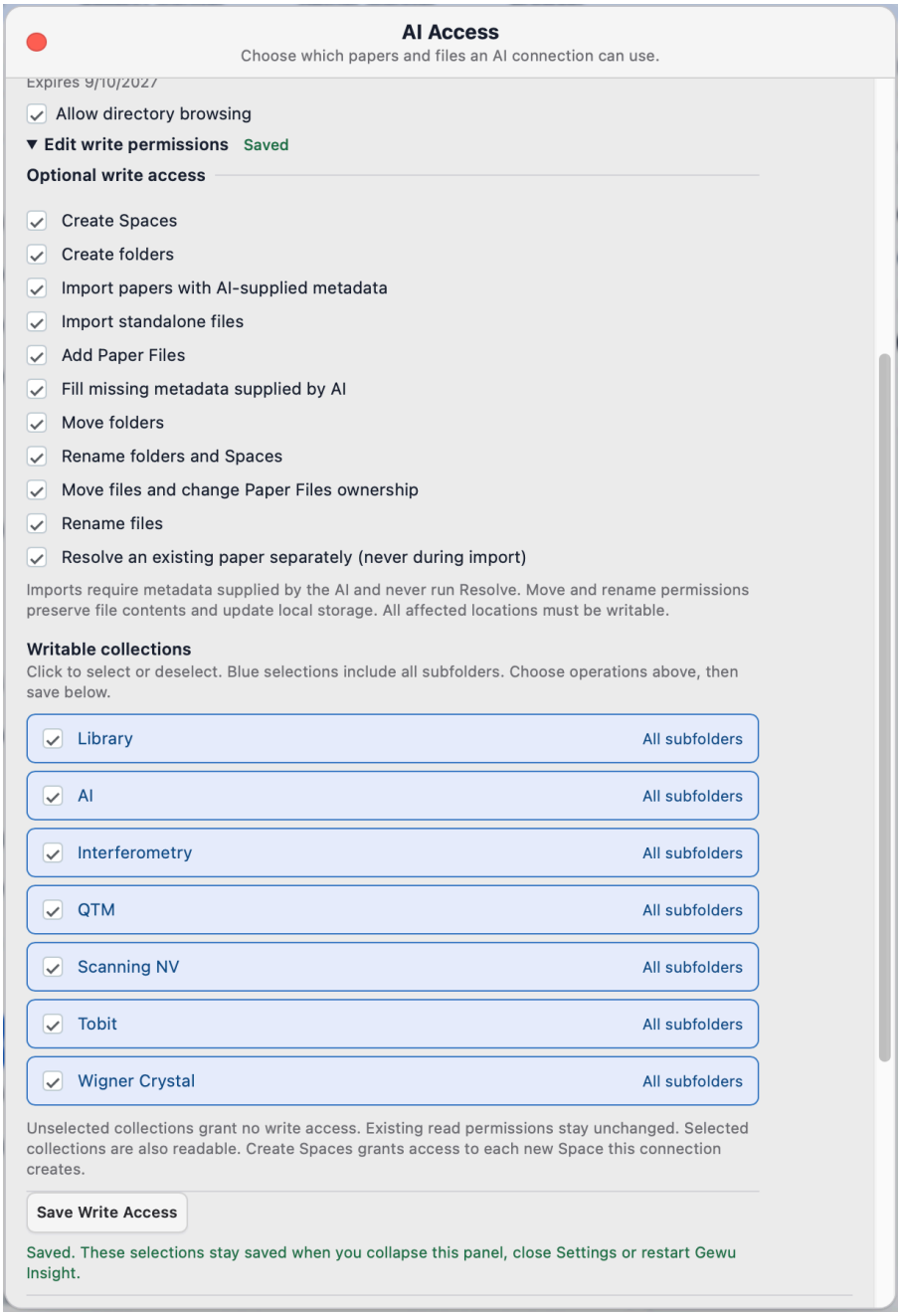
New analyses are registered as Markdown notes: **File type: Other**, **Annotation: Unannotated**, and **Source: Other** with **AI analysis** and the connection name as source detail. File size, content hash, creation time and analysis provenance are recorded. These categories are assigned by Gewu; the AI cannot relabel its note as a published paper.

Let an AI import a prepared collection

Open **Setting > AI Access** and expand **Edit write permissions** for the intended connection. Choose operations, then click **Library** or root Space names under **Writable collections**. Selected rows stay blue and include all active subfolders; click again to deselect. My Publications is not a write destination in this editor. Choose **Save Write Access**, then reconnect the AI client to refresh its tools.

Saved remains beside the editor title even when collapsed. Saved selections survive closing Settings and restarting the App. **Unsaved changes** requires saving; **Not saved** retains your draft for retry. Earlier child-only grants show **Some subfolders only** and stay limited until you explicitly select their root. No write permission is added automatically.

Selected write collections also become readable; other existing read permissions and tokens stay unchanged. An empty selection grants no existing write destination. **Create Spaces** grants access only to each new Space this connection creates. Choose directory browsing separately if the AI needs to explore the tree.



Saved write permissions. The expanded editor shows operations, blue writable collections, Save Write Access and the saved result. These are this existing connection's choices, not the defaults for a new connection. My Publications is absent from writable collections. Only enable the operations and locations needed for your task.

Tool	Purpose
`create_space`	Create a new Space and return its folder ID.
`create_folder`	Create a nested folder under a writable Library or Space folder.
`move_folder` / `rename_folder`	Move a complete folder subtree or rename a folder/Space, synchronizing the actual directories.
`move_file` / `rename_file`	Move an item or Paper File, or rename its placement/filename under the existing file ownership rules.
`upload_file`	Transfer one prepared client-local file to temporary staging and return its upload ID.

Tool	Purpose
`import_paper`	Save a paper with AI-supplied metadata in the chosen folder, with or without a PDF.
`import_file`	Import an independent non-PDF file in any format, such as a direction overview in Markdown.
`add_paper_file`	Append a file of any format to the specified existing Gewu Key.
`update_paper_metadata`	Fill missing fields from AI-supplied facts and a source URL; preserve existing values and report disagreements.
`get_upload_status` `release_upload`	Check upload capacity or release an unused staged upload.
`resolve_paper` `get_resolve_status`	Start or check a separately requested Resolve job for one existing paper. Never part of an import.

The AI must bring the paper metadata. A paper import requires title, authors, publication type, journal or repository, known publication date or year, DOI and arXiv status, and the source URL. A genuinely absent identifier uses an explicit null value. If required facts are missing, the AI must find them using its own sources and retry. Import does not search a PDF for identifiers, fetch metadata online, call Resolve or queue enrichment. Supplied facts retain their source and connection provenance; Gewu does not claim to have verified them. A PDF is optional: omit `upload_id` and `file_type` to save the paper and receive its Gewu Key without a placeholder file or automatic download. Later, upload the PDF and call `add_paper_file` with that key. Supplied PDFs still receive integrity checks; annotation status initially remains Unknown.

Choose ownership explicitly. Import an independent Markdown overview with `import_file`. Attach a paper-specific analysis, spreadsheet, image or code file with `add_paper_file` and the owning Gewu Key. An ordinary non-PDF file retains its internal storage key but has no public Gewu Key field in the App. It does not become a scholarly paper simply because the AI imported it.

Reuse the existing paper identity. A matching DOI or arXiv identity reuses the existing Gewu Key and adds its placement in the requested folder. A different incoming PDF is retained as another Paper File. A valid manual Default remains; otherwise eligible versions use the normal priority: Corrected published version, Published version, Accepted manuscript, then Preprint. This also applies to `add_paper_file`. The automatic Default becomes the managed main file; other versions stay in Associated Files and folder aliases are synchronized. Supplements and analyses do not take automatic Default. Identical bytes reuse the file and its classification and record the import source. Similar titles alone never merge papers. Uncollected provider/Monitor records, including managed download PDFs, reuse the same Key through a writable destination and the normal archive workflow. Unreadable private collections, notes or external files still require user access review. DOI and arXiv pointing to different records remain an identity conflict.

Publication notices belong to the original article. Corrections, errata and retraction notices retain their own DOI, title and date as associated notice metadata. They do not create another research-paper Gewu Key or become the automatic Default. For MCP import, the AI must supply `publication_notice` with the notice kind, original article DOI and an evidence URL; import the original first if absent. Missing or ambiguous relations require review and never trigger Resolve. Paper Files shows linked notices while retaining the original article and its files.

When a file is present, the main file goes in the selected Library or Space folder. Its other files go in the existing mirrored Associated Files bundle. Multiple placements share that bundle and use the normal aliases. The AI supplies folder IDs and file ownership; Gewu calculates and maintains the physical paths according to the same storage rules used by the App.

Complete one file at a time. Upload one file, immediately register it with the appropriate import or attachment tool, then proceed to the next. Each successful file import immediately releases that file's staging; it does not wait for an entire folder or batch. Do not upload a whole collection before registering files. Capacity limits apply to currently retained files, not the total number or size of a literature collection.

Upload staging allows up to 256 MiB per file and 512 MiB or 16 currently staged files per connection. If it fills, register ready uploads or release unused ones, then continue without reconnecting. Failed transfers remove their partials; unused ready uploads expire after 24 hours, while active transfers are retained. Read copies have their separate limits below and must be released only after all readers finish. Actual free disk space and single-file limits still apply.

Keep a stable request ID for every write and a separate ID for its upload. Retry an interrupted response with the same arguments and ID to reuse its result. Existing files are never overwritten. If the result says alias synchronization is pending, retry that same import to finish the folder shortcuts.

Resolve remains independent. Enable its separate permission only when needed. It operates on one existing writable paper, requires an explicit standalone purpose, and returns a job immediately. Known DOI/arXiv is used first; PDF identifier extraction also needs file-reading permission. Results fill missing fields while preserving existing values and Default. At most two jobs run concurrently, and their network lookup does not block imports. The AI must never use this tool to prepare or complete import metadata.

Let an AI organize files and folders

Enable **Move folders**, **Rename folders and Spaces**, **Move files and change Paper Files ownership** or **Rename files** separately. Select the source and destination collections, save, and reconnect the client. Every affected location must be writable, including other placements of the same record. My Publications remains outside these write permissions.

Use a folder ID to move a whole subtree or rename a folder/Space. The actual directory and its mirrored Associated Files move together; empty folders and existing contents are retained. Library, Unsorted and Recycle are protected. A folder cannot move into itself or a descendant. A Space root may be moved into Library using the same conversion rules as the App.

For files, select either a membership ID from directory browsing or an exact Gewu Key and numeric file ID from Paper Files. A membership moves the whole item to another folder, or attaches its files to another record. A specific Paper File moves to another record or becomes an independent item in a folder. Independent non-PDF files keep their ordinary-file behavior. Renaming a membership changes its label and disk projection; renaming a Paper File changes that filename and must retain its extension. The bibliographic title stays unchanged.

File contents, IDs, classifications and provenance stay intact. The main file stays in its owning directory; other files use Associated Files. Transfers follow the App's existing ownership rules: the source chooses a remaining Default if necessary, the target keeps its existing Default, and an emptied source may retire. Final-file transfers from records with multiple placements are rejected. Moving into a folder that already contains the same record reuses that membership without losing files.

Operations do not run Resolve, upload existing files again or consume transfer staging. Use identical arguments and the same request ID to retry an interrupted response or finish pending shortcut synchronization. Unrelated files are never overwritten. External originals cannot be renamed through MCP; a permitted ownership transfer of an externally linked file keeps the original and uses the App's managed-copy rule.

Browse a directory and its papers

Directory browsing is a separate, read-only permission. For a new connection, select **Browse authorized directories and paper memberships**. For an existing connection, select **Allow directory browsing** beside its name. This adds `folders:read` without replacing the token, changing the paper collection or granting file reading or analysis writing. Existing connections are not expanded automatically. Reconnect the AI client's MCP connection to discover the two new tools.

These directories are the logical folders in Library and Spaces, not disk paths. A Space uses a stable `folder_id` just like any folder. Shortcuts lead to the original target; they do not duplicate its membership. My Publications-only papers remain available through search and `get_paper`. Recycle and trashed folders or items are excluded.

Selected collection	Visible directories and memberships
Library, Spaces and My Publications	All active Library and Space folders, including empty ones, and their active paper placements.
One folder and its subfolders	Only that folder and its active descendants. The displayed path starts there; parents, siblings and outside placements stay hidden.
Specific Gewu Keys	Only folders containing those papers and the parent paths needed to reach them. Enabling directory browsing allows these folder names to be read; unrelated branches and their counts stay hidden.

For example, ask the AI to list the papers in your AI Space and preserve their folder classifications:

1. Call `list_folders` without a parent to discover visible roots. Follow `next_cursor` until it is `null`, and identify AI by its name, kind and path.
2. Use AI's returned `folder_id` as `parent_id` in `list_folders` to inspect its children. Repeat as needed to map the folder hierarchy.
3. Call `list_folder_papers` with that `folder_id`. By default it returns direct members; set `recursive` to `true` to include descendants. Follow every page. Each page defaults to 50 entries, with a maximum of 100.
4. Keep the returned `membership_id`, `folder_id` and `gewu_key` together. One paper can have several placements; use the Gewu Key to avoid reading it repeatedly while retaining its classifications. Call `get_paper` for the selected key before reading files.

Names can repeat, so use folder IDs and the returned ID/name path segments instead of treating a display path as a unique identifier. Each call states its directory explicitly; navigating in another task does not change a shared current directory.

> **Keep a multi-page collection walk consistent**
> Pass the first returned `revision` with subsequent directory calls. A visible rename, move, deletion or membership change returns `collection_changed`; discard the incomplete walk and start again without the old cursor or revision. Folder IDs survive renames and moves, while deleted and recreated folders get new IDs. A folder grant follows its root ID, but descendants moved outside it lose access. Ordinary metadata refreshes do not interrupt membership pagination; bibliography fields remain current, not a historical snapshot.

Give the original PDF to the AI client

Use `get_paper_file` with the Gewu Key and a file ID from `get_paper`, or `default`. When no Default exists, choose an explicit file ID. Gewu checks the connection and transfers the original bytes. The client receives a read-only temporary copy, title, authors, DOI, file classification and actual SHA-256. Its own tools can read the PDF, inspect pages and figures, or run OCR. Gewu does not render pages, run OCR, upload the file to a model or fetch a replacement from the DOI.

The delivered copy is separate from Paper Files and never syncs changes back. Write rendered pages and intermediate results separately. A file is limited to 64 MiB, with up to 256 MiB or 64 retained copies per MCP session. Each delivery returns a `delivery_id` and `quota` showing used and remaining bytes and file slots. Use `get_file_delivery_status` to check capacity and find active delivery IDs.

Remaining capacity is reported to the AI client in these tool results; it is not a separate Library or AI Access status meter. Ask the connected assistant to check file-delivery status when you need the current amount.

After every reader, rendering and OCR task finishes with a copy, call `release_paper_file` with its delivery ID. This frees its temporary quota immediately; original Paper Files, saved analyses and derived outputs remain unchanged. Keep the source file ID, SHA-256 and page references for citations and saving analyses. Active repeat delivery shares the same copy and ID, so finish all its users before release. Repeating a release is safe. No files are automatically evicted: when quota is full, release finished copies and retry.

Unreleased copies last until normal MCP process exit. The two local management tools also work offline or after grant expiry; they cannot retrieve new files. Reconnect MCP once after upgrading to discover them, after finishing work with old copies. A client needs access to the adapter's local filesystem; a returned path is not a cloud PDF attachment. If unavailable, report that limitation rather than reading the source Library directly.

Revocation, expiry or disabling AI Access prevents new deliveries. Copies already delivered to a client cannot be recalled. File transfer uses local disk and memory; PDF processing done by a client on the same computer can still compete for resources. Gewu adds no automatic whole-document OCR workload.

A repeatable research workflow

1. **Identify.** Give the AI an authorized Gewu Key, or use `search_papers` to find candidates and confirm the intended paper with `get_paper`.
2. **Read.** Inspect the file inventory, then use `get_paper_file` to obtain the original PDF and read its temporary copy with the AI client's tools. For text-only requests, `read_paper` remains available. Retain the returned file ID, SHA-256 and page references.

3. **Analyze and save.** Ask the AI to produce a Chinese Markdown analysis. With analysis-writing permission, call `save_analysis` under the same key and include the source references. Reuse the request ID only for an identical retry; changed source bytes require a fresh read before saving.
4. **Release.** Once all work using a temporary copy finishes, call `release_paper_file` with its delivery ID. Check the returned remaining quota, then proceed to the next paper without restarting MCP.
5. **Review and continue.** Inspect the note in Paper Files. In another task, retrieve that same paper and explicitly read the saved note before continuing. Each session works with the current authorized state.

These steps can be combined into a scripted workflow or repeated over a selected collection. Pagination and bounded reads help control how much material is retrieved; they do not guarantee model accuracy or a particular speed. Review the analysis against its cited sources.

Keep AI connections after an App or database move

Codex, DeepSeek Harness, Claude Code and other compatible clients connect to the same running Gewu service. Moving research data through a verified migration preserves connection records, their scope and permissions; it does not require a new token merely because the database folder changed.

Keep the updated App open on the intended database. If the adapter's code path, Python executable or service port changed, update the existing client's `command`, `args` and `GEWU_API_URL` from the installed App's configuration, preserving its own connection token and other client settings. If those values did not change, they can remain as they are. Do not point the MCP command at the database folder.

After reconnecting, list the available tools and perform a read-only lookup of a familiar authorized paper. Check its Gewu Key and available Paper Files. This verifies the existing collection without creating a new paper, changing grants or saving a test analysis.

Example: connect Codex on macOS

This example connects a local Codex client to Gewu Insight on the same Mac. Keep Gewu running. Create a connection named **Codex** using the steps above. Enable **Read PDF, attachment and note contents** to read full text; metadata-only access cannot read PDFs. Enable **Add new Markdown analyses to Paper Files** only when you also want Codex to save analyses.

1. Open the desktop client's **Settings > MCP servers > Add server**.
2. Name the server `gewu-insight` and choose **STDIO**. This starts our adapter as a local process.
3. Fill the command, one argument and two environment variables shown below. Use the values from your own **Copy MCP Configuration**.
4. Save, then choose **Restart** to reload the MCP connection.

These fields follow OpenAI's MCP configuration documentation (checked October 5, 2026). The same local Codex host shares MCP configuration across the desktop app, CLI and IDE extension.

Codex field	Example value
Name	`gewu-insight`
Type	`STDIO`
Command	`<GEWU_ROOT>/.venv/bin/python`
Arguments: one entry	`<GEWU_ROOT>/scripts/gewu_mcp.py`

What <GEWU_ROOT> means

`<GEWU_ROOT>` in these source-installation examples is the absolute path to the code folder containing `scripts/gewu_mcp.py`. It is not `Gewu Database` or the Library. A standalone package can supply its runtime and adapter from inside the App; use the generated paths instead of inventing a project `.venv`. This setup does not require one fixed installation location. Replace the entire placeholder, including the angle brackets, before saving; Codex will not expand it automatically. The command example assumes a project-local `.venv`. If your Python executable is elsewhere, use the complete `command` from **Copy MCP Configuration** instead. Its first `args` entry supplies the actual script path. Keep a path containing spaces as one field. If you move the installation later, update both paths from a newly generated configuration.

Fill in Environment variables

Under **Environment variables**, enter the first row below: the left **Key** box holds the variable name and the right **Value** box holds its value. Click **Add environment variable** to add the second row. Enter each name and value separately, without quotation marks or an equals sign.

Key (left box)	Value (right box)
`GEWU_API_URL`	`http://127.0.0.1:8765`, or the exact address under `env` in your generated configuration if the port differs.
`GEWU_API_TOKEN`	Paste the token just created in Gewu Insight using **Copy Token**.

Leave **Environment variable passthrough** empty. It is for forwarding variables already present in the client's environment; this example supplies both values explicitly above. The API address belongs in these variables while the server type remains **STDIO**. Copy the token into settings, not the chat.

Alternative: edit Codex's configuration file

Copy MCP Configuration currently produces generic JSON. Codex's file uses TOML, so do not paste that JSON directly into it. If you prefer editing the file, merge this equivalent block into `~/.codex/config.toml`. Replace `<GEWU_ROOT>` as explained above and `<GEWU_TOKEN>` with your copied token; use the generated command and port if they differ. Preserve your other settings; update an existing `gewu-insight` block instead of adding a duplicate. Keep the quotation marks in this TOML version and reload the connection after saving.

```
[mcp_servers.gewu-insight]
command = "<GEWU_ROOT>/.venv/bin/python"
args = ["<GEWU_ROOT>/scripts/gewu_mcp.py"]

[mcp_servers.gewu-insight.env]
GEWU_API_URL = "http://127.0.0.1:8765"
GEWU_API_TOKEN = "<GEWU_TOKEN>"
```

Type `/mcp` in Codex to inspect the connected server, then use the shared verification steps below.

Example: connect DeepSeek Harness on macOS

Create a Gewu connection named **DeepSeek Harness**, choose its paper collection and permissions, and copy its MCP configuration. Keep Gewu running. DeepSeek Harness can help configure its own MCP client: ask it to translate the generated settings into its local profile, preserving your other settings. Supply the token through private local configuration; the setup instructions and verification prompts do not need to contain the secret.

> Configure gewu-insight as a local stdio MCP server using the Gewu MCP configuration I provide locally. Preserve existing client settings and use the supplied command, arguments and environment values. Keep the token in local configuration and do not repeat it in messages or logs. Then list the available Gewu tools and run a read-only search. Use MCP for Gewu records and files.

The default Desktop profile uses `~/.dsh/profiles/desktop/cordis.patch.yml`. If your Harness home is customized, use that home's `profiles/desktop/cordis.patch.yml`. The generated Gewu configuration is JSON; the profile patch uses YAML. You or the client can merge the following entry into that file. For a new entry, retain the outer `- insert:`; if Gewu is already configured, update its existing entry instead of creating a duplicate.

```
- insert:
    - id: mcp-gewu-insight
      name: "@deepseek-ai/dsh-mcp-client"
      config:
        serverName: gewu-insight
```

```
transport: stdio
command: "<GEWU_ROOT>/.venv/bin/python"
args:
  - "<GEWU_ROOT>/scripts/gewu_mcp.py"
env:
  GEWU_API_URL: "http://127.0.0.1:8765"
  GEWU_API_TOKEN: "<GEWU_TOKEN>"
failOnStartupError: false
```

Replace both path placeholders with the generated command and script path, and use this connection's token and port. Keep each full path as one quoted value. After saving, let the client reload the connection; restart Harness if the tools do not appear. Look for tools such as `mcp__gewu-insight__search_papers`, then verify an authorized paper as described below. A running client alone does not confirm that its Gewu connection is ready.

With the relevant permissions, DeepSeek Harness can search and read papers, create a Space, add papers and append files through this same MCP interface. The example follows the official MCP client reference, Desktop profile documentation and profile-patch example.

Use the same settings in another client

For Claude Code or another compatible client, add a local stdio server named `gewu-insight` using the same generated command, argument list and environment variables. Follow the client's own configuration syntax; Claude Code's MCP documentation includes local stdio setup. A model subscription or API key belongs to the AI client; the Gewu access token separately controls access to your papers.

Verify a real paper, then request an analysis

These steps work in Codex, DeepSeek Harness and other connected clients. First ask for a read-only `search_papers` call; an empty result can simply mean the authorized collection has no matching papers. Replace `YOUR_GEWU_KEY` below with the exact key of an authorized paper that has a local PDF, then send:

> Use the gewu-insight MCP tools to get the paper with Gewu Key YOUR_GEWU_KEY. Show its title and Default filename. Use get_paper_file to obtain the original PDF, then read the first three pages of its delivered copy with your PDF tools and summarize them in Chinese. Preserve the file ID and SHA-256; do not access the source Library directly.

This checks both metadata access and PDF reading. When analysis-writing permission is enabled, you can follow with:

> Read the relevant remaining PDF pages, then save a Chinese Markdown analysis to this same paper through save_analysis. Include the source file ID, SHA-256 and page references, and use a unique request ID.

The saved analysis appears as an associated Markdown file under that paper's Paper Files; the Default PDF stays unchanged. Check the returned Gewu Key and file ID, then inspect the result in the App. When requesting a new Space or a prepared collection, enable the corresponding write operations first and specify the destination. For example:

> Create a Space named Reading Project. Add the authorized paper YOUR_GEWU_KEY to it using its existing DOI or arXiv identity and complete metadata. Preserve its Gewu Key and existing files. Report the destination folder ID and saved paper key, and stop if a required permission or metadata field is missing.

This example writes to your collection; run it only when you want that new Space and placement. A PDF is optional for the import, so no upload is needed just to add an existing paper to the destination. A retry of the same write uses the same request ID and identical arguments; a different operation needs a new request ID. See the prepared-collection instructions above for file uploads and classification.

If a tool is missing, check that connection's permissions and reload MCP. If the service is unavailable, start Gewu; if access is rejected, check the AI switch, expiry, revocation and selected paper collection. A replacement connection supplies a new token, which must also be updated in the AI client's settings. Client changes do not transfer permissions from another connection.

Pause, revoke and review

Turn off **Enable AI access** to pause all connections. **Revoke** stops one connection and removes it from the active Connections list; expired connections also stop working. Re-enabling access resumes connections that are still valid. Expand **Recent Activity** to review requests and outcomes as separate rows in an independently scrolling list, and choose **Refresh** to update it. Tokens, prompts and full file text are not included in that activity log.

Limits and troubleshooting

The AI client must run on the same computer, and Gewu Insight's local service must be running. The local MCP adapter supports protocol 2025-11-25 and earlier negotiated versions. There is no remote MCP endpoint in this version. Developers can retrieve the authenticated API v1 specification at `/api/v1/openapi.json`.

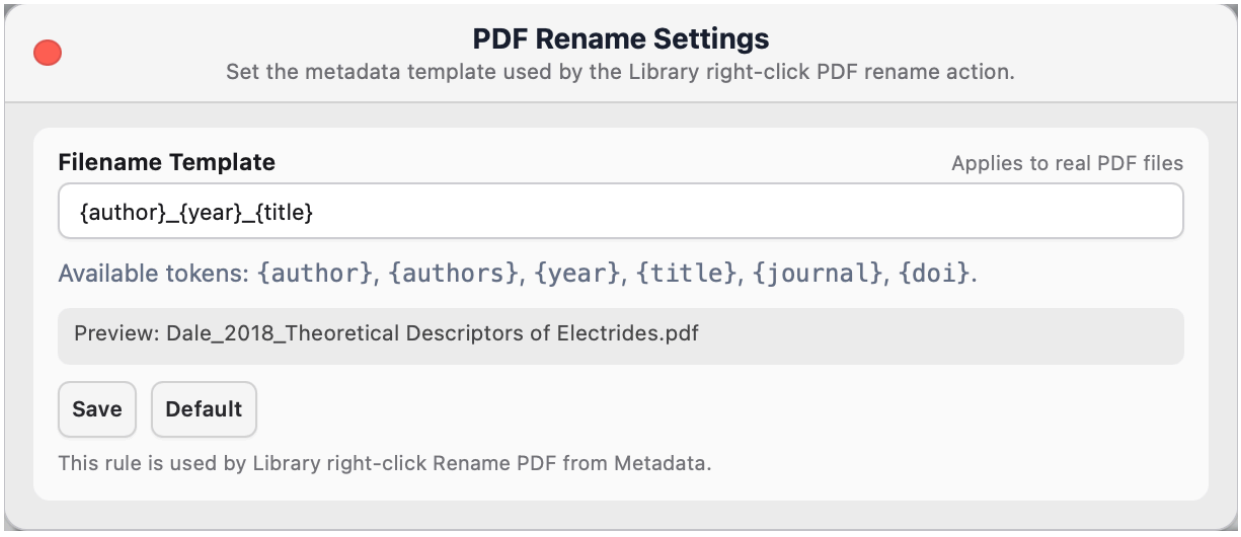
Original file delivery has a 64 MiB file limit and includes every page. PDF text extraction through `read_paper` is limited to ten pages per request and a bounded extraction time; text attachments are returned in chunks. Scanned PDFs may need OCR in the AI client; no OCR or model runs automatically in Gewu. Analysis notes are limited to 256 KiB. A busy response means retry later. An unavailable paper or file may be outside the connection's collection, missing, or no longer accessible.

Source editing, building and temporary tests require no AI token; calling this API against your collection does require one. Configure each client with the collection and operations its task needs. The interface and filesystem permission boundary is explained above.

See also: Understand Paper Files · Preview Files with Quick Look · Local Data and Privacy

3-9.8 Naming and PDF Rename Rules

Display naming and physical PDF renaming are separate choices. Review metadata before renaming files.



PDF Rename Settings. Edit the filename template and inspect its preview before saving. Available tokens appear beside the template.

Choose the Library naming rule

Choose **Setting > Naming Rule > Use Title** or **Use Citation**. The choice is saved immediately and supplies the default name for a newly added Library placement that has no name yet. It does not rename the existing collection. The default rule is Use Citation; physical PDF renaming uses the separate template below.

Configure PDF renaming

Open **Setting > PDF Rename Settings**. A template can use tokens such as:

- `{author}` or `{authors}`
- `{year}`
- `{title}`
- `{journal}`
- `{doi}`

Edit **Filename Template** and inspect the sample **Preview**, then choose **Save** or press Return. **Default** immediately saves the standard `{author}_{year}_{title}` template. Saving a template does not rename existing PDFs; apply it through the Library rename action below.

Rename one item from metadata

1. **Confirm its DOI, title, author, journal, and year.**
2. **Control-click the item.**
3. **Choose Auto rename from Metadata.**
4. **Verify the result in the Library and Finder.** To rename several PDFs, select only local PDF paper rows, Control-click, and choose **Auto rename n selected**. Each file uses its own metadata; the progress message reports renamed, already named, and skipped items.

> **Note**
> Choosing a Default PDF can move it to the main file folder and archive the previous main PDF in Associated Files. This does not apply the filename template or rewrite file contents; name conflicts use a unique filename. Use a naming or rename action to apply your preferred naming rule.

Long filenames retain their extension

Managed import and attachment names preserve the file extension, such as `.pdf`, when shortening a long basename to fit filesystem limits. A long title does not require losing the suffix. File type, Gewu Key and title are separate from the filename; renaming a file does not create another work or identify its scholarly version. Use the App's file controls for managed names so recorded paths and aliases stay consistent.

See also: Choose File Types and the Default File · Folders, Shortcuts, and Spaces

3-9.9 How Gewu Insight Stores Files

The visible Library stays simple: each file-backed paper has a main managed file at one canonical location, while its other privately managed resources live in Associated Files. Paper Files presents these resources together as one flat list.

One work, several representations

Each managed paper has one database identity and a stable **Gewu Key**. DOI and arXiv are external identifiers attached to that identity; neither is used as the sole storage identity.

Where the files go

- The Default is the main file in the visible Library, Space, or My Publications tree. It may be a PDF or another file type.
- Other PDF versions belong in the associated bundle, usually under `Versions`.
- Supporting information goes to `Supplements`.
- Other resources use the bundle's `Figures` or `Files` areas as appropriate; the file-type label remains separate from its physical subfolder.

The associated bundle mirrors the paper's managed path under data/AssociatedFiles. When you move the paper in the UI, Gewu Insight moves the associated bundle and refreshes live paths.

Why the UI and Finder look different

The UI presents a paper as one item with a flat Paper Files list inside it. Finder shows the Default as the main file in the normal tree and the remaining managed resources in Associated Files. Choosing a different Default synchronizes both locations, including for non-PDFs, while preserving file identity, classification and bytes. With no manual Default, the automatic priority and fallback rules still select the main file when an eligible resource exists.

Copies and filesystem shortcuts

When the same Gewu Key appears in several folders, it shares one Paper Files bundle. The real main file stays at one canonical location; secondary locations contain a native Finder alias on macOS. The corresponding secondary path under Associated Files is a directory alias to the canonical associated bundle, rather than another copy of its contents. A paper without associated resources does not need an empty bundle. Other supported platforms use filesystem links. **Make Copy Desktop** resolves these pointers into real files for a portable export.

A surviving normal Library, Space or My Publications placement keeps the original outside the local Recycle directory. Deleting or dragging a folder checks each paper; restoring or re-adding a recycled paper migrates its original back to normal storage. SQLite resource paths and secondary shortcuts follow. See Delete and Recover Items.

The canonical location stays stable while it is still an active placement. If that location is removed or moved, Gewu Insight uses a surviving Library, Space or My Publications placement and updates its aliases. External originals, files shared with another paper, and occupied destinations are protected; do not replace an unrelated file to force a layout change.

> **Important**
> Do not manually move an Associated Files bundle or replace an alias in Finder. Use Gewu Insight so the database and physical paths change together.

See also: Storage Reference · Make a Portable Desktop Copy · Understand Paper Files

3-9.10 Local Data and Privacy

Gewu Insight runs a local interface and keeps its authoritative library database on your Mac. Network requests occur only for the features that need external metadata, pages, profiles, or files.

Local components

- The UI server listens on the local loopback address, normally `127.0.0.1:8765`.
- SQLite stores papers, identities, relationships, settings, and operational state.
- Browser bookmarks, folders, history, and sidebar preferences also live in SQLite. Browser cookies and cache use a separate persistent WebKit profile; they are not shared with Safari or exported with the database.
- Managed PDFs, attachments, photos, and generated reports stay in your selected Gewu Database. The App, source code, browser profile and machine preferences have separate locations.
- JSON files are exports or diagnostics; they are not the authoritative database.

External requests

Metadata refresh, citation counts, journal/subject/author discovery, profile images, website opening, and downloads can contact their named services. The app paces services that require strict request limits.

Protect credentials

Optional service credentials belong only in Literature Data Source or approved environment variables. They must not appear in a screenshot, support message, Git commit, or exported diagnostic file.

Apple Passwords stays under macOS control. Gewu's refusal to persist webpage WebCrypto keys is a separate Browser compatibility policy; it does not turn off normal webpage encryption or clear cookies. See Website keys and macOS authorization for the effect on sites that rely on saved keys.

See also: Literature Data Source · Back Up or Move to Another Mac · Publisher and Network Access

3-10 Zotero Integration

3-10.1 One Workspace, Two Interfaces

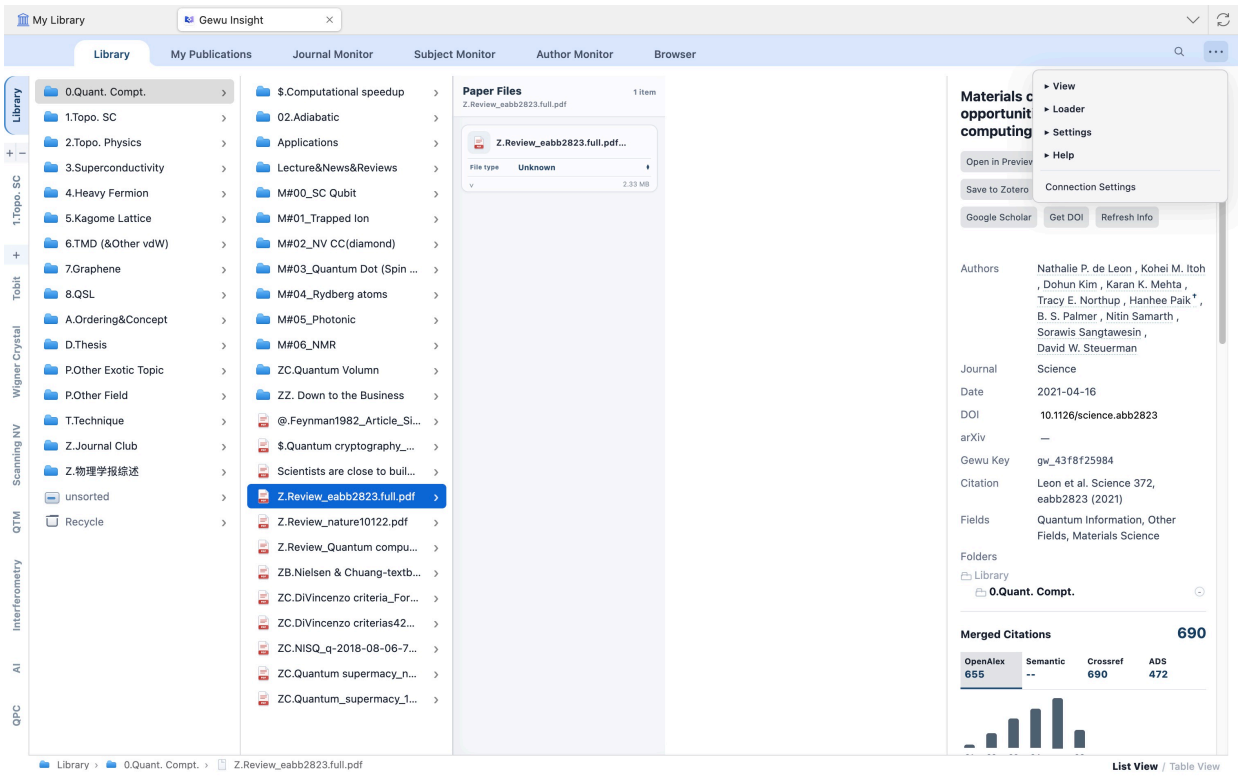
Keep one Gewu Insight installation and one selected Gewu Database on your computer. Choose whether to work in the standalone Gewu App or in its Zotero tab.

One Gewu, not two copies
The plugin is a companion interface to your existing Gewu Insight. Zotero displays the workspace and adds actions such as Save to Zotero; Gewu's local background service still runs the workspace. Installing the plugin does not create another Gewu Library, duplicate your Paper Files, or require a second set of monitor settings. Keep Gewu Insight installed even if you normally use it through Zotero.

What stays shared

Part of your work	What happens when you change interfaces
Gewu Library, Spaces and Paper Files	Both interfaces read and change the same records, folder placements and files in the selected Gewu Database.
Monitors, My Publications and settings	Saved journals, subjects, authors, profiles, resource API settings and preferences remain available in either interface.
AI Access and MCP	Existing connections and their permissions remain attached to the same Gewu workspace. The plugin does not require an AI connection or grant additional access.
Browser page tabs and website logins	These belong to each interface. You may need to sign in to a publisher again inside Zotero.
Zotero's own library	This is a separate collection. Save to Zotero copies a Gewu paper into it; Import from Zotero copies selected Zotero content into Gewu. Neither is continuous synchronization.

For example, add a journal in Gewu's Journal Monitor inside Zotero, then close that tab and open the standalone App: the journal remains monitored. Change a saved setting in the standalone App, then open the same workspace in Zotero: that setting remains in effect. These are changes to the same data, not a synchronization process between two Gewu databases.



The same workspace in a Zotero tab. Library, My Publications, all three monitors and Browser remain available. The upper-right three-dot menu contains View, Loader, Settings, Help and Connection Settings. Zotero's My Library tab is separate.

Open and switch interfaces

1. After installation, choose **Plugin > Zotero Integration > Open in Zotero** in the standalone App. This prepares the connection to the database you are using.
2. Once Zotero confirms that the workspace has opened, the standalone App quits. The Gewu background service keeps running, so files, settings and monitors remain available. If opening fails, the App stays open.
3. For later sessions, use **Tools > Gewu Insight** in Zotero. With background startup enabled, the plugin starts the installed Gewu service when needed; you do not need to open the standalone window first.
4. To return to the standalone App, close the **Gewu Insight** tab in Zotero, then open Gewu Insight. You can leave Zotero itself running.

Why the standalone window may be paused

While a connected Gewu tab is open in Zotero, the standalone main window and its editing menus pause. This prevents editing through a stale view in two windows. Closing the Gewu tab refreshes the standalone view before restoring interaction. Switching to Zotero's My Library tab is not the same as closing the Gewu tab.

This pause affects the interface; background monitoring and jobs can continue. Closing the Zotero tab also leaves the shared service running. It does not delete your workspace or require exporting and importing your data.

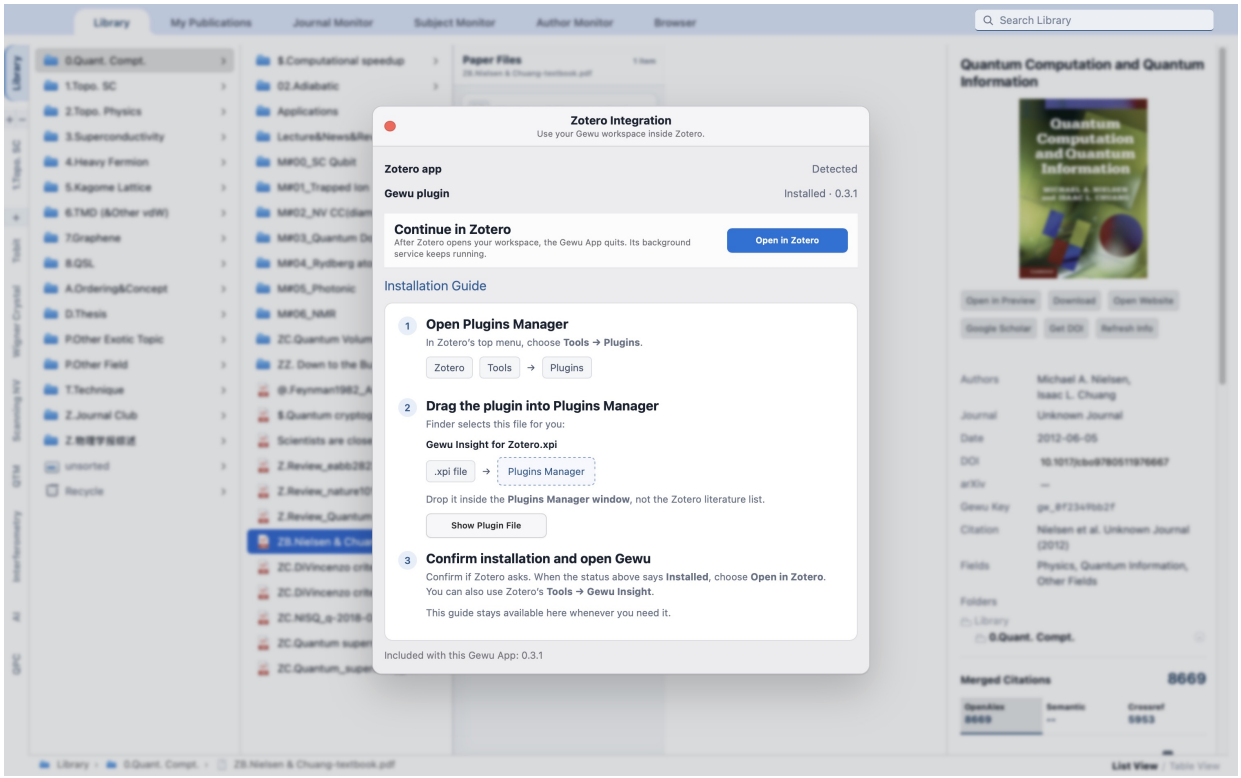
See also: Install, Update, and Remove the Plugin · Save Papers and Preview PDFs · Connection Settings and Troubleshooting

3-10.2 Install, Update, and Remove the Plugin

Start in the standalone Gewu App: choose Plugin > Zotero Integration. The installation guide stays available in that window whenever you need it.

Install for the first time

1. Install and open Zotero. In Gewu, open **Plugin > Zotero Integration**. **Zotero app: Detected** means that Zotero is present; it does not mean the Gewu plugin is installed.
2. Click **Install in Zotero**. Gewu opens its persistent guide and reveals **Gewu Insight for Zotero.xpi** in Finder. This XPI file is the plugin package bundled with your Gewu App.
3. In Zotero's top menu, choose **Tools > Plugins** to open **Plugins Manager**.
4. Drag the selected XPI file from Finder into the **Plugins Manager window**. Do not drop it into Zotero's literature list: that adds a file attachment rather than installing a plugin. If Finder is no longer open, click **Show Plugin File** in Gewu's guide.
5. Confirm Zotero's installation prompt. Check that **Gewu Insight** is enabled in Plugins Manager. When Gewu's integration window reports **Installed**, choose **Open in Zotero**.



The persistent installation guide. This example already has plugin 0.3.1 installed, so the primary button is Open in Zotero. Before installation it is Install in Zotero. The three steps and Show Plugin File remain available afterwards.

Understand the primary button

Button	When to use it
Install in Zotero	No installed plugin is detected. Follow the guide and confirm installation in Zotero.
Open in Zotero	The plugin is installed and ready for everyday use with the current Gewu database.
Update Plugin	Your Gewu App includes a newer plugin. Replace the installed companion with that bundled version.

Update an existing plugin

After updating Gewu Insight, check this window. If it offers **Update Plugin**, click it and follow any displayed instructions. For an installed 0.2 or later plugin, the update is applied without dragging the XPI again. Previously saved Zotero items and Gewu data are retained.

Update Plugin uses the package already included with your Gewu App; it is not an online update check and is not required every time you open Zotero. Reinstalling after removal, or replacing the earlier 0.1 preview, uses Plugins Manager again. A future Gewu change may require a companion update; compatibility is checked when the plugin connects.

Disable or remove it yourself

In Zotero, open **Tools > Plugins**, find **Gewu Insight**, and use its menu to disable or remove the plugin. Close any remaining Gewu tab. Removing the plugin leaves your Gewu Database and previously saved Zotero items in place. To install it again, follow the first-time steps above.

Installation and updates are managed from the standalone App. The embedded Gewu menu in Zotero does not contain another Plugin > Zotero Integration window.

> **Preview availability**
> This chapter documents companion 0.4.0, validated with Zotero 10.0.6 on macOS. Zotero 7/8 and Windows companion workflows have not been validated. An older Gewu installer may not include this preview; the installed App's integration window is the authority for its bundled plugin version.

See also: One Workspace, Two Interfaces · Connection Settings and Troubleshooting

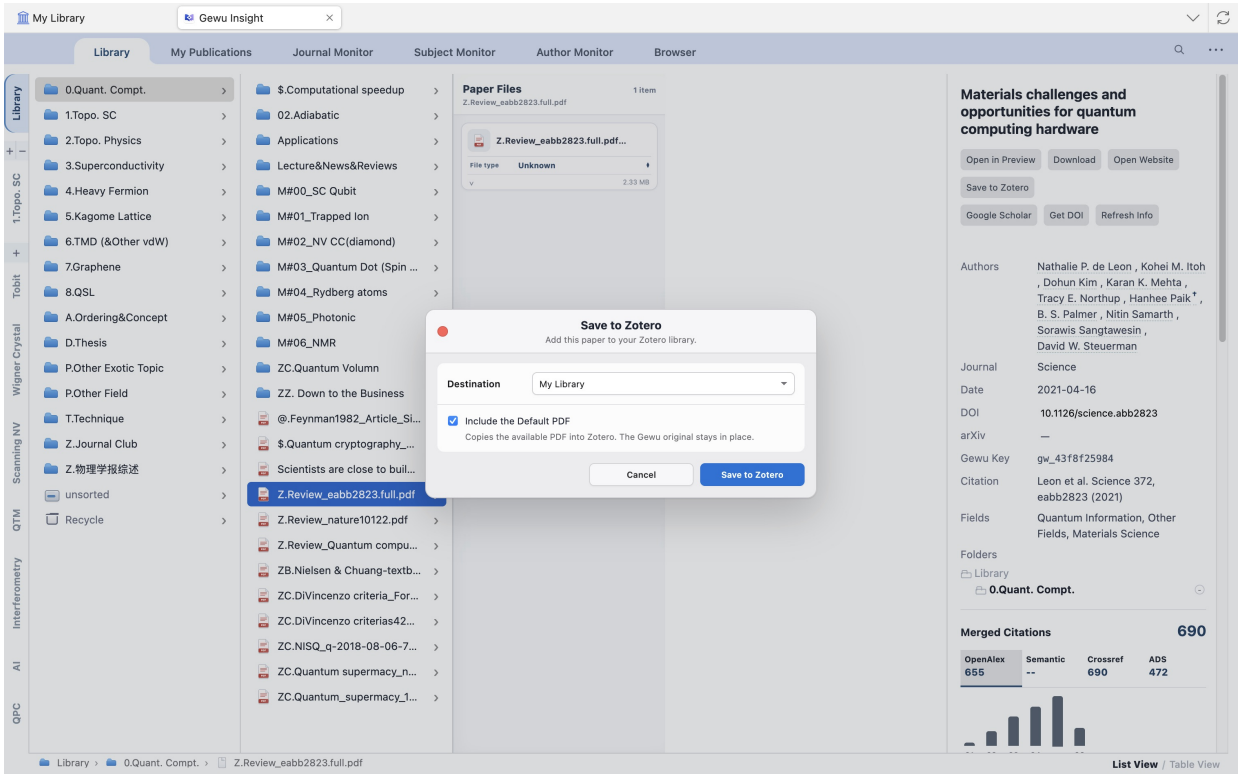
3-10.3 Save Papers and Preview PDFs

Choose where to keep a paper. Saving to Gewu uses your existing Gewu Library; Save to Zotero creates or reuses an item in Zotero's own library.

Action	Result
Save to Gewu / Add to Library	Use the normal Gewu destination and Paper Files workflow. The paper is available in the standalone App because it uses the same database.
Save to Zotero	Copy the paper's metadata into a selected writable Zotero library or collection, optionally with its available Default PDF.

Save a paper to Zotero

1. Find a paper in a Gewu monitor or select an existing paper in the Gewu Library, then click **Save to Zotero**.
2. Choose the destination library or collection in the dialog. Only destinations you can write to are available.
3. Leave **Include the Default PDF** checked to copy the existing local Default PDF, or uncheck it to save metadata only.
4. Confirm **Save to Zotero**. Read the result notification; **Show in Zotero** reveals the saved item in Zotero's library.



Save to Zotero. Choose a destination in Zotero's own library. Include the Default PDF copies the available file and leaves the Gewu original in place. Cancel closes the dialog without saving.

The PDF is copied, not moved. Gewu retains its original and the rest of its Paper Files. This option does not copy every attachment, supplementary file or analysis note. If no local Default PDF is available, the item is saved with metadata only. A copy failure is reported separately so you can retry it.

To obtain a missing PDF, use Gewu's normal download workflow first, then save again. Publisher access requirements still apply. Repeated saves try to reuse a matching Zotero item and add it to the selected collection while preserving existing Zotero metadata edits and PDFs. Conflicting matches are reported instead of automatically merged.

> **Two libraries are not automatically synchronized**
> Save to Zotero is an explicit, one-way copy. Later edits, annotations, notes or attachment changes in Zotero do not flow back into Gewu, and later Gewu edits do not continuously update the copied Zotero item. Opening Gewu inside Zotero still uses the shared Gewu database; it does not merge the two products' libraries. To copy existing Zotero collections, files and notes into Gewu, follow Import a Zotero Library. That explicit import is also separate from continuous synchronization.

Read without importing into Zotero

In the Gewu Library or Paper Files, select a PDF and press `Space` for Quick Look. Scroll through the pages, click thumbnails to jump, or select text. Press `Space` or `Esc` to close the preview. **Open in Preview** opens the macOS reader.

The companion renders PDFs inside Gewu's preview using Zotero's PDF components. Previewing does not create a Zotero item or copy the file into Zotero storage. The standalone Gewu App keeps its own native preview behavior.

See also: Understand Paper Files · Preview Files with Quick Look · One Workspace, Two Interfaces

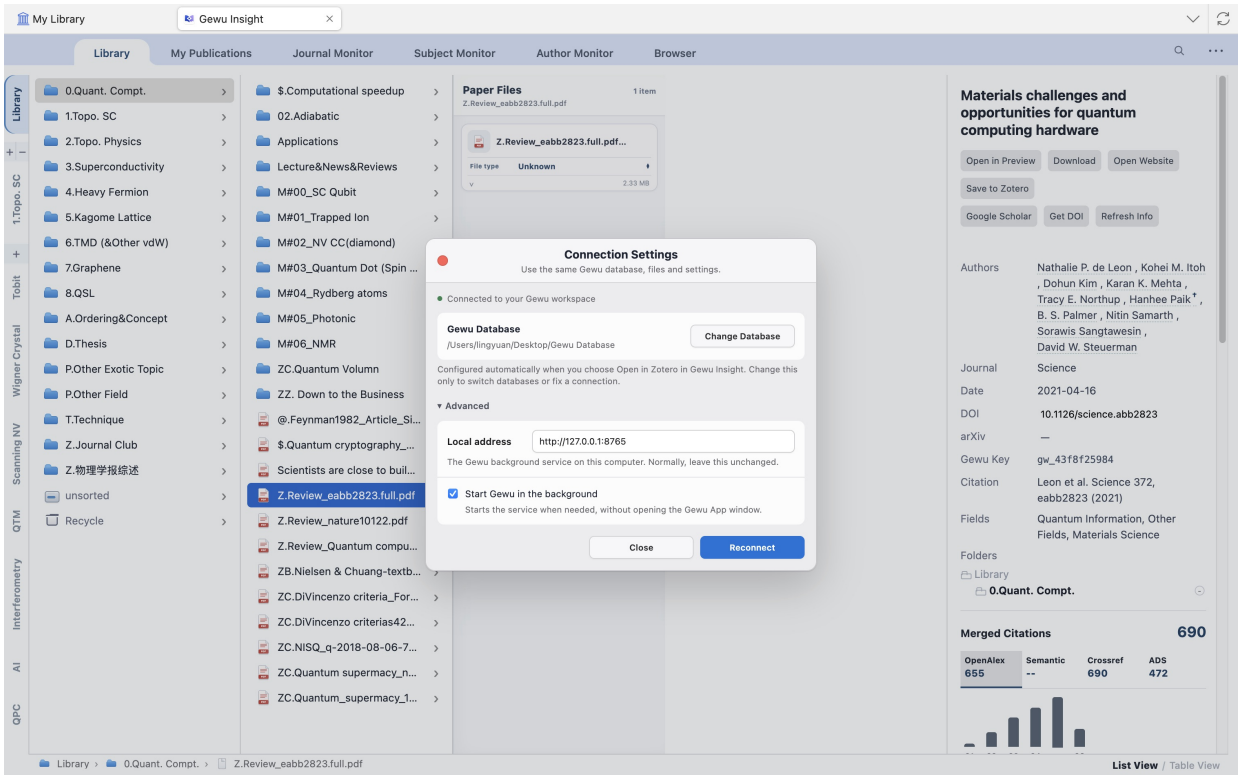
3-10.4 Connection Settings and Troubleshooting

Connection Settings tells the Zotero companion which existing Gewu workspace to open and how to reach its background service on this computer. Normal use does not require changing these settings.

To bring Zotero's own library into Gewu, use Loader > Import from Zotero. Its Connect Zotero action connects a source library for import; this Connection Settings dialog selects which existing Gewu workspace the tab displays.

Open the connection dialog

In Zotero's Gewu tab, click the upper-right **…** menu, then **Connection Settings**. The standalone App's **Open in Zotero** action normally configures this connection for you.



Connection Settings. This connected example has Advanced expanded to show the local service address and background startup option. The displayed database path belongs to this Mac; use your own existing Gewu Database. Routine use does not require changing these fields.

Control	Meaning and use
Gewu Database	The folder containing the Gewu papers, Paper Files and settings you want to use. Change Database selects an existing database root, not its data subfolder. This does not copy or migrate a database.
Advanced > Local address	The address of Gewu's service on this computer, commonly http://127.0.0.1:8765. It is not a Zotero account or cloud-server address. Leave it unchanged unless troubleshooting a known local service configuration.
Start Gewu in the background	Allow the plugin to start the installed Gewu service when needed, without opening the standalone App window. Keep this enabled for routine use through Zotero.
Reconnect	Retry the connection using these settings. It does not import papers, merge libraries, or create an empty replacement database.

If something looks wrong

- **The Library is empty or different.** Compare Gewu Database in this dialog with **Gewu Insight > Database Location…** in the standalone App's macOS menu. Close the Gewu tab, open the App with your usual database, then use **Plugin > Zotero Integration > Open in Zotero** again. Do not create a fresh database to repair a connection.
- **The workspace does not connect.** Check that Gewu Insight is still installed, the selected database folder is accessible, and background startup is enabled. Try Reconnect; use the standalone App to inspect a startup error if necessary.
- **The Gewu tab remains but its menu is missing.** Check Tools > Plugins for an enabled Gewu Insight plugin. Close an old tab left after uninstalling, then open a new one through Tools > Gewu Insight. Reinstall through the guide if the plugin is absent.
- **The standalone window cannot be edited.** Close the connected Gewu tab in Zotero and allow the App to refresh. Merely switching to another Zotero tab does not end the companion session.
- **A PDF asks to download instead of previewing.** Check the installed plugin version and update from the current Gewu App. The current companion provides an embedded PDF preview; no global Zotero PDF setting needs to be changed.

Existing source API settings and AI Access permissions belong to the Gewu database. Reconnecting does not require entering them again, creating a new MCP connection, or widening its permissions.

See also: Install, Update, and Remove the Plugin · Startup, Monitor Cache, and Text Size · Back Up or Move to Another Mac

Part IV.

Product Identity

4-1 Visual Identity and Cultural Elements

The name Gewu Insight (格物洞见) expresses a way of doing research shaped by the practice of physics: begin with the real world, and seek to see what lies behind its phenomena.

Gewu (格 物) means investigating things. Here, it emphasizes starting from concrete phenomena, experimental observations, and questions that nature presents to us. Research should remain in contact with the world it seeks to understand. Abstract ideas and theories gain their purpose from that contact; they should not become castles in the air, sustained only by further abstraction.

Insight (洞见) means seeing through phenomena to the patterns and mechanisms beneath them. It values physical pictures, intuition, imagination, and the ability to make an insightful guess—to recognize what may be happening before every step has been expressed in equations. A useful physical picture can reveal which features matter, suggest a mechanism, and open a direction that formal derivation alone might not make apparent. Mathematics serves as a tool for thinking, expression, and calculation. Absolute formal rigor, axiomatic completeness, and mathematical elegance are not ends in themselves, nor should they constrain the search for physical understanding.

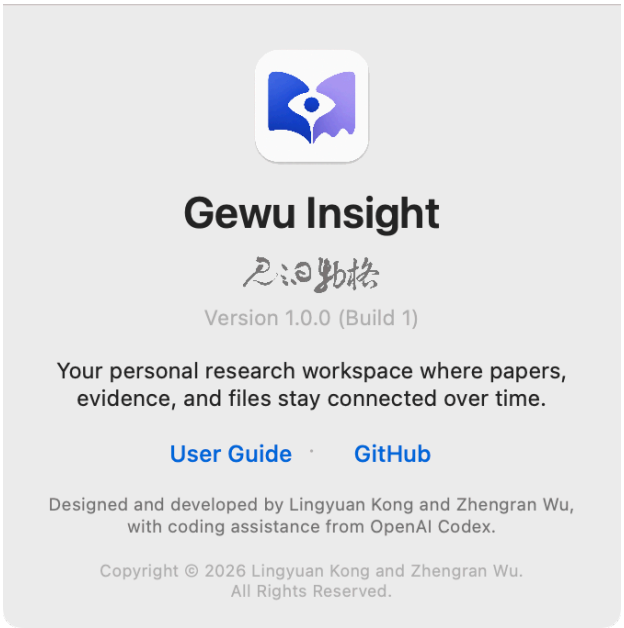
This outlook also shapes the purpose of the software. Research literature brings us into contact with other researchers' experiments, observations, interpretations, and ideas. Gewu Insight keeps these materials connected: a paper retains its identity across versions, its figures and supporting files remain accessible, and related work and research notes can be organized together and revisited over time.

These functions support the everyday work from which insight can emerge: returning to an experimental result, comparing competing physical pictures, noticing an unexpected connection, or recording a conjecture worth pursuing. The researcher brings the questions, intuition, and judgment; the workspace helps keep the material close at hand and the developing line of thought connected to its sources.

> Investigate nature through experimental phenomena; seek insight through physical pictures and intuition.（从实验现象中格物，以物理图像和直觉求洞见。）

The calligraphic Chinese name, the app icon, and the seal-script watermark give this research outlook a visual identity.

The marks do not decorate every surface. Each has a different job: the icon identifies the application, the calligraphic name gives the product a human signature, and the quiet seal-script pattern belongs to transitional or otherwise empty space.



About Gewu Insight. Open the application menu and choose About Gewu Insight to see the product name, version and calligraphic signature.

4-2 The calligraphic name

The transparent production artwork used in the About panel. From left to right the picture contains 見、洞、物、格; read in the traditional direction from right to left, it says 格物洞見.

This is a 祝枝山行草集字 composition, assembled from separately acquired character images attributed to the Ming-dynasty calligrapher **Zhu Yunming (祝允明, 号枝山)**. It is a modern collection and arrangement of character forms, not "a title written by Zhu Zhishan." No surviving single inscription titled 格物洞見 is being claimed.

4-2.1 Historical sources of the four characters

Reading order	Character	Attributed hand and script	Original work
1	格	祝允明 · 行书	Original work not yet identified.
2	物	祝允明 · 草书	Original work not yet identified.
3	洞	祝允明 · 草书	草书前后赤壁赋卷, Shanghai Museum.
4	見	祝允明 · 草书	Original work not yet identified.

The selected 洞 character matches the form in 客有吹洞簫者 in the first *Red Cliff Rhapsody* of Zhu Yunming's 草书前后赤壁赋卷, held by the Shanghai Museum. This identification follows comparison with the complete scroll, rather than the mere occurrence of the same character in a text. The original works for 格、物、見 remain to be established.

The selected glyphs were optically balanced into a common line: scale, spacing, ink tone, edge cleanup, and transparent alpha were normalized while the characteristic strokes were retained. The production preparation used a Codex-assisted image workflow. The interface and accessibility text use the contemporary simplified product name 格物洞见; the artwork deliberately preserves the traditional character 見 and traditional right-to-left composition.

> **Provenance and reuse**
> Historical authorship and the modern arrangement are distinct. The museum reference identifies an underlying work; it does not claim that the production image was extracted directly from the museum's photograph. The remaining original-work identifications are still open. Gewu's software License grants only rights held by the project, not independent rights in external reproductions.

4-3 The app icon and the seal-script watermark

4-3.1 The app icon

The icon's two blue-violet fields read simultaneously as an open book and two sides of a field of view. The white eye and central point suggest observation and focus; the tapered center also recalls a pen nib or the spine of a book.

This description is the product's visual reading of the finished mark, not a claim that every interpretation was specified in an original design brief.

4-3.2 The seal-script watermark

The seal-style motif was generated with AI assistance for Gewu Insight. Read down the right column and then down the left, its four characters are 格、物、洞、見. The selected letterforms form a modern interpretation of seal-script composition, with the traditional character 見.

The motif appears as a faint gray watermark in startup transitions and quiet empty space. It is a contemporary screen graphic rather than a reproduction of a historical carved seal.

4-4 Why the three marks are used differently

Mark	Role
Icon · recognition	Use at application scale: Dock, startup identity, About, and platform packaging.
Calligraphy · signature	Use where the Chinese name can be read deliberately, especially the About panel and identity materials.
Seal script · atmosphere	Use at very low opacity in startup transitions or quiet lower-right space, never where it competes with paper details.

Together the three registers establish the product's visual identity: the icon is contemporary and operational; the calligraphy is personal and gestural; the seal-script watermark is historical in reference but restrained in use. Their common purpose is to make the name visible without turning the research workspace into a themed illustration.

See also: Welcome · The Personal Research Workspace · Startup, Monitor Cache, and Text Size

Part V.

Technical and Operational Reference

5-1 Keyboard Shortcuts

Use standard macOS editing commands throughout the app. The shortcuts below are specific to navigation and Gewu Insight commands.

Shortcut	Action
↑ / ↓ / ← / →, Enter	In Browser chrome, move between the top workspace tab, toolbar, webpage tabs, Favorites, and visible sidebar; Enter activates the focused control. Native webpages retain scrolling and editing keys. Up at an ordinary document's top can return to Browser chrome; ⌘L then Escape also returns to the toolbar.
↑ / ↓	Move through the focused Library or Paper Files list; on the source rail, move among Library, shortcuts, and Spaces and open the highlighted source.
→	Enter the highlighted source from the vertical rail, enter a folder, or move from a selected paper into Paper Files.
←	Leave Paper Files for the selected paper, return to a parent folder, or move from the leftmost Library column to the source rail.
Space	Open or close Quick Look for the selected local paper/Paper Files item, including available monitor PDFs; not while editing text.
Esc	Dismiss an active transient control or return focus where supported.
⇧⌘V	Toggle List View and Table View.
⌘L	Focus the address/search field in Browser; otherwise open Folder Loader.
⌘D	In Browser, bookmark the current page.
⌘R	In Browser, reload the current page.
⌘F	In Browser, Find on Page. In Library, My Publications, and the three Monitors, focus the current workspace search.
⌘T	In Browser, open a new webpage tab at Home.
⌘[	Return from Browser to the source workspace; keep its position and webpage tabs.
⌘,	Show Literature Data Source.
⌘W	In Browser, close the selected webpage tab; closing the last creates Home. Outside Browser, dismiss active Quick Look first, otherwise close the current window.
⌘Q	Quit Gewu Insight. Save webpage drafts first; open Browser tabs are not restored after quitting.

5-1.1 Monitor keyboard navigation

In a Journal or Subject calendar, Left/Right moves one day and Up/Down moves one week. Enter selects the date and enters the papers. In the paper area, Up/Down selects papers, Left/Right changes the journal or subject, and Escape returns to the calendar. Space opens an available local PDF in Quick Look.

In Author Monitor, Left selects the author rail and Right selects its papers; Up/Down moves through the active pane. Up from the first item enters the main-tab navigation level. At that level, Left/Right chooses a main tab and Down or Enter returns to its content. These commands do not apply while typing in an input.

5-1.2 Browser keyboard context

Find on Page uses Enter for the next match, Shift-Enter for the previous match, and Escape to close. In address suggestions, Up/Down selects a result and Escape restores your query. A focused webpage tab uses Left/Right to change tabs. These contexts are separate from Library navigation; ⌘L in Browser does not open Folder Loader.

Paper Files mouse behavior: single-click selects, double-click opens, and Control-click opens the context menu.

5-1.3 Selection and unread state in Discovery

In Journal and Subject Monitor, Up/Down selection marks only the selected paper read, just as clicking that card does. Entering a date or journal tab is not a Read All action. The tab's blue dot disappears when no currently visible unread papers remain; unread papers hidden by filters keep their status.

See also: Navigate with Mouse and Keyboard · Understand Paper Files

5-2 Publisher and Network Access

Publishers can rate-limit shared VPN addresses or automated-looking traffic. Gewu Insight reduces overlap and request rate, but it cannot remove a publisher block.

5-2.1 If a publisher page is blocked

1. **Stop repeated refreshes.** Additional requests can extend a temporary block.
2. **Try your direct connection.** If the page works with VPN off, investigate the VPN route, quota or exit address before treating it as a product or certificate failure.
3. **Use Open Website or an external browser.** Complete any normal institutional login or publisher interaction yourself.
4. **Download only when authorized.** Add a legitimately obtained PDF with **Add File**.

5-2.2 Why figures may arrive later

Figure and article-page retrieval is queued behind discovery work and uses caching. This reduces simultaneous RSS, page, and image requests to the same publisher. A paper can appear before its representative image.

5-2.3 Browser and your system network

The Browser tab uses native WebKit requests and does not proxy pages through the Python server. System proxy or TUN capture can send its traffic into Clash; only then do the active rules determine DIRECT or proxy routing. A DIRECT rule uses your current physical network, which must itself provide institutional access. Gewu Insight neither changes these rules nor guarantees a particular route; use Clash connection records to verify the actual route. See Browse, Bookmark, and Revisit Websites for downloads, sign-in limits, and the separate browser profile.

5-2.4 Check access to the actual main PDF

An article's title and abstract can be public while its PDF requires a subscription. A working webpage, a DIRECT rule, or a successful open-access download does not establish access to every article. Away from campus, direct traffic uses your present network; use your institution's authorized access method when needed. Gewu does not force Science through a proxy or change Clash/VPN rules.

Article pages, PDF services and content-delivery hosts may follow different network routes. APS Harvest is the first direct APS download route; a denied or unavailable response can fall back to background browsing. Science, Nature and ACS also have supported background fallbacks in the native App. These attempts share Gewu Browser's session, but cannot supply missing institutional rights or complete human verification. A clear access-required or timeout message with preserved files is preferable to saving an HTML page as a PDF.

5-2.5 Secure-connection failures and VPN interruptions

A secure-connection error such as `NSURLErrorDomain −1200` means WebKit could not complete a secure connection; it does not by itself establish that the site's certificate has expired. Check the network, VPN quota or route, and the Mac's date and time. After restoring the connection, reload the page once and retry the failed download if needed. A change of VPN node can restore the route without any change to Library data. Certificate verification stays enabled.

5-2.6 Service rate limits

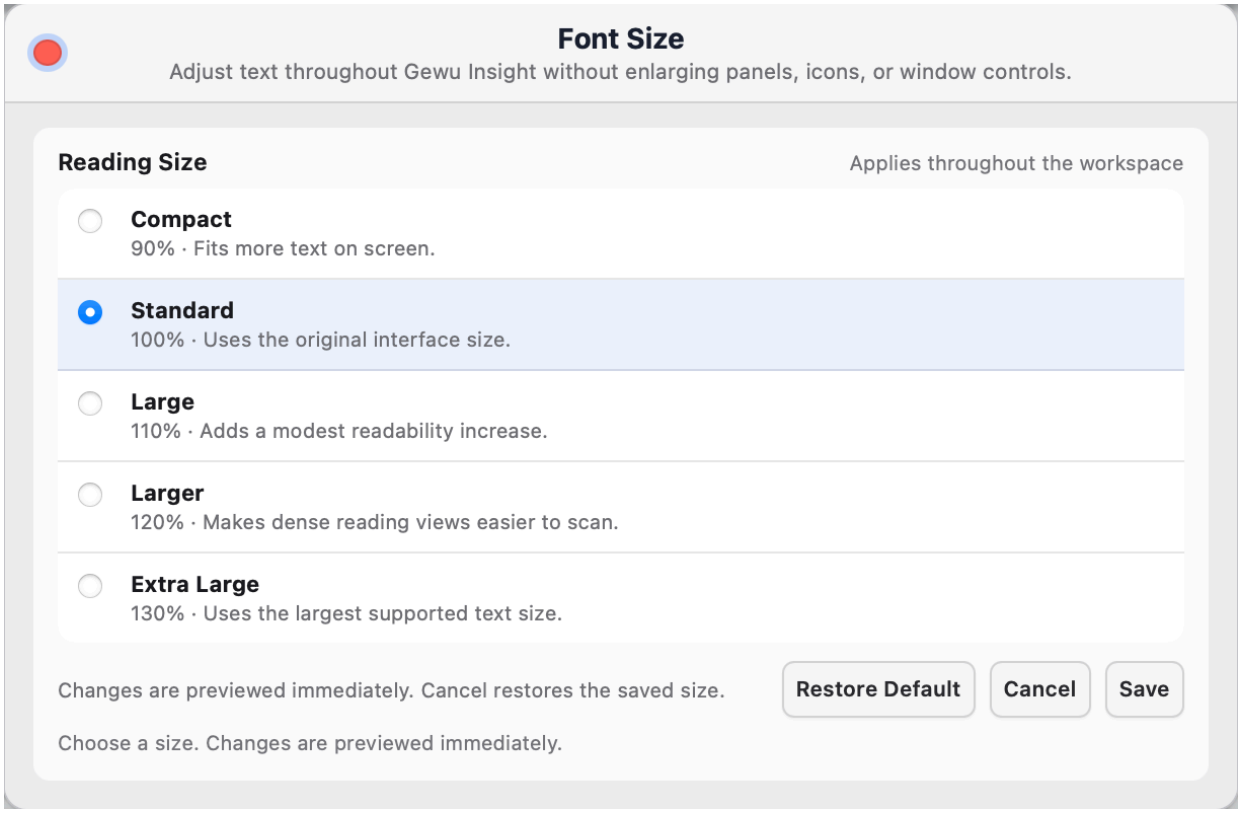
Semantic Scholar, Google Scholar, and some publisher sources use shared pacing, cooldowns, or exponential backoff. Let the bottom status area settle before starting the same action again.

> **Important**
> Do not attempt to bypass a paywall, CAPTCHA, browser verification, or access control. Use your authorized browser session or attach a file you obtained legitimately.

See also: Journal Monitor · Download and Sync Versions · Troubleshooting

5-3 Startup, Monitor Cache, and Text Size

The macOS app shows its branded startup view immediately, starts or connects to the local UI service, and reveals the workspace when Library is ready.



Font Size. Preview one of five reading sizes. Save keeps the selected size; Cancel restores the previous preference.

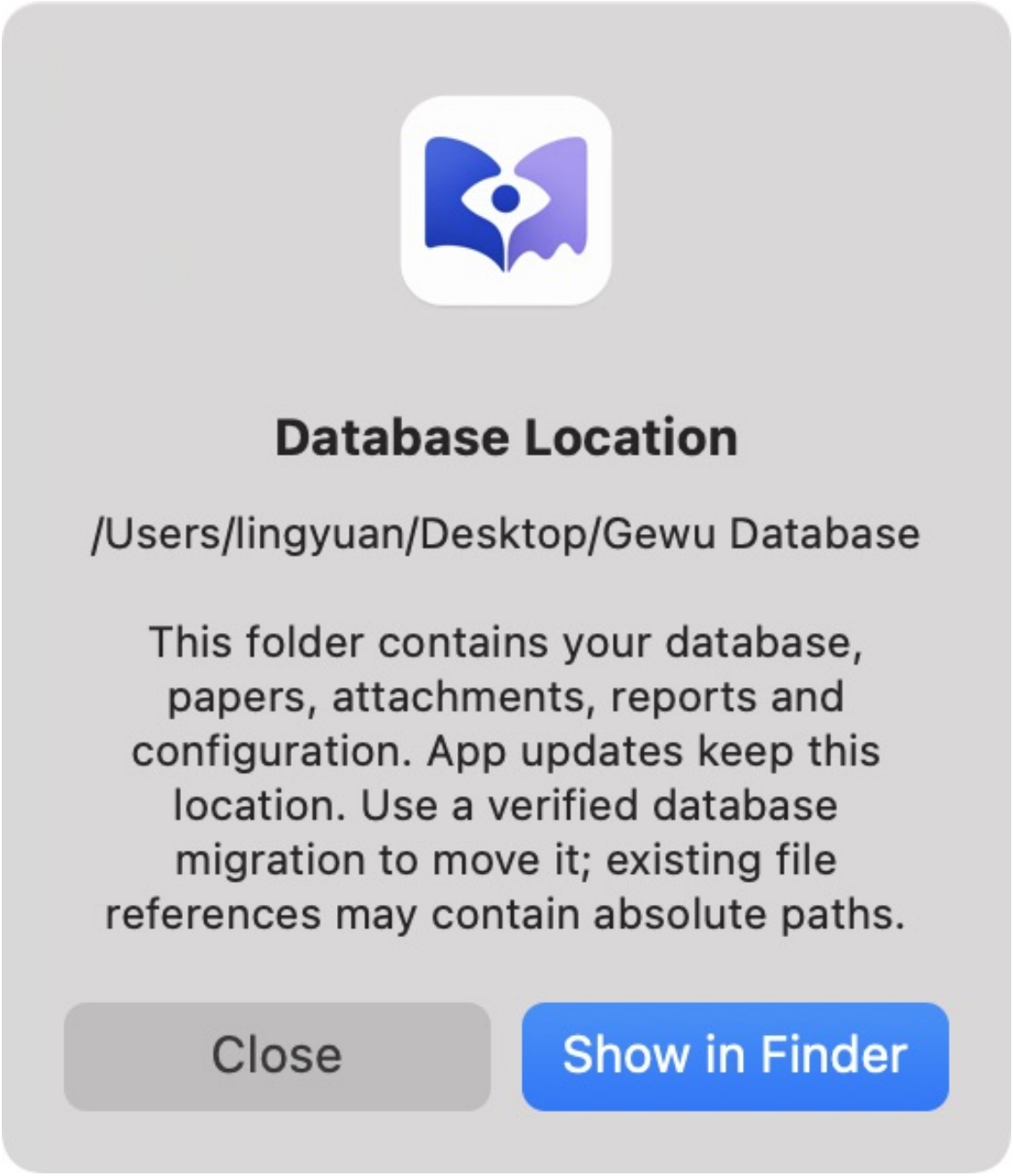
5-3.1 Before the first launch

The native App requires macOS 14 or later. Open the installed `/Applications/Gewu Insight.app`. A standalone package includes Python and its required dependencies; a source-backed installation uses its configured source checkout and Python environment, which must remain available. Follow the instructions supplied with your installation. An unconfigured source-build App cannot become standalone merely by moving its bundle.

On first launch, choose **Open Existing Database** or **Create New Database** as described in Your First Session. Gewu remembers the selected research folder separately from the App. Network access is needed for online discovery and downloads; it is not needed simply to choose an existing local database.

5-3.2 Find the database you are using

Choose **Gewu Insight > Database Location…** in the macOS menu bar. The dialog shows the active research folder; **Show in Finder** opens it. **Close** returns to your work. This dialog displays the location; it is not a Move or Switch Database command.



Database Location. The original App shows the current research folder. This desktop path is an example from one Mac; use the path displayed by your own installation. The dialog explains that moving a populated database requires verified migration.

If the remembered folder is missing, reconnect its disk or use **Locate Your Gewu Database** to choose the existing workspace. The saved location is retained until another database opens successfully. A mismatched database identity, a database requiring a newer App, another process holding it open or an occupied local service port stops startup with an explanation. Preserve the intended database and resolve the stated condition.

5-3.3 Keep the current App in the Dock

After moving from the old project App to Applications, the Dock may still show the old shortcut alongside the running App. Right-click the old shortcut and choose **Options > Remove from Dock**. Open Gewu Insight from Applications, then right-click its running icon and choose **Options > Keep in Dock**. Removing the shortcut does not delete the App, database or backup. Confirm Database Location before using the new shortcut for daily work.

For an existing installation, follow Back Up and Upgrade before replacing files. The manual's edition date and companion-plugin version can change independently of the App version; a newer manual does not require a fresh Library.

5-3.4 Normal startup

The first window frame contains the warm background, packaged icon, and Starting Gewu Insight panel. The native startup view is installed before the window is shown, so an empty white web background does not precede the branding. The panel stays visible while the local service and first Library view prepare, then fades into the workspace.

The first launch after an additive database upgrade can take longer because the existing SQLite records are migrated and older provenance or file metadata is backfilled. The native launcher allows up to 120 seconds for this preparation. Keep the single startup panel open rather than launching a second copy; the migration preserves the existing database and managed files.

The Dock tile, native startup panel, and in-window loading state use the icon packaged with the same app build. Startup has a native launch phase and a local-web bootstrap phase, but the visual handoff should not introduce a second icon frame or fall back to an older product mark.

5-3.5 Startup is not a Library cleanup

Ordinary startup does not reorganize the whole Library, merge historical duplicate files, or make a new full database backup. A pre-schema SQLite snapshot is created only when a schema upgrade is needed; it does not include the research files. Managed layout and duplicate protections apply when you import or change the affected paper and files. Keep your own regular backups.

If a file move was interrupted before its database and disk changes finished together, startup recovers only the recorded unfinished operation before background workers start. A launch with no unfinished move goes directly to normal preparation. An ambiguous conflict is reported and preserved for repair rather than overwriting a destination or triggering a full-library sweep.

5-3.6 Automatic local-server recovery

If the local UI server exits unexpectedly while the macOS window remains open, the launcher starts it again. The window rechecks local health and reloads the workspace when the service becomes ready. A startup-failure sheet is nonblocking; if the server recovers after that sheet appears, the stale sheet is dismissed automatically instead of continuing to interrupt a healthy workspace.

A brief reload during recovery does not mean the SQLite database was rebuilt or that another copy of Gewu Insight should be opened. If repeated recovery never reaches a healthy Library view, use the launcher log steps below.

Recovery reloads the workspace, so Browser's memory-only tabs and unsubmitted forms can be lost. Bookmarks and history already saved in SQLite remain. Save work on websites regularly; recovery is not a web-session or draft backup.

If the local workspace rendering process exits while the service is healthy, the native App can reconnect its private workspace session and rebuild the interface without restarting background jobs. Recovery attempts are bounded. This addresses a blank workspace or an “Open Gewu Insight to access this workspace” page; it does not grant a website access to the Library or reset your database. If recovery repeats, preserve the launcher log and report the displayed version/build.

5-3.7 When automatic updates run

Automatic monitoring runs inside the local UI service, with startup delays, periodic checks, and saved completion/retry state. Keep the Mac awake and the service running. Closing a window does not necessarily stop that service. Quitting the App stops a server it started; a separately launched server must be stopped separately. A late launch checks work that is due rather than guaranteeing a run at an exact minute.

Monitor	Current automatic schedule
Journal Monitor	Daily work becomes due from 09:00 in the Mac's local time zone; stored completion and pending retry state determine what still needs work.
Subject Monitor	Daily work becomes due from 09:00 local time; incomplete sources have their own bounded retry state.
Author Monitor	Checks due authors from 10:00 local time; already checked authors are not blindly fetched again on every scheduler tick. Failed transport checks use a separate paced retry queue, including before 10:00. Unrecognized Scholar pages use an hourly delay and a daily failure cap; access cooldowns still apply. Opening an author does not force a full profile fetch each time.
arXiv	Uses its separate announcement-ready schedule and buffer, not the journal's 09:00 rule. Read the displayed update/countdown state.

These are local scheduler rules, not guarantees of remote source completion or a hosted monitoring service. Manual Refresh remains available for the selected scope; repeated broad refreshes can worsen a source's rate limit.

5-3.8 If the app cannot start

1. **Quit any old Gewu Insight window.** An earlier local server may still own port 8765.
2. **Open the launcher log.** For the installed App, check `~/Library/Logs/Gewu Insight/paper_monitor_launcher.log` for the first error. A legacy source launcher may use its configured log directory.
3. **Allow the first startup to finish.** After an upgrade, wait through the migration window before treating the preparation panel as a failure.
4. **Start the app again once if recovery does not complete.** Avoid opening several copies while the first is still preparing.
5. **Preserve the database.** Do not delete SQLite as a startup fix.

5-3.9 Clear a stale interface

Choose **Setting > Clear Monitor Cache** if a monitor still shows stale downloaded images or rendered web assets after an app update. This clears disposable monitor presentation cache, not the SQLite database, saved paper records, or managed PDFs.

This is not Browser's Clear History command and does not clear Browser sign-in cookies or Apple Passwords. Use **Browser > History > Clear History** for visit records only.

Monitor cache and macOS app identity are separate. Clearing Monitor Cache cannot repair a stale Dock tile. After replacing the app bundle, quit every running copy and open the rebuilt **Gewu Insight.app**. If a permanently pinned Dock tile still points to an older bundle at another path, remove that tile and add the current app again.

5-3.10 Choose a comfortable text size

Choose **Setting > Font Size**, then select **Compact**, **Standard**, **Large**, **Larger**, or **Extra Large**. The workspace previews the choice immediately while keeping panel dimensions, icons, and window controls unchanged.

Cancel, the Escape key, or clicking outside the sheet restores the previously saved size. **Restore Default** previews Standard and still requires Save. Choose **Save** to keep the preference in the local SQLite settings. New app sessions apply the saved size automatically.

See also: Troubleshooting · Local Data and Privacy

5-4 Troubleshooting

Start with the visible status and the selected work. Avoid broad repair actions until you know whether the problem is network, metadata, database state, or a local path.

5-4.1 An upgrade appears to open a different or empty Library

Check **Gewu Insight > Database Location…** and the App opened by the Dock. A new icon can simply mean that the daily App moved to Applications; it does not establish that your data changed. If the displayed folder is not your intended workspace, stop before importing or recreating records. Preserve both folders and locate the original database. **Create New Database** starts a separate empty collection and is not a recovery action.

5-4.2 The database cannot be opened

For a missing folder, reconnect its disk or locate the existing workspace. For a newer schema, use a compatible newer App. For a database identity mismatch, check that the folder is the intended one rather than replacing the saved identity manually. For an already-open database or busy port, quit the other Gewu process normally. Do not remove a lock file, reset the database or terminate unrelated processes as a shortcut. See Startup for logs.

5-4.3 A PDF cannot be found after a move

1. Reselect the paper and try **Show in Finder**.
2. If the move happened in Gewu Insight, close and reopen the item to refresh the view.
3. If the file was moved manually in Finder, return it if possible, then organize it through the app.
4. Check Recycle before importing another copy.

5-4.4 Paper Files shows Missing File

This label is a path-state warning, not a failed metadata lookup: SQLite still knows the resource, but the recorded file path does not currently exist. Check **Show in Finder**, the corresponding data / AssociatedFiles bundle, and Recycle. If the file was moved outside Gewu Insight, restore it before reorganizing it through the app. Do not delete the row merely because its filename resembles the default PDF; explicitly typed attachments and the default resource remain visible for recovery.

5-4.5 Quick Look is slow on the first open

A large or previously unseen PDF may need an initial structural read and thumbnail render. The main document should appear before the full thumbnail strip. Leave the preview open briefly instead of pressing Space repeatedly; later opens can reuse page information and cached thumbnails. If the viewer never produces a page, use **Open in Preview** to check whether the file itself is valid.

5-4.6 Markdown stays on Loading Markdown

The note is read and formatted locally. Unchanged notes may reopen from a validated memory cache; first opens and changed files need a fresh render. A stalled preview shows an error after about 15 seconds. Close and reopen it to retry, or open the file directly in its normal application. Very large notes show a bounded, marked preview. No metadata refresh, DOI lookup or reimport is needed just to preview Markdown.

5-4.7 Markdown begins with unexpected metadata

Collapse **Source record** and **Properties** to read the body. If a header is malformed, the reader may show its escaped source instead of properties. The preview does not rewrite the original note. Wide tables and code blocks have their own horizontal scrolling.

5-4.8 PDF switching flashes in built-in Quick Look

The built-in preview can flash as it loads another PDF. Open **Setting > Quick Look Settings**, choose **macOS Quick Look for all files**, and Save to use the system panel for rapid comparison. Up and Down still follow the selected files. This changes the preview method; it does not alter your PDF files or default-file choices.

5-4.9 The wrong file is the default

1. Open Paper Files and verify every file type.
2. Correct a mislabeled Corrected published version, Published version, Accepted manuscript, or Preprint; expand the size footer to set annotation separately.
3. Choose **Set as Default** for the desired valid file, or choose **Use Automatic Selection** to resume automatic selection.

5-4.10 A duplicate appears after download

Compare file size and classification. New downloads use exact SHA-256 and PDF content fingerprints. A copy with annotations may be retained intentionally. Exact byte duplicates are reused, while clean and annotated peers can both remain. Keep the intended default before deleting any redundant non-default row.

5-4.11 A corrected DOI still shows old metadata

Press Return in the DOI field and wait for **Identifier updated**. Then choose **Refresh Info**. Hover over the button to check whether the exact corrected identifier was found or a service is waiting to retry. Saving an identifier and retrieving its bibliography are separate steps. Do not erase files or create another paper merely to bypass a metadata conflict; see Edit DOI and arXiv Identifiers.

5-4.12 Refresh complete reports partial results

Green **Refresh complete** means the operation ended. Successful source updates are retained even if another source was unavailable. The hover report and batch task window retain the detailed outcomes and cooldowns. Repeating a broad task immediately is usually unnecessary.

5-4.13 A normal Library item points into Recycle

With an active Library, Space or My Publications placement, app-managed operations keep or return its original to normal storage. A recycled occurrence may instead be a shortcut to that retained original. For a historical discrepancy, inspect the selected Key, file path and surviving locations before taking action. Do not empty Recycle or manually delete the path as a diagnostic shortcut.

5-4.14 Search misses the phrase you expected

Try a title fragment, surname or exact identifier. Library supports spelling tolerance and word variants, but does not automatically translate languages or invent synonyms. It searches saved metadata and names, not a semantic index of PDF content. Clear the search to return to normal folder browsing.

5-4.15 A monitor shows no papers

Read the source status. An empty successful source is not the same as a failed fetch. In Subject Monitor, allow scheduled source-level retries to finish, or use the selected subject's refresh command when a manual retry is needed. In Journal Monitor, confirm the date and filters. If a VPN is involved, test a direct connection without repeatedly refreshing.

5-4.16 Author Monitor shows a red Scholar pause

Check the stated retry time and full status message. A saved access cooldown survives restart and network changes; it is separate from a temporary connection failure or the all-author metadata counter. See Author Monitor status and recovery. Repeated refreshes do not remove a cooldown. A completed refresh that says papers need identity review instead concerns those records' identifiers or metadata, not the network.

5-4.17 A journal count and the visible cards disagree

This is usually expected when **Field Filter** is active. Journal counts describe all stored papers for the selected journal and day; field choices narrow the visible cards. Clear or broaden the field selection before diagnosing a missing fetch.

5-4.18 A journal disappeared from the configured list

Open **Journal Settings** and inspect the disabled history before editing configuration files. Removing a journal disables its subscription but preserves previously collected papers and history. Add it again to restore monitoring. Because the subscription list is SQLite state, replacing source files should not be used as a journal-list repair.

5-4.19 LaTeX remains visible as source text

On a cold launch, allow the math renderer a moment to become ready. The active day should re-render automatically. If raw `$...$` text remains, check network availability for the renderer, then reload the same view once; switching calendar dates should not be required.

5-4.20 Folder Loader looks incomplete

Open the live or previous Loader log and the result guide. Confirm the metadata mode, source path, and any stop action. Re-running the same source should reuse existing files rather than overwrite the primary PDF.

5-4.21 A default file cannot be deleted

The Default is kept as the protected main file, including for non-PDFs. Choose another eligible resource with **Set as Default** before removing the old main file from Paper Files. The old resource remains in Associated Files until you explicitly remove it. To remove the paper itself, use the normal Library reference/Recycle workflow, which preserves shared references. Do not bypass a main-file warning in Finder. An **Invalid file** remains an audit record and is never eligible to become Default.

5-4.22 Change where website links open

Open Setting > Website Links and save **System Default Browser** to use the browser selected in macOS. The default, **Gewu Insight Browser**, opens a new built-in webpage tab or reuses its unused first tab. This covers website actions in all five research views, including publisher arrows, DOI/arXiv links, scholarly profiles, citation-source links, and links inside Note. Existing web pages stay open. Local PDF, Preview, Finder, download, and file-import actions are unchanged.

5-4.23 Home works offline but a website will not load

Home displays local Favorites without a Google request. Missing site icons fall back to initials; they do not prevent opening Home. Search results and account actions require a connection. Read the load error, check the destination and Clash route, and try Reload once. A website may reject embedded browsers or have its own session policy; an unsupported-browser banner alone does not prove why a sign-in ended. Open in Safari is a manual compatibility fallback, with a separate sign-in session.

5-4.24 A publisher stays on security verification

A verification loop does not establish that you were identified as AI. Browser supports standard internal child documents used by embedded widgets, but a publisher's verification can still reject an embedded browser. Try Reload once; if the loop persists, use **Open in Safari**. Safari has a separate session. For publisher support, note the public URL, time, and displayed Ray ID, without sharing cookies or passwords. Do not repeatedly retry or bypass the challenge.

5-4.25 AutoFill does not appear or does not fill

Automatic triggering is deliberately limited to an explicit click in a clearly marked login field. Ordinary email, search, registration, verification-code, and uncertain fields do not qualify. Cancellation uses up the automatic offer for that field kind/site during the App session. Right-click the actual input and choose **AutoFill > Passwords** to retry manually.

Unlock and select the appropriate entry yourself. If the item says **Passkey** and shows a creation date, it is not a text password. If a real Password entry closes without filling, macOS or that website may not support the return path in this embedded view. Gewu does not read your vault, relay credentials, or submit forms to work around it. Use the website's supported login flow or Safari; do not send passwords or unmasked credential screenshots in a support report.

> **Important**
> Before any manual filesystem repair, quit the app and make a full backup of the database and managed files together.

5-4.26 A WebCrypto keychain prompt keeps appearing

The current Browser declines persistent webpage-key access before WebKit asks Keychain for the **Gewu Insight WebCrypto Master Key**. If that exact prompt still appears, quit Gewu and reopen the updated App from Applications; an older Dock shortcut may still point elsewhere. Record the App version/build and the dialog's exact wording if it persists. Do not delete the keychain item, reset the Browser profile or allow all applications as a repair. A macOS Passwords or AutoFill authentication request is a separate operation.

5-4.27 Science displays a PDF but Save shows Article only

Wait for the Science ePDF reader to finish loading, then reopen **Save to Gewu > Details** and check the title, DOI and PDF status. Choose **Use Loaded PDF** if offered. A changed document, incomplete transfer or file above 128 MB can prevent the original from being read. Choose a saved local PDF, or save the article alone and attach its PDF later under the same Gewu Key. See Save a Science ePDF; a failed fresh download does not prove that the loaded copy is unusable.

5-4.28 The in-app guide and Print Guide look like different editions

The searchable guide is served by the running Gewu backend; Print Guide opens the fixed-layout PDF packaged in the App. A source-backed backend reads its configured checkout, while a standalone backend uses its packaged source. If one guide is older, the running code and bundled resources are out of sync. Follow Back Up and Upgrade to install a complete matching version. For a source build, regenerate the PDF and package it before reopening the App. Clearing Monitor Cache does not replace a bundled guide.

5-4.29 A long task keeps returning as Continue

That task was paused or interrupted, so its remaining queue is saved. Choose the footer's **Details**, or open **Loader > Refresh All Metadata** and select the task. Choose **Stop** to end it permanently. Completed work remains. Closing the window merely hides it; **Pause** deliberately keeps it resumable.

5-4.30 Two Paper Files cards seem to be the same file

Two folders showing the same Gewu Key normally share a single Paper Files bundle. Within that bundle, different edited PDFs can have similar names. Use **Show in Finder** and inspect each resource before deleting anything. Exact unchanged reimports are reused, but changing a DOI, restarting, or clearing Monitor Cache is not a historical duplicate-repair command. If two cards reveal the same physical file, report the Gewu Key and the steps that produced it through **Help > Contact Us**.

5-4.31 Contact support

Help > Contact Us opens your default email application's draft to **info@gewu-insight.com**, with subject **Gewu Insight — Feedback and Support**. If no email handler is configured, use that address in your preferred mail service. Include the version/build, operation, displayed error and relevant screenshot; never include tokens or passwords. The app does not send the message automatically.

5-4.32 Download reports supplements but no main PDF

This reports two separate outcomes: some supplementary PDFs were saved, but the main article did not arrive as a readable PDF. MP4, ZIP, spreadsheets and other non-PDF supplements are skipped by automatic PDF collection. They do not turn a successful main-PDF download into a failure. Check access to the actual main PDF in Gewu Browser, then retry Download or use Save to Gewu with an already-loaded original. See Choose the right save action.

5-4.33 A journal's unread dot returns after changing fields

This is expected when the newly visible field contains unread papers. A tab counts only its currently visible unread results. Click or use Up/Down on each paper you read; opening the tab or changing fields does not acknowledge the whole list. Read history is not reset by the updated rule.

5-4.34 A correction or retraction is missing

For a saved original, check its DOI and request Refresh Info; inspect the source outcomes and Paper Files. In Journal Monitor, choose the notice's publication date and check the Journal and Field filters before expanding Publication updates. Saved papers do not receive a continuous retraction watch. Missing provider relations, delayed indexing or unavailable sources can leave a notice undiscovered. Verify the publisher record when necessary; do not create a duplicate article simply to store its notice.

5-4.35 A mathematical note shows formula source

Built-in Markdown math renders locally. Check paired formula delimiters and keep formulas outside code blocks. Unsupported or oversized expressions remain readable as source. This does not require refreshing the paper's metadata or downloading its PDF again.

See also: Back Up or Move to Another Mac · Startup, Monitor Cache, and Text Size · Publisher and Network Access

5-5 Storage Reference

The relative paths below are inside your selected Gewu Database, not inside the App or source checkout. Find the active folder through Gewu Insight > Database Location…. The remaining machine locations are listed separately for backup and diagnosis.

Location	Purpose
data/paper_monitor.sqlite	Authoritative database for papers, identities, relationships, settings, operational state, Browser bookmarks, and visit history. May contain saved metadata API keys; treat backups as private.
data/Library	Visible main Library root, canonical main files and secondary Finder aliases.
data/Spaces	Independent Space roots.
data/MyPublications	Managed local files for My Publications.
data/AssociatedFiles	Mirrored bundles for versions, supplements, and attachments.
data/pdfs	Legacy managed PDF location retained for compatibility where needed.
`reports`	Generated daily reports, exports and legacy analysis artifacts. New AI analyses saved to Paper Files belong to the paper's managed associated bundle.
`config`	Workspace configuration retained with the research data. Saved per-user settings and authoritative journal subscriptions also live in SQLite.
`backups`	Pre-schema SQLite snapshots, migration reports and retained recovery material when present. Backups in this folder are on the same disk and do not replace a separate full backup.

Quick Look page thumbnails use a separate, automatically trimmed temporary cache. It is rebuildable preview data, not part of the Library, Associated Files, or the SQLite record, so it does not belong in a research backup.

5-5.1 App and machine locations

- `/Applications/Gewu Insight.app`
  Daily App and packaged resources. Research files are stored in the selected workspace.
- `~/Library/Application Support/Gewu Insight/database-location.plist`
  This Mac's remembered workspace path and database identity. It is not the database itself.
- `~/Library/Logs/Gewu Insight/paper_monitor_launcher.log`
  Installed-App startup diagnostics.
- `~/Library/Caches/com.zhengranwu.gewuinsight`
  Default rebuildable cache for the standalone App. A source-backed installation can retain its configured workspace data/cache, with journal logos under data/cache/journal_logos.

Here ~ means the current Mac user's home folder. Browser website data and its disposable favicon cache remain in a separate macOS-managed profile. They are distinct from SQLite bookmarks/history and monitor caches. See Browser and credential backup boundaries before moving to another Mac.

5-5.2 Associated Files layout

An associated bundle follows the managed scope and path, then a work-specific bundle. Its children separate `Versions`, `Supplements`, and `Files`. The exact physical name can change as the paper moves; the Gewu Key remains stable.

5-5.3 Database and exports

SQLite is the only source of truth. JSON, HTML, Markdown, Word, LaTeX, and PDF outputs are exports, diagnostics, or reading artifacts. Editing one of them does not edit the Library record.

> **Important**
> Do not delete, rebuild, or "clean" the database because a generated JSON file looks stale. Diagnose the selected record and local path first.

See also: How Gewu Insight Stores Files · Back Up or Move to Another Mac · Local Data and Privacy

Copyright and product notice

Gewu Insight is source-available research software. Copyright © 2026 Lingyuan Kong and Zhengran Wu. All Rights Reserved, except for the limited permissions in the Gewu Insight Academic and Non-Commercial Research License.

The License permits academic/non-commercial research use, including downloading, installing, building, running and modifying the software. Original and modified versions, derivative works and patches may be shared free of charge for those purposes under the same research-use terms, with retained notices and clearly identified modifications. Public research forks and pull requests within those terms need no separate permission. Commercial use, commercial redistribution, sublicensing and integration into commercial products or services require separate prior written authorization. This is not an open-source license. The complete LICENSE in the distribution governs; the same text is included in the App resources. Contact: info@gewu-insight.com.

Third-party software, fonts, artwork and research content retain their own terms. The Gewu License does not grant rights in those materials. See THIRD_PARTY_NOTICES.md and the dependency and asset audit before redistribution. Independent research outputs do not inherit this software license merely through use of Gewu.

This manual describes the product as shipped at the version printed on the title page. Online services, publisher access, and third-party metadata may change independently of the application.